

Shifted Catalan Hankel Polynomials

Exact coefficients, sharp recurrences,
and a denominator conjecture

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With Part II (28 September 2026): root collisions and sharp recurrences for arbitrary polynomial multipliers,
and Parts III and IV (29 September 2026): cyclotomic resonances of one repeated root, and two-endpoint confluence

Abstract

For the Catalan numbers C_n , consider the size- N determinant $D_N^{(m)}(a, b) = \det(aC_{i+j+m} + bC_{i+j+m+1})_{0 \leq i, j < N}$. We give a self-contained proof of the denominator-exponent conjecture stated in Paul Barry's 2020 note on this family. With $d = \binom{m}{2}$, the generating function is $P_m(t)/(1 - (a + 2b)t + b^2t^2)^{d+1}$. The denominator is minimal whenever $b \neq 0$, $a \neq 0$, and $a \neq -4b$. We determine every exceptional case, prove that $\deg P_m = (m-1)^2$ for $b \neq 0$, and establish a reciprocal symmetry of the numerator. An explicit positive-integer coefficient formula also identifies and repairs a one-unit shift in the source's separate Conjecture 2. The essential degree reduction is a Vandermonde cancellation in confluent Christoffel minors. It yields an exact sum of two exponentials with polynomial coefficients, a finite full asymptotic expansion, and an explicit first correction. Independent integer and rational computations, including direct determinants and polynomial-identity certificates at fixed sizes, accompany the proofs.

Addition of 28 September 2026 (batch 39). Part II, from a later manuscript, replaces $x^m(a + bx)$ by an arbitrary nonzero multiplier $q(x) = \gamma x^m(x-4)^\ell \prod_i (x - c_i)^{t_i}$. Each exponential sector has a polynomial coefficient of exact degree $\binom{m}{2} + \binom{\ell+1}{2} + \sum_i k_i(t_i - k_i)$ with a nonzero product leading coefficient, so off resonance the minimal recurrence order is $\prod_i (t_i + 1)(\binom{m}{2} + \binom{\ell+1}{2} + 1 + \frac{1}{6} \sum_i t_i(t_i - 1))$; a secondary Hankel determinant detects exactly when it drops, and a certified finite-prefix algorithm handles the resonances.

Addition of 29 September 2026 (batch 42). Part III gives every characteristic multiplicity for one repeated root $(x - c)^t$, $c = 2 + z + z^{-1} \notin \{0, 4\}$; it depends on the sign of z^L , not only on t and $L = \text{ord}(z^2)$.

Addition of 29 September 2026 (batch 53). Part IV lets roots approach the endpoints 0 and 4 on the scale N^{-2} : the normalized determinant is an even convergent series in $1/(N + \deg q/2)$, and outward displacements are negligible exactly when they are $o(N^{-2})$.

Main result. For all nonexceptional complex parameters,

$$\text{minimal constant-coefficient recurrence order} = m(m-1) + 2.$$

The proof is uniform in the determinant size and the Catalan shift.

Status and scope. This is an unrefereed, AI-assisted mathematical manuscript, not a formalized proof. Its theorems are supported by explicit arguments, not inferred from finite tests. The target is a conjecture in a specified source version; no claim of first discovery or exhaustive verification of present-day open status is made. Classical Christoffel formulas and the known general rationality theory are credited, rather than presented as new.

Reproducibility. The accompanying archive contains this L^AT_EX source, the PDF, a standard-library-only Python implementation, an independently organized verification suite, and exact machine-readable data. No external mathematical software is needed to rerun the tests.

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1 The problem and the contribution

Let

$$C_n = \frac{1}{n+1} \binom{2n}{n} \quad (n \in \mathbb{N}_0) \quad (1.1)$$

be the Catalan numbers. We use determinant size, rather than the last row index, throughout:

$$D_N^{(m)}(a, b) = \det_{0 \leq i, j < N} (aC_{i+j+m} + bC_{i+j+m+1}), \quad D_0^{(m)} = 1. \quad (1.2)$$

Here $m, N \in \mathbb{N}_0$, and the parameters may be indeterminates or complex numbers. Define the ordinary generating function

$$F_m(t; a, b) = \sum_{N \geq 0} D_N^{(m)}(a, b) t^N. \quad (1.3)$$

All generating-function identities below first make sense as identities of formal power series. Assertions about poles concern the associated rational functions.

Barry's preprint [1], version 1, explicitly conjectures on page 4 that the exponent of the quadratic generating-function denominator is $\binom{m}{2} + 1$. The same page contains a more complicated coefficient conjecture. These are distinct statements: the denominator claim is correct in the uniform sense proved here, whereas the printed coefficient formula uses a shift one smaller than its accompanying determinant. Appendix A makes the translation precise.

There is substantial prior theory. Dougherty, French, Saderholm, and Qian [2] studied recurrences for linear combinations of Catalan numbers. Krattenthaler [3] gives a detailed account of the Christoffel-type determinant identity, confluence, and constant-coefficient recurrences; his Section 8 also records Elouafi's prior proof of general Catalan recurrence assertions. For a multiplier of degree r , that general construction supplies an order at most 2^r . Consequently, rationality by itself is *not* the research claim here. For the special multiplier $x^m(a + bx)$, the objective is the exact, much smaller order, together with its sharp parameter classification.

The article separates two useful calculations. Jacobi orthogonality immediately exposes the individual coefficients. A confluent determinant, on the other hand, exposes the dependence on N and explains the precise recurrence order. Their independent implementations provide a useful check against indexing and normalization errors.

1.1 A notation guide

We shall repeatedly use

$$d = \binom{m}{2}, \quad e = d + 1, \quad K = (m - 1)^2, \quad q(t) = 1 - (a + 2b)t + b^2 t^2. \quad (1.4)$$

The shifted pure Catalan determinant and its leading coefficient are

$$H_N^{(m)} = \det_{0 \leq i, j < N} (C_{i+j+m}) = \prod_{1 \leq j \leq i < m} \frac{2N + i + j}{i + j}, \quad (1.5)$$

$$h_m = \prod_{1 \leq j \leq i < m} \frac{2}{i + j}. \quad (1.6)$$

Every empty product and empty determinant equals 1. In particular, $H_N^{(0)} = H_N^{(1)} = 1$ and $h_0 = h_1 = 1$. We prove (1.5) rather than assuming it. As a polynomial in N , $H_N^{(m)}$ has degree d and leading coefficient h_m . The notation $(u)_k = u(u+1) \cdots (u+k-1)$ denotes a rising factorial, with $(u)_0 = 1$.

2 Statements of the main theorems

Theorem 2.1 (Explicit coefficients). *For every $N, m \in \mathbb{N}_0$,*

$$D_N^{(m)}(a, b) = \sum_{k=0}^N c_{N,m,k} a^k b^{N-k}, \quad c_{N,m,k} = H_N^{(m+1)} \binom{N}{k} \frac{(N+m+1)_k}{4^k (m+\frac{1}{2})_k}. \quad (2.1)$$

All the coefficients $c_{N,m,k}$ are positive integers. Equivalently,

$$c_{N,m,k} = H_N^{(m+1)} \binom{N}{k} \frac{(N+m+k)!(2m)!(m+k)!}{(N+m)!(2m+2k)!m!}. \quad (2.2)$$

They can be generated without evaluating any determinant:

$$c_{N,m,0} = H_N^{(m+1)}, \quad \frac{c_{N,m,k+1}}{c_{N,m,k}} = \frac{(N-k)(N+m+1+k)}{2(k+1)(2m+2k+1)}. \quad (2.3)$$

Theorem 2.2 (Uniform rational form and sharp denominator). *For every $m \in \mathbb{N}_0$ there is a polynomial $P_m \in \mathbb{Z}[a, b, t]$ such that*

$$F_m(t; a, b) = \frac{P_m(t; a, b)}{q(t)^{d+1}}. \quad (2.4)$$

It satisfies $P_m(0; a, b) = 1$ and $\deg_t P_m \leq K$. If $b \neq 0$, then $\deg_t P_m = K$ and

$$P_m(t; a, b) = s_m (bt)^K P_m\left(\frac{1}{b^2 t}; a, b\right), \quad s_m = (-1)^{(m-1)(m-2)/2}. \quad (2.5)$$

In particular, $[t^K]P_m = s_m b^K$. If $b \neq 0$, $a \neq 0$, and $a \neq -4b$, the representation (2.4) is reduced over $\mathbb{C}[t]$. The sequence has minimal constant-coefficient recurrence characteristic polynomial

$$(X^2 - (a+2b)X + b^2)^{d+1}, \quad (2.6)$$

and hence minimal order $2d+2 = m(m-1)+2$.

Here “minimal” refers even to recurrences allowed to start after a finite initial segment. Thus minimality is not an artifact of the empty-determinant convention. The polynomial numerator in (2.4) remains the *uniform, possibly unreduced* numerator at exceptional parameters.

Theorem 2.3 (Every exceptional denominator). *For fixed complex a, b , the reduced denominator of F_m , normalized to have constant term 1, is exactly the following:*

Parameter condition	Reduced denominator
$a = b = 0$	1
$b = 0, a \neq 0$	$(1 - at)^{\binom{m}{2}+1}$
$a = 0, b \neq 0$	$(1 - bt)^{\binom{m+1}{2}+1}$
$a = -4b, b \neq 0$	$(1 + bt)^{\binom{m}{2}+2}$
$b \neq 0, a(a+4b) \neq 0$	$q(t)^{\binom{m}{2}+1}$

Theorem 2.4 (Finite asymptotic expansion). *Let $a, b > 0$ be fixed and put*

$$r = \frac{a/b + 2 + \sqrt{(a/b)(a/b+4)}}{2} > 1, \quad \lambda_+ = br, \quad \lambda_- = b/r. \quad (2.7)$$

There are explicit degree- d polynomials $B_+(N), B_-(N)$ such that

$$D_N^{(m)}(a, b) = B_+(N)\lambda_+^N + B_-(N)\lambda_-^N \quad (N \geq 0). \quad (2.8)$$

In particular,

$$D_N^{(m)}(a, b) = A_m N^d \lambda_+^N \left(1 + \frac{d \left(\frac{m}{2} - \frac{1}{r-1} \right)}{N} + O(N^{-2}) + O(r^{-2N}) \right), \quad (2.9)$$

$$A_m = h_m \frac{r^{m+1}}{(r-1)^m(r+1)}. \quad (2.10)$$

The entire algebraic expansion terminates after N^{-d} ; the second exponential is also multiplied by a polynomial of degree d . For $d = 0$, there are no algebraic correction terms at all.

2.1 The scale of the improvement

For a degree- $(m+1)$ multiplier the general 2^{m+1} construction need not be minimal. The theorem determines the actual order in this family. The first values are:

m	Exponent e	Exact order $2e$	Numerator degree K	2^{m+1}
0	1	2	1	2
1	1	2	0	4
2	2	4	1	8
3	4	8	4	16
4	7	14	9	32
5	11	22	16	64
6	16	32	25	128
8	29	58	49	512
10	46	92	81	2048

The reduction is not obtained by fitting a shorter recurrence to a long initial segment. It comes from a cancellation of polynomial degree in a family of minors, proved in Section 5.

3 Moments, orthogonality, and the coefficient formula

3.1 Catalan moments and a normalized Jacobi polynomial

For $m \in \mathbb{N}_0$ define a positive measure on $(0, 4)$ by

$$d\mu_m(x) = \frac{1}{2\pi} x^{m-1/2} (4-x)^{1/2} dx. \quad (3.1)$$

The beta integral, after $x = 4t$, gives

$$\int_0^4 x^n d\mu_m(x) = \frac{4^{n+m+1}}{2\pi} B(n+m+\frac{1}{2}, \frac{3}{2}) = C_{n+m}. \quad (3.2)$$

Consequently every moment matrix is positive definite, so its determinant is nonzero and its monic orthogonal polynomials are well-defined.

Lemma 3.1 (The monic polynomial). *The degree- N monic orthogonal polynomial for μ_m is*

$$p_N^{(m)}(x) = (-1)^N \frac{4^N (m + \frac{1}{2})_N}{(N + m + 1)_N} {}_2F_1 \left(\begin{matrix} -N, N + m + 1 \\ m + \frac{1}{2} \end{matrix}; x/4 \right). \quad (3.3)$$

The hypergeometric series terminates after $N + 1$ terms.

Proof. Write $\alpha = m - \frac{1}{2}$, $\beta = \frac{1}{2}$, so both are greater than -1 . A Rodrigues representation on $0 < t < 1$ is

$$\frac{t^{-\alpha}(1-t)^{-\beta}}{(\alpha+1)_N} \frac{d^N}{dt^N} \left(t^{N+\alpha}(1-t)^{N+\beta} \right) = {}_2F_1 \left(\begin{matrix} -N, N + \alpha + \beta + 1 \\ \alpha + 1 \end{matrix}; t \right). \quad (3.4)$$

Leibniz's rule gives the polynomial on the right, including its normalization at $t = 0$. Multiplying the left side by $t^j(1-t)^\beta$ and integrating by parts N times yields zero for $0 \leq j < N$. Every boundary term vanishes: after at most $N - 1$ derivatives, the endpoint exponents that can occur are at least $\alpha + 1 > 0$ and $\beta + 1 > 0$. The final derivative of t^j is zero.

The coefficient of t^N on the right is $(-1)^N (N + \alpha + \beta + 1)_N / (\alpha + 1)_N$. Replacing t by $x/4$ and normalizing this leading coefficient gives (3.3). This also identifies the polynomial with the standard Jacobi family; see the conventional hypergeometric representation in DLMF [4], Eq. 18.5.7. $\square$

3.2 A characteristic-polynomial determinant identity

The following elementary identity will be used twice.

Lemma 3.2. *Let v_j be the moments of a positive measure, let $H_N = \det(v_{i+j})_{0 \leq i, j < N}$, and let p_N be its monic orthogonal polynomial. Then*

$$\det_{0 \leq i, j < N} (v_{i+j+1} - z v_{i+j}) = (-1)^N H_N p_N(z). \quad (3.5)$$

Proof. The monic polynomial is represented by

$$p_N(z) = \frac{1}{H_N} \det \begin{pmatrix} v_0 & v_1 & \cdots & v_N \\ v_1 & v_2 & \cdots & v_{N+1} \\ \vdots & \vdots & & \vdots \\ v_{N-1} & v_N & \cdots & v_{2N-1} \\ 1 & z & \cdots & z^N \end{pmatrix}.$$

Its leading coefficient is one, and integrating it against any power z^j , $j < N$, creates a repeated row. Thus it is indeed p_N . Starting with the last column and moving left, replace col_j by $\text{col}_j - z \text{col}_{j-1}$. The last row becomes $(1, 0, \dots, 0)$. Expansion along it gives (3.5), with sign $(-1)^N$. $\square$

Applying the lemma to μ_m and $z = -a/b$ gives, for $b \neq 0$,

$$D_N^{(m)}(a, b) = (-b)^N H_N^{(m)} p_N^{(m)}(-a/b). \quad (3.6)$$

3.3 A proof of the shifted Catalan product

For $m = 0$, (3.3) reduces to

$$p_n^{(0)}(x) = \sum_{j=0}^n (-1)^{n-j} \binom{n+j}{2j} x^j. \quad (3.7)$$

Its initial values and recurrence are

$$p_0^{(0)} = 1, \quad p_1^{(0)} = x - 1, \quad p_{n+2}^{(0)} = (x - 2)p_{n+1}^{(0)} - p_n^{(0)}. \quad (3.8)$$

These

$p_n^{(0)}$, the coefficient 1 of $p_{n-1}^{(0)}$ in $x p_n^{(0)}$ implies $\eta_n = \eta_{n-1}$. Indeed, the two ways of writing $\langle x p_n^{(0)}, p_{n-1}^{(0)} \rangle$ give η_{n-1} and η_n , respectively. Since $\eta_0 = C_0 = 1$, all these norms equal 1. Changing from monomials to monic orthogonal polynomials is unitriangular, and therefore $H_N^{(0)} = \prod_{n=0}^{N-1} \eta_n = 1$.

Next set $z = 0$ in (3.5) and use (3.3). We obtain

$$\frac{H_N^{(m+1)}}{H_N^{(m)}} = (-1)^N p_N^{(m)}(0) = \frac{4^N (m + \frac{1}{2})_N}{(N + m + 1)_N} = \prod_{j=1}^m \frac{2N + m + j}{m + j}. \quad (3.9)$$

For completeness, the middle expression equals

$$\frac{(2m + 2N)! m! (N + m)!}{(m + N)! (2m)! (2N + m)!} = \frac{(2N + 2m)! m!}{(2m)! (2N + m)!},$$

which is the last product. Induction on m , beginning with $H_N^{(0)} = 1$, proves (1.5).

3.4 Completion of the coefficient proof

Insert (3.3) into (3.6). The constant factor is $H_N^{(m+1)}$ by (3.9). In each summand, $(-N)_k (-1)^k / k! = \binom{N}{k}$. This gives exactly (2.1). The formula is a polynomial identity for $b \neq 0$, so it holds for $b = 0$ as well. The duplication identity for a rising half-integer factorial, or direct cancellation, gives (2.2). Taking adjacent coefficient ratios gives (2.3).

All factors in (2.1) are positive rational numbers. The left side is a determinant with entries in $\mathbb{Z}[a, b]$, so its coefficients are integers. This proves positivity and integrality simultaneously and completes Theorem 2.1. In particular,

$$c_{N,m,0} = H_N^{(m+1)}, \quad c_{N,m,N} = H_N^{(m)}. \quad (3.10)$$

3.5 Real roots and coefficient inequalities

Corollary 3.3. *For $N \geq 1$, the polynomial $a \mapsto D_N^{(m)}(a, 1)$ has exactly N simple real roots, all in $(-4, 0)$. For $N \geq 2$, its normalized coefficients $u_k = c_{N,m,k} / \binom{N}{k}$ satisfy*

$$u_k^2 > u_{k-1} u_{k+1} \quad (1 \leq k < N). \quad (3.11)$$

Proof. A degree- N orthogonal polynomial for a strictly positive density on $(0, 4)$ has N simple roots in that interval. One elementary proof is to multiply it by the product of its distinct sign-changing roots in $(0, 4)$. If there were fewer than N such roots, that product would have degree below N , while the integral of its product with the orthogonal polynomial would have constant nonzero sign. This contradicts orthogonality. Equation (3.6) now negates the roots.

Alternatively, the coefficient inequality follows without any root theory. Theorem 2.1 gives

$$\frac{u_{k+1}}{u_k} = \frac{N + m + 1 + k}{4(m + \frac{1}{2} + k)}.$$

This is strictly decreasing in k , because the numerator's constant term exceeds the denominator's unscaled constant term by $N + 1/2$. The decrease is equivalent to (3.11). $\square$

4 A fixed-size confluent determinant

The explicit coefficient formula grows in length with N . To understand the recurrence order, we need a representation whose determinant size depends only on m . Define

$$R_n(y) = (-1)^n p_n^{(0)}(-y) = \sum_{i=0}^n \binom{n+i}{2i} y^i. \quad (4.1)$$

Thus

$$R_0(y) = 1, \quad R_1(y) = y + 1, \quad R_{n+2}(y) = (y + 2)R_{n+1}(y) - R_n(y). \quad (4.2)$$

Extend R_n to all integer indices using this recurrence. Since $R_{-1} = R_0 = 1$, it also satisfies the reflection identity

$$R_{-n-1}(y) = R_n(y) \quad (n \in \mathbb{Z}). \quad (4.3)$$

The reversed form of the polynomials is convenient because it eliminates unnecessary signs from Christoffel's identity.

Lemma 4.1 (Christoffel determinant in the present normalization). *For pairwise distinct $y_1, \dots, y_r$,*

$$\det_{0 \leq i, j < N} \int_0^4 x^{i+j} \prod_{\ell=1}^r (x + y_\ell) d\mu_0(x) = \frac{\det[R_{N+j}(y_i)]_{\substack{1 \leq i \leq r \\ 0 \leq j < r}}}{\prod_{1 \leq i < j \leq r} (y_j - y_i)}. \quad (4.4)$$

The quotient extends polynomially when some y_i coincide.

Proof. Here is a proof that also verifies the normalization. Write $\Delta(x_1, \dots, x_N) = \det[x_i^{j-1}]_{1 \leq i, j \leq N}$. Expanding two determinants and integrating termwise gives the identity

$$\frac{1}{N!} \int \cdots \int \det[f_i(x_j)] \det[g_i(x_j)] \prod_j d\mu_0(x_j) = \det \left[\int f_i(x) g_j(x) d\mu_0(x) \right].$$

No analytic interchange difficulty occurs here: the sums are finite and all polynomials are integrable on a bounded interval.

For external variables $z_1, \dots, z_r$ we have

$$\Delta(x, z) = \Delta(x) \Delta(z) \prod_{i=1}^N \prod_{\ell=1}^r (z_\ell - x_i).$$

In the determinant defining $\Delta(x, z)$, replace monomials by the monic polynomials $p_j^{(0)}$; the determinant is unchanged. Expand along its first N rows and integrate against $\Delta(x)$. By orthogonality, every choice of columns vanishes except columns of degrees $0, \dots, N-1$. That surviving integral equals $N! H_N^{(0)} = N!$. The remaining minor is $\det[p_{N+j}^{(0)}(z_i)]$. Dividing by $\Delta(z)$ yields the identity for the multiplier $\prod_\ell (z_\ell - x)$.

Finally put $z_\ell = -y_\ell$ and use $p_{N+j}^{(0)}(-y) = (-1)^{N+j} R_{N+j}(y)$. The signs from the multiplier, column factors, and Vandermonde cancel, giving (4.4). The numerator is alternating in the y_i , hence divisible by their Vandermonde as a polynomial. This proves the extension statement. This is the classical fixed-size identity discussed in [3]; its inclusion here is a normalization proof, not a novelty claim. $\square$

Take $r = m + 1$, let the first m variables tend to zero, and put the last one equal to $y = a/b \neq 0$. Dividing by the Vandermonde and taking the limit replaces those first rows by Taylor coefficients. Since $[y^i] R_{N+j}(y) = \binom{N+j+i}{2i}$, we obtain:

Proposition 4.2 (Confluent representation). *If $ab \neq 0$, then*

$$D_N^{(m)}(a, b) = b^N y^{-m} \det \begin{pmatrix} \binom{N+j+i}{2i} & (0 \leq i < m, 0 \leq j \leq m) \\ R_{N+j}(y) & (0 \leq j \leq m) \end{pmatrix}, \quad y = a/b. \quad (4.5)$$

The upper block has m rows and the lower block has one row. For $m = 0$ the formula simply reads $D_N^{(0)} = b^N R_N(a/b)$.

Define the $m \times m$ minors

$$M_{m,\ell}(N) = \det_{\substack{0 \leq i < m \\ j \in J_\ell}} \binom{N+j+i}{2i}, \quad J_\ell = \{0, 1, \dots, m\} \setminus \{\ell\}, \quad (4.6)$$

where columns are in increasing order and $0 \leq \ell \leq m$. Expansion of (4.5) along its last row gives

$$D_N^{(m)}(a, b) = b^N y^{-m} \sum_{\ell=0}^m (-1)^{m+\ell} M_{m,\ell}(N) R_{N+\ell}(y). \quad (4.7)$$

Although a naive degree bound for $M_{m,\ell}$ is $m(m-1) = 2d$, its actual degree is only d . This is the key point.

5 The degree cancellation and its first correction

Lemma 5.1 (Sharp minor degree). *For $0 \leq \ell \leq m$, $M_{m,\ell}(N)$ is a polynomial of degree $d = \binom{m}{2}$ with leading coefficient*

$$[N^d]M_{m,\ell}(N) = h_m \binom{m}{\ell}. \quad (5.1)$$

For $d \geq 1$, the next coefficient is

$$[N^{d-1}]M_{m,\ell}(N) = h_m \binom{m}{\ell} \left(\frac{d(m+2)}{2} - \frac{(m-1)\ell}{2} \right). \quad (5.2)$$

Proof. Let $f_i(x) = \binom{x+i}{2i}$, so f_i has degree $2i$ and leading coefficient $1/(2i)!$. The determinant $\det[f_i(x_j)]_{0 \leq i, j < m}$ is alternating in its m independent variables. It is divisible by $\Delta(x) = \prod_{p < q} (x_q - x_p)$, whose degree is d . The determinant has total degree at most $\sum_i 2i = 2d$; its quotient by Δ therefore has total degree at most d . Substituting $x_j = N + j$ for $j \in J_\ell$ makes Δ a nonzero constant. Hence the degree in N is at most d .

To obtain the leading coefficient, expand $f_i(N + j)$ in Taylor powers of j . The first possible nonzero determinant uses distinct Taylor orders $0, 1, \dots, m-1$; these have the least possible sum d . The coefficient of N^d is

$$\frac{\det[\binom{2i}{r}]_{0 \leq i, r < m}}{\prod_{i=0}^{m-1} (2i)!} \Delta(J_\ell) = \frac{2^d \Delta(J_\ell)}{\prod_{i=0}^{m-1} (2i)!}. \quad (5.3)$$

To see the second equality, regard $\binom{2i}{r}$ as a polynomial of degree r in i with leading coefficient $2^r/r!$ and evaluate its determinant at $i = 0, \dots, m-1$. The Vandermonde is $\prod_{r=1}^{m-1} r!$, so these factorials cancel. Removing ℓ from $0, \dots, m$ also gives

$$\Delta(J_\ell) = \binom{m}{\ell} \prod_{r=1}^{m-1} r!. \quad (5.4)$$

Since

$$h_m = \frac{2^d \prod_{r=1}^{m-1} r!}{\prod_{i=0}^{m-1} (2i)!} \quad (m \geq 1),$$

we obtain (5.1). It is nonzero, proving the degree assertion. The case $m = 0$ is the empty determinant and is immediate.

For the next coefficient, observe the symmetric factorization

$$f_i(x) = \frac{1}{(2i)!} \prod_{r=1}^i \left(\left(x + \frac{1}{2}\right)^2 - \left(r - \frac{1}{2}\right)^2 \right). \quad (5.5)$$

Thus $f_i(x) = (x + \frac{1}{2})^{2i} / (2i)!$ plus terms of degree at most $2i - 2$. Replacing one or more rows by those lower-degree terms reduces the total degree by at least two. The same alternating argument then bounds their degree after translation by $d - 2$. The leading two terms are consequently obtained from

$$\det \left[\frac{(N + j + \frac{1}{2})^{2i}}{(2i)!} \right]_{0 \leq i < m, j \in J_\ell} = \frac{\prod_{\substack{p < q \\ p, q \in J_\ell}} (q - p)(2N + p + q + 1)}{\prod_{i=0}^{m-1} (2i)!}. \quad (5.6)$$

The relative coefficient of N^{d-1} is

$$\frac{1}{2} \sum_{\substack{p < q \\ p, q \in J_\ell}} (p + q + 1) = \frac{1}{2} \left((m-1) \left(\frac{m(m+1)}{2} - \ell \right) + d \right),$$

which simplifies to (5.2). If $d = 1$, the putative remainder of degree $d - 2 = -1$ is identically zero. $\square$

It is useful to collect the minors in a polynomial in an auxiliary variable r :

$$Q_m(N; r) = \sum_{\ell=0}^m (-1)^{m+\ell} M_{m,\ell}(N) r^\ell. \quad (5.7)$$

By the binomial theorem,

$$[N^d] Q_m(N; r) = h_m(r - 1)^m. \quad (5.8)$$

When $r \neq 1$ and $d \geq 1$, the relative next coefficient is

$$\frac{[N^{d-1}] Q_m(N; r)}{[N^d] Q_m(N; r)} = \frac{d(m+2)}{2} - d \frac{r}{r-1} = d \left(\frac{m}{2} - \frac{1}{r-1} \right). \quad (5.9)$$

Indeed, applying $r \, d/dr$ to $(r - 1)^m$ sums the terms weighted by ℓ in (5.2). The exact value d , rather than an upper bound $2d$, is what makes the denominator exponent sharp.

6 Two exponentials and the minimal recurrence

Suppose first that $b \neq 0$ and $a(a + 4b) \neq 0$. Set $y = a/b$ and choose one ordering of the two roots

$$r_{\pm} = \frac{y + 2 \pm \sqrt{y(y+4)}}{2}, \quad r_+ r_- = 1. \quad (6.1)$$

They are distinct and nonzero. Solving the recurrence (4.2) with its two initial values gives

$$R_n(y) = \frac{(r_+ - 1)r_+^n - (r_- - 1)r_-^n}{r_+ - r_-}. \quad (6.2)$$

Write

$$\lambda_{\pm} = b r_{\pm} = \frac{a + 2b \pm \sqrt{a(a+4b)}}{2}. \quad (6.3)$$

Substitution in (4.7) gives the exact identity

$$D_N^{(m)}(a, b) = \frac{(r_+ - 1)Q_m(N; r_+)\lambda_+^N - (r_- - 1)Q_m(N; r_-)\lambda_-^N}{y^m(r_+ - r_-)}. \quad (6.4)$$

In particular, both polynomial coefficients in this two-exponential representation have degree *exactly* d : neither r_+ nor r_- can equal 1, since $y \neq 0$.

Lemma 6.1 (Minimal annihilator of an exponential polynomial). *If $\lambda_1, \dots, \lambda_s$ are distinct nonzero complex numbers and the nonzero polynomials $A_i(N)$ have degrees d_i , then $u_N = \sum_i A_i(N)\lambda_i^N$ has minimal constant-coefficient annihilator*

$$\prod_{i=1}^s (E - \lambda_i)^{d_i+1}, \quad Eu_N = u_{N+1}. \quad (6.5)$$

Its generating function has poles of orders $d_i + 1$ at $t = 1/\lambda_i$.

Proof. On $A(N)\lambda^N$, the operator $E - \lambda$ is $\lambda^{N+1}(A(N+1) - A(N))$; it lowers the polynomial degree by exactly one until the result vanishes. For $\mu \neq \lambda$, $E - \mu$ preserves that degree, multiplying the leading coefficient by $\lambda - \mu$. The generalized eigenspaces for distinct eigenvalues are independent: polynomial projectors from the Chinese remainder identity isolate each one. Thus every annihilator needs the stated multiplicity at every eigenvalue. This argument remains valid after any finite shift in N .

Equivalently, applying $(z d/dz)^k$ to $1/(1-z)$ shows that $\sum_{N \geq 0} N^k z^N$ has a pole of exact order $k+1$ at $z = 1$, with nonzero leading pole coefficient $k!$. Distinct pole locations cannot cancel. This proves the generating-function assertion too. $\square$

Applying the lemma to (6.4) proves the minimal characteristic polynomial (2.6). It also proves that $q^{d+1}F_m$ is a polynomial of degree at most $2d+1 = 2e-1$ for these parameters. Section 7 sharpens its degree to K .

6.1 Why the identity extends to all parameters

Each $D_N^{(m)}$ is a polynomial in a, b with integer coefficients. The coefficients of $q^{d+1}F_m$ are therefore such polynomials too. Every coefficient that vanishes for all $b \neq 0$ and $a(a+4b) \neq 0$ vanishes identically: a polynomial cannot be nonzero and vanish on that nonempty open set. Hence the uniform rational identity, and its coefficient integrality, extend to every a, b , even where its displayed denominator is not reduced. This argument is coefficientwise in a formal series and does not require a convergence limit in t .

6.2 Proof of the exceptional classification

When $b = 0$,

$$D_N^{(m)}(a, 0) = a^N H_N^{(m)}. \quad (6.6)$$

For $a \neq 0$, the polynomial $H_N^{(m)}$ has degree d , so Lemma 6.1 gives denominator $(1-at)^{d+1}$. When $a = 0$ and $b \neq 0$,

$$D_N^{(m)}(0, b) = b^N H_N^{(m+1)}, \quad (6.7)$$

whose polynomial degree is $\binom{m+1}{2}$.

At the remaining collision $a = -4b$, with $b \neq 0$, the recurrence gives

$$R_n(-4) = (-1)^n (2n+1). \quad (6.8)$$

Inserting this into (4.7) and tracking the signs produces

$$D_N^{(m)}(-4b, b) = (-b)^N E_m(N), \quad (6.9)$$

$$E_m(N) = 4^{-m} \sum_{\ell=0}^m (2N + 2\ell + 1) M_{m,\ell}(N). \quad (6.10)$$

By (5.1), its degree is exactly $d + 1$, with

$$[N^{d+1}]E_m(N) = 4^{-m} 2h_m \sum_{\ell=0}^m \binom{m}{\ell} = 2^{1-m} h_m \neq 0. \quad (6.11)$$

Thus the reduced denominator is $(1 + bt)^{d+2}$. Finally, if $a = b = 0$, every nonempty determinant vanishes and $F_m = 1$. Together with the nonexceptional result, this proves Theorem 2.3.

The two root-collision lines have different pole orders. This is not a contradiction: $x = 0$ and $x = 4$ are inequivalent endpoints of the weight $x^{m-1/2}(4-x)^{1/2}$. In particular, blindly substituting $a = 0$ into a generic minimal-order statement would be incorrect.

7 The numerator degree and reciprocal symmetry

A rational function can have a denominator of large degree and a surprisingly short numerator. Here the explanation is a string of zero values in the natural bilateral extension of the determinant.

For the moment retain the nonexceptional hypotheses. Define D_N for all $N \in \mathbb{Z}$ by the right side of (4.5). The upper rows are interpreted through their polynomial binomial coefficients; this agrees with the Taylor coefficients of R_n at zero, including negative n , because

$$\binom{-n-1+i}{2i} = \binom{n+i}{2i}.$$

It also agrees with the unique bilateral extension supplied by (6.4).

Lemma 7.1 (Reflection and a zero interval). *Let $\sigma_m = (-1)^{\binom{m+1}{2}}$. Then*

$$D_{-N-m-1} = \sigma_m b^{-2N-m-1} D_N \quad (N \in \mathbb{Z}), \quad (7.1)$$

$$D_{-1} = D_{-2} = \cdots = D_{-m} = 0. \quad (7.2)$$

The second statement is empty for $m = 0$.

Proof. The index change $N \mapsto -N - m - 1$, followed by $R_{-n-1} = R_n$, reverses all $m + 1$ columns of the determinant. The sign of that reversal is σ_m . Comparing its prefactor b^N on the two sides gives (7.1). If $N = -s$ with $1 \leq s \leq m$, columns $j = s - 1$ and $j = s$ correspond to R_{-1} and R_0 , respectively, which are the same polynomial. Their values and all needed Taylor coefficients coincide, so the determinant vanishes. $\square$

Lemma 7.2 (The expansion at infinity). *For a sequence of exponential-polynomial form with nonzero characteristic roots and its bilateral extension, its proper rational generating function satisfies*

$$F(t) = - \sum_{r \geq 1} D_{-r} t^{-r} \quad (7.3)$$

as a formal Laurent expansion at infinity.

Proof. For $D_N = \lambda^N$ the assertion is simply $(1 - \lambda t)^{-1} = - \sum_{r \geq 1} \lambda^{-r} t^{-r}$. Apply powers of $t \, d/dt$ to obtain the assertion for $D_N = N^k \lambda^N$, and then use linearity. $\square$

Combining (7.2), (7.1), and $D_0 = 1$ yields

$$F_m(t) = -\sigma_m b^{-m-1} t^{-m-1} + O(t^{-m-2}) \quad (t \rightarrow \infty). \quad (7.4)$$

Since q^e has degree $2e$ and leading coefficient b^{2e} , the numerator has degree

$$2e - m - 1 = m(m - 1) + 2 - m - 1 = (m - 1)^2 = K \quad (7.5)$$

and leading coefficient $-\sigma_m b^{2e-m-1} = s_m b^K$. The identity $s_m = -\sigma_m$ follows because the difference of the two exponents $\binom{m+1}{2}$ and $(m - 1)(m - 2)/2$ is $2m - 1$, an odd integer. This calculation proves the sharp degree assertion, including $m = 0$.

The full reflection identity yields more than the leading term. Putting $1/(b^2 t)$ into (7.3) and reindexing with (7.1), we find

$$F_m\left(\frac{1}{b^2 t}\right) = s_m b^{m+1} t^{m+1} F_m(t). \quad (7.6)$$

But $q(1/(b^2 t)) = q(t)/(b^2 t^2)$. Substitution into (7.6) gives exactly (2.5). After denominators in b are cleared, each of these numerator identities is polynomial in a, b . Thus they extend to the exceptional parameters; for $b \neq 0$ the leading coefficient remains nonzero. This completes the proof of Theorem 2.2.

7.1 An explicit finite prescription for the numerator

No uncomputed infinite series is needed to determine P_m . Expand

$$q(t)^e = \sum_{j=0}^{2e} q_j t^j. \quad (7.7)$$

For example, its coefficients have the finite expression

$$q_j = \sum_{v=\max(0, j-e)}^{\lfloor j/2 \rfloor} \frac{e!}{(j-2v)! v! (e-j+v)!} (-(a+2b))^{j-2v} b^{2v}. \quad (7.8)$$

Then

$$P_m(t) = \sum_{\ell=0}^K \left(\sum_{j=0}^{\min(\ell, 2e)} q_j D_{\ell-j}^{(m)}(a, b) \right) t^\ell, \quad (7.9)$$

where every determinant in the right side is given by Theorem 2.1. Equations (7.8) and (7.9) therefore constitute a finite sum/product formula for every numerator coefficient. All coefficients of $q^e F_m$ after degree K vanish.

The usual recurrence is

$$\sum_{j=0}^{2e} q_j D_{N-j}^{(m)}(a, b) = 0 \quad (N \geq 2e). \quad (7.10)$$

In fact the truncated convolution already vanishes for $N > K$ when negative-index terms are taken as zero for the purpose of multiplying ordinary power series. Those artificial zeros should not be confused with the distinct bilateral extension used in the proof above.

8 Exact exponential expansions and asymptotics

Equation (6.4) supplies every coefficient of the two polynomials in Theorem 2.4. Explicitly,

$$B_+(N) = \frac{(r_+ - 1)Q_m(N; r_+)}{y^m(r_+ - r_-)}, \quad B_-(N) = -\frac{(r_- - 1)Q_m(N; r_-)}{y^m(r_+ - r_-)}. \quad (8.1)$$

These formulas are valid for arbitrary nonexceptional complex parameters; the assumption $a, b > 0$ is only needed to select a strictly dominant real exponential for the asymptotic statement.

For $a, b > 0$, write $r = r_+$, so $r_- = r^{-1}$ and

$$y = r + r^{-1} - 2 = \frac{(r-1)^2}{r}. \quad (8.2)$$

Using (5.8), the coefficient of N^d in B_+ is

$$\frac{h_m(r-1)^{m+1}}{y^m(r-r^{-1})} = h_m \frac{r^{m+1}}{(r-1)^m(r+1)} = A_m.$$

Its relative next coefficient is exactly (5.9). Also $\lambda_-/\lambda_+ = r^{-2}$. Division of (6.4) by $A_m N^d \lambda_+^N$ proves (2.9).

One may write the complete answer in the finite form

$$D_N^{(m)} = \lambda_+^N \sum_{j=0}^d \beta_j^+ N^{d-j} + \lambda_-^N \sum_{j=0}^d \beta_j^- N^{d-j}, \quad (8.3)$$

where

$$\beta_j^+ = \frac{r_+ - 1}{y^m(r_+ - r_-)} [N^{d-j}] Q_m(N; r_+), \quad \beta_j^- = -\frac{r_- - 1}{y^m(r_+ - r_-)} [N^{d-j}] Q_m(N; r_-). \quad (8.4)$$

Thus the “full asymptotic expansion” is particularly strong here: it is an exact expression, with no uncomputed remainder once both finite sectors are retained. No assertion of asymptotic convergence or summability is needed. The leading coefficient ratio of the two sectors is

$$\frac{\beta_0^-}{\beta_0^+} = (-1)^m r^{-(m+1)}. \quad (8.5)$$

These estimates are for fixed m, a, b . They are not claimed to be uniform when $a \rightarrow 0$, when the two roots coalesce, or when m grows with N . In such regimes the exact finite formula remains valid away from the collision itself, but a different uniform asymptotic analysis would be required. At a collision, use the exact exceptional formulas from Section 6 instead.

9 Worked examples and integer sequences

9.1 Small shifts

The first uniform generating functions are

$$F_0(t) = \frac{1-bt}{q(t)}, \quad F_1(t) = \frac{1}{q(t)}, \quad (9.1)$$

$$F_2(t) = \frac{1+bt}{q(t)^2}, \quad F_3(t) = \frac{(1-b^2t^2)(1+(a+6b)t+b^2t^2)}{q(t)^4}. \quad (9.2)$$

They can be obtained directly from (7.9). Notice that $m = 0$ is a genuine boundary case: its numerator has degree 1, not 0. For $m = 3$, the numerator is anti-reciprocal; for $m = 2$, it is reciprocal. This is the pattern described uniformly by s_m .

For $m = 2$, put $A = a + 2b$. The recurrence becomes

$$D_N = 2AD_{N-1} - (A^2 + 2b^2)D_{N-2} + 2Ab^2D_{N-3} - b^4D_{N-4} \quad (N \geq 4). \quad (9.3)$$

The initial polynomials are

$$\begin{aligned} D_0 &= 1, \\ D_1 &= 2a + 5b, \\ D_2 &= 3a^2 + 14ab + 14b^2, \\ D_3 &= 4a^3 + 27a^2b + 54ab^2 + 30b^3. \end{aligned}$$

The recurrence is minimal outside the exceptional lines; displaying only a valid order-four relation would not by itself prove that fact.

9.2 The specialization $a = b = 1$

The first sequences, all beginning with the empty determinant, are

$m \backslash N$	0	1	2	3	4	5	6
0	1	2	5	13	34	89	233
1	1	3	8	21	55	144	377
2	1	7	31	115	390	1254	3893
3	1	19	170	1075	5580	25529	107036

The first two are F_{2N+1} and F_{2N+2} , respectively, as follows from their common recurrence $u_{N+2} = 3u_{N+1} - u_N$ and initial values. They connect the family to the odd and even Fibonacci subsequences in OEIS A001519 and A001906 [6]; Catalan numbers themselves are A000108 [5]. The higher rows and their coefficient triangles are natural OEIS-style objects, but the archive does not assert that any particular row is absent from OEIS.

For every fixed shift, the dominant exponential is

$$\lambda_+ = \frac{3 + \sqrt{5}}{2},$$

while its polynomial prefactor has degree $\binom{m}{2}$. Thus increasing the shift changes the power of N , not the exponential growth constant. For $m = 2$, the first relative asymptotic correction equals $1 - 1/(r - 1) = (3 - \sqrt{5})/2$ with $r = (3 + \sqrt{5})/2$.

9.3 Rational characteristic roots: $a = 1, b = 2$

This specialization gives $r_+ = 2, r_- = 1/2$ and $\lambda_+ = 4, \lambda_- = 1$. Therefore the exact formulas have rational polynomial coefficients and no radicals:

$$D_N^{(0)}(1, 2) = \frac{2 \cdot 4^N + 1}{3}, \quad (9.4)$$

$$D_N^{(1)}(1, 2) = \frac{4 \cdot 4^N - 1}{3}, \quad (9.5)$$

$$D_N^{(2)}(1, 2) = \frac{8N 4^N + N + 3}{3}, \quad (9.6)$$

$$D_N^{(3)}(1, 2) = \frac{(32N^3 + 48N^2 + 16N + 96)4^N - (2N^3 + 21N^2 + 73N + 78)}{18}. \quad (9.7)$$

For example, the $m = 2$ sequence starts 1, 12, 87, 514, 2733, 13656, ... Its dominant polynomial is $(8/3)N$ with no constant term. This vanishing agrees with the first-correction formula $d(m/2 - 1/(r - 1)) = 1(1 - 1) = 0$; the entire correction is the smaller exponential sector $(N + 3)/3$. For $m = 3$, the relative coefficient of N^{-1} in the dominant sector is $3/2$, again exactly as predicted.

9.4 An endpoint example

For $m = 2$, (6.10) simplifies to

$$E_2(N) = \frac{(N+1)(N+2)}{2}.$$

Hence

$$D_N^{(2)}(-4b, b) = (-b)^N \frac{(N+1)(N+2)}{2}, \quad F_2(t; -4b, b) = \frac{1}{(1+bt)^3}. \quad (9.8)$$

This also follows by cancelling $1+bt$ from $(1+bt)/(1+bt)^4$, the uniform form of F_2 . At the other collision, $a = 0$, one instead obtains $F_2(t; 0, b) = (1+bt)/(1-bt)^4$. These explicit cases illustrate why the reduced denominator needs a separate classification.

10 Algorithms, exact experiments, and reproducibility

10.1 Implementation choices

The file `code/catalan_hankel.py` implements the coefficient formula, pure shifted product, homogeneous polynomial evaluation, fixed-size determinant, and uniform and reduced denominators. All numerical parameters in its public computation interface are integers; internal fractions use Python's exact `Fraction`. The mathematical theorems have the broader complex-parameter scope stated above. No floating-point arithmetic enters the tests.

For a coefficient row, first compute the triangular product $H_N^{(m+1)}$ and then apply (2.3) successively. The number of arithmetic operations is $O(m^2 + N)$; this is an *arithmetic-operation count*, not a bit-complexity claim. Every division in the update is checked for exactness. Homogeneous Horner evaluation avoids division by a or b , so zero parameters are handled without special numerical limits.

The independent determinant path uses a fraction-free Bareiss algorithm on the original matrix of Catalan moments, including pivoting when a diagonal pivot vanishes. This matters for signed parameters: an implementation tested only on positive definite examples can conceal pivot failures. The source-indexing formula is implemented separately, not obtained by calling the corrected coefficient routine.

For many consecutive values at fixed m , it is also possible to use (7.10) after producing the first $2e$ values. For very large isolated indices, powering the companion matrix or reducing X^N modulo (2.6) provides standard recurrence-based methods. These methods are consequences of the theorem; the supplied reference implementation prioritizes transparent exact verification rather than optimized large-integer throughput.

10.2 Verification that was actually run

The included `data/verification.json` records PASS for all checks in the following table. The report is generated by the test program, not manually written.

Check	Count
Direct determinant versus coefficient evaluation	1,183
Additional direct signed/degenerate parameter evaluations	810
Printed source formula versus corrected shift	990
Confluent fixed-size determinant evaluations	364
Minor polynomials exactly interpolated and checked	36
Generating-function/recurrence coefficient identities	2,816
Numerator reciprocal coefficient identities	604
Nonexceptional numerator–denominator gcd checks	44
Exceptional reduced-denominator checks	44
Exact two-exponential evaluations	168
Strict normalized log-concavity inequalities	5,655

The first 1,183 evaluations cover every pair $0 \leq N, m \leq 12$, using $a = 0, 1, \dots, N$ and $b = 1$. At each fixed (N, m) , both sides are polynomials of degree at most N in a . Equality at $N + 1$ distinct points certifies that particular polynomial identity exactly. Thus this part certifies 169 full coefficient polynomials, not merely 169 evaluations. It still does not prove the result for untested values of N or m ; the proof in Section 3 does that.

For $0 \leq m \leq 7$, the minor tests interpolate using $2d + 2$ points, more than enough for the elementary degree bound $2d$. They verify the drop to degree d , both leading coefficients from Lemma 5.1, and additional negative-index values. The recurrence checks include $0 \leq m \leq 10$ and four nonexceptional parameter pairs, while separate tests examine every exceptional class. The exact two-exponential tests use $(a, b) = (1, 2)$, $m \leq 7$, and $0 \leq N \leq 20$. These tests check different parts of the derivation, but are not represented as formal proof certificates for arbitrary parameters.

10.3 How to reproduce the package

From the archive's top directory, run:

```
python3 code/verify.py --out data
python3 code/catalan_hankel.py 8 3 --a 2 --b 3 --direct
latexmk -pdf -interaction=nonstopmode -halt-on-error article.tex
```

Use Python 3.10 or later and do not pass -0 , because the verification suite deliberately uses assertions. The script writes coefficient rows, generating functions, minor polynomials, exact exponential-sector polynomials, and integer sequence tables in JSON or CSV. Rational polynomial coefficients are stored as strings such as "8/3", in ascending power order, to avoid any loss of exactness when imported into other systems. The included README documents every file and its indexing convention. [Added 28 September 2026, batch 39: the first command regenerates the shipped files in data/; in the repository, run it on a copy (see the report README). Part II's separate suite, and why its delivered commands must not be run in this directory, are described in Section 27.]

11 What has been resolved, and what has not

The all-shift denominator formula is proved here, with a stronger minimality assertion and a complete description of the degenerate parameters. The coefficient expression is likewise proved for all N, m ; comparison with the exact formula printed in the target source requires the explicit indexing correction in the next appendix. The reciprocal numerator and finite exponential expansions are proved from the same structural determinant calculation.

These are substantive mathematical conclusions about the selected conjectural family. They are not evidence that the Christoffel identity, Jacobi representation, Catalan shifted product, or general rationality

are new. A literature search was performed against the source title, arXiv identifier, denominator phrase, and related recurrence work. The cited version still labels the sharp exponent conjectural, and the retrieved sources did not supply an explicit statement of this full sharp classification. That is a bounded search result, not a proof that no earlier equivalent theorem exists. Before a priority claim or submission, the exact formulas should be checked against the wider orthogonal-polynomial and determinant literature and by a human referee.

Several natural extensions remain separate from the theorem. One may seek a direct combinatorial interpretation of the individual positive integers $c_{N,m,k}$, or a product formula for the minors $M_{m,\ell}(N)$ beyond their first two coefficients. Multiple distinct linear factors in the multiplier lead to more exponential sectors and new collision patterns. A joint limit with growing m or a parameter approaching $a = 0$ would require uniform, rather than fixed-parameter, asymptotics. None of these extensions is assumed in the present proofs.

[Added 28 September 2026, batch 39: the preceding paragraph describes the report at the state fbba58593 against which Part II was written. The extension to several distinct linear factors, and to repeated factors, is treated in Part II: for every nonzero polynomial multiplier the exponential sectors, their exact degrees and nonzero leading coefficients (Theorems 14.2 and 18.1), the minimal order off resonance (14.12), an exact detector of the parameters where it drops (Theorem 14.5), and a certified finite algorithm on that resonance locus (Theorems 20.1 and 20.2). Part II gives no short closed classification of the resonant cancellations; it poses that as a research direction (Section 24). The other extensions named here remain open: Part II does not address a combinatorial interpretation of $c_{N,m,k}$, a product formula for the minors $M_{m,\ell}(N)$, or joint limits with growing m ; and its asymptotics (Section 22) are for fixed parameters, so a parameter approaching $a = 0$ (a root reaching the endpoint 0) is among the uniform endpoint limits it poses, not solves.]

[Added 29 September 2026, batch 42: for a single repeated root away from the endpoints, Part III gives that classification (Theorems 30.1, 30.4 and 37.1); resonant cancellations among several distinct roots, or with endpoint factors, remain unclassified.]

[Added 29 September 2026, batch 53: for fixed m the limit $a \rightarrow 0$ is treated in Part IV. For $b \neq 0$, $x^m(a + bx)$ is b times $x^m(x + a/b)$, a cluster of $m + 1$ roots at the endpoint 0; for real $a/b \geq 0$ the ratio $D_N^{(m)}(a, b)/D_N^{(m)}(0, b)$ tends to 1 exactly when $a/b = o(N^{-2})$, to an explicit profile when $N^2 a/b$ converges, and to infinity when $N^2 a/b \rightarrow \infty$ (Theorem 47.4 and Section 44.4). Joint limits with growing m remain open.]

Part II. Root collisions: arbitrary polynomial multipliers

Sections 1–11 and Appendices A–B form the original report (Part I) of 19 September 2026, unchanged except for dated pointers to this addition, a line on the title page, contents entries for the two parts, and two bibliography entries used only by Part II. Part II was added on 28 September 2026 in batch 39 of ProveIt’s incoming-report intake, from one manuscript written against the pinned state of Part I. It answers the extension that Part I’s Section 11 named but left open, “multiple distinct linear factors in the multiplier” and their “new collision patterns”: for every nonzero polynomial multiplier it determines the exponential sectors, their exact polynomial degrees and leading coefficients, the minimal recurrence off resonance, and a finite algorithm on the resonance locus. Section 12 records the provenance and fixes the notation of the addition.

12 Provenance and conventions of Part II

12.1 One manuscript, one addition

Part II is batch-39 manuscript 03, *Root Collisions and Sharp Recurrences for Polynomially Weighted Catalan Hankel Determinants*, a proof-focused research article prepared with OpenAI ChatGPT for Vladimir Reshetnikov’s ProveIt research program and dated 28 September 2026 (23 pages as delivered). It arrived as the archive `ProveIt_Catalan_Root_Collisions.zip` (inner directory `ProveIt_Catalan_Root_Collisions/`, main file `article.tex`) in ProveIt commit `74f7f5bdb` and was placed in `e2b1f016a`. It was written against ProveIt commit `fbba58593dc0622aa914972896150d4848f935b5`, reading Part I’s `article.tex` and `README.md` at that commit (consulted 28 September 2026 through the pinned repository tree). Part I’s files are unchanged between that commit and the placement commit, so every statement the manuscript makes about “the repository report” refers to Part I as printed above.

The manuscript’s code and data are shipped beside Part I’s files with the prefix `02-root-collisions-` (Part I’s unprefix files count as 01), and its provenance notes, delivered as `notes/provenance.md`, as `02-root-collisions-provenance.md` at the report root. Its manuscript, delivery README and PDF are not shipped; they survive in the arrival commit. Section 27 lists the shipped files.

The manuscript contributes, in the order printed here: its research question and a statement of what is classical (Section 13); the definitions and principal results, namely the exact sector-degree law, the minimal recurrence off resonance and the secondary-Hankel factorization (Section 14); the orthogonal polynomials and the confluent Christoffel identity in its normalization (Section 15); the local jets at interior points and at both endpoints (Section 16); the determinant degree calculus (Section 17); the canonical sectors and the exact leading coefficient (Section 18); minimality, the order formula and the secondary factorization (Section 19); resonance, finite-prefix certification and a root-free algorithm (Section 20); worked examples, including the recovery of Part I’s theorem (Section 21); exact asymptotics (Section 22); its verification record, implementation interface and a formalization route (Section 23); eight research directions (Section 24); a conclusion; a normalization and sign audit (Section 26); and its reproduction instructions (Section 27). Every result, proof, example, remark, limitation and question of the manuscript is printed. Where it re-derives something Part I already proves, the overlap is marked in the text (Section 12.4).

The manuscript’s own abstract reads:

For a nonzero polynomial q , let $H_N(q)$ be the size- N Hankel determinant of the Catalan moments multiplied by q . General constant-coefficient recurrences for this family are classical. We determine their sharp multiplicities. Write $q(x) = \gamma x^m (x-4)^\ell \prod_i (x-c_i)^{t_i}$, with distinct $c_i \notin \{0, 4\}$, and put $c_i = 2 + z_i + z_i^{-1}$. The branch-labelled exponential with rate $(-1)^{\ell+\sum t_i} \gamma \prod_i z_i^{2k_i-t_i}$ has a polynomial

coefficient whose degree is exactly $\binom{m}{2} + \binom{\ell+1}{2} + \sum_i k_i(t_i - k_i)$. A nonvanishing confluent-Vandermonde product gives its leading coefficient. Consequently, when the rates are distinct, the minimal recurrence order is $\prod_i (t_i + 1) \left(\binom{m}{2} + \binom{\ell+1}{2} + 1 + \frac{1}{6} \sum_i t_i(t_i - 1) \right)$. We prove an exact secondary-Hankel factorization that detects every loss of this maximal order, and give a finite, root-free algorithm determining the minimal recurrence at all rational or algebraic specializations. For one repeated nonendpoint root the generic order is cubic in the multiplicity, rather than the classical exponential bound. The example $q = (2x - 9)^4$ has order 15, while resonant fourth powers can have order 5. The proofs extend the repository's linear-multiplier result to arbitrary root-multiplicity patterns, including both endpoints, and are accompanied by exact-arithmetic verification code.

Status (manuscript 03). This is an unrefereed, AI-assisted mathematical manuscript. The universal results are proved in the text; the finite computations are an independent audit, not a substitute for those arguments. No Lean formalization or claim of first discovery is made. General polynomial-multiplier rationality, the Christoffel formula and its confluent form are classical and are not presented as new; the proposed contribution is the exact branch degree, its nonvanishing leading constant, the resulting minimality classification off resonance, and their assembled structural and algorithmic consequences. No novelty is claimed for the secondary-Hankel identity of an arbitrary exponential polynomial (Lemma 19.2), an elementary confluent-Vandermonde calculation. The literature comparison was targeted (the sources cited in Section 13) and does not establish that the theorem has never appeared in another notation; Schur-function, Toeplitz/Hankel determinant and orthogonal-polynomial literature may contain equivalent specializations, and expert review and a broader priority search would be needed before any first-discovery claim. No current global open-problem status is asserted for anything, including the eight research directions of Section 24. The theorems are over characteristic zero and are not to be transported unchanged to small positive characteristic. Nothing in Part II is formalized in any proof assistant, and its placement beside a Lean project confers no formal status. (Sources: the manuscript and its provenance notes, 02-root-collisions-provenance.md.)

12.2 Reading conventions

Inside Sections 13–27, “this article” and “the present article” mean manuscript 03 as printed in Part II, “the repository report” and “the ProveIt report” mean Part I at the pin fbba58593, and “the archive” or “the package” means the delivered package, whose code and data are shipped under the prefixed names listed in Section 27. The manuscript’s two appendices are printed as Sections 26 and 27, so that Part I’s appendices keep their letters. The manuscript cites Krattenthaler’s Ramanujan Journal paper in its preprint form arXiv:2101.04225v5 (10 September 2021), and points to its Theorem 1, Proposition 5 and Section 8, Corollary 9; that is the same paper as Part I’s reference [3]. The manuscript cites Barry’s note [1] for the denominator-exponent assertion in its introduction and does not reassert the printed coefficient conjecture without the index correction of Appendix A.

12.3 Notation of Part II

Part II uses the manuscript’s symbols, which differ from Part I’s in many places. Within Sections 13–27 every symbol has its Part II meaning, except where a sentence names Part I or cites one of its equations. Part I’s parameters a, b appear only where Part I’s multiplier $x^m(a + bx)$ is discussed: the first subsection of Section 13 and the first subsection of Section 21. Elsewhere a and b are local letters (roots, indices, block counts, a leading coefficient). Table 1 lists the symbols that a reader of Part I is likely to misread. One symbol was renamed: the manuscript’s maximal order R_0 is written $\mathcal{R}_0$, because Part I’s $R_0(y) = 1$ is the first of its polynomials (4.1). No normalization changed.

Table 1: Symbols of Part II. The last column gives Part I's meaning of the same letter, or the tempting false reading.

Symbol	Meaning in Part II	Not to be confused with
$q(x)$; subscripts in $\Delta_q, \chi_q, \Psi_q, B_q$	the nonzero multiplier polynomial (14.2)	Part I's $q(t) = 1 - (a + 2b)t + b^2 t^2$ (1.4), the quadratic denominator; Part II's generating-function variable is u , not t .
$H_N(q)$	the weighted Hankel determinant (14.1)	Part I's $H_N^{(m)} = H_N(x^m)$ and $D_N^{(m)}(a, b) = H_N(x^m(a + bx))$ are special cases; in Theorem 20.2, $h = (h_N)$ is an arbitrary sequence, not Part I's h_m .
m	multiplicity of the root 0 of q	agrees with Part I's shift m when $q = x^m(a + bx)$.
ℓ	multiplicity of the root 4 of q	Part I's column index in $M_{m,\ell}$ and Q_m .
$s; s(z), s_i$	number of distinct roots outside $\{0, 4\}$; the scalar $1/(z - z^{-1})$ of (16.2)	Part I's sign s_m in (2.5), and the source shift s of Appendix A.
c_i, z_i, t_i	roots outside $\{0, 4\}$, spectral parameters $c_i = 2 + z_i + z_i^{-1}$, multiplicities	Part I's coefficients $c_{N,m,k}$ and its variable t ; for $q = x^m(a + bx)$, $-z^{\pm 1}$ are Part I's $r_{\pm}$.
T, D	$\sum_i t_i$ and $\deg q$	Part I's $T_{n,k,s}$ (Appendix A) and its determinant $D_N^{(m)}$.
$\mathcal{K}, \mathbf{k}, \kappa$	index set (14.4), a label in it, and $\binom{m}{2} + \binom{\ell+1}{2}$	Part I's numerator degree $K = (m - 1)^2$.
λ_k	rate of the sector $\mathbf{k}$ (14.6)	for $q = x^m(a + bx)$ the two rates are Part I's $\lambda_{\pm}$ (6.3).
d_k, r_k	exact sector degree (14.7) and $d_k + 1$	Part I's $d = \binom{m}{2}$ and $e = d + 1$ (the linear case), its roots $r_{\pm}$ and the variable r of $Q_m(N; r)$.
$\mathcal{R}_0$ (renamed from R_0)	the maximal recurrence order (14.12)	Part I's polynomial $R_0(y) = 1$ (4.1); Part II's plain R is another order or size (an assumed bound in Theorem 20.2, a matrix size in Lemma 19.2, the root-free bound (20.6)).
$P_k(N), L_k$	sector polynomial in the size N (18.4), and its leading coefficient	Part I's numerator $P_m(t; a, b)$, a polynomial in t ; Part I's sector polynomials are $B_{\pm}(N)$.
$A_{\pm}(z), A_{r,\pm}(n; z)$	branch amplitudes and jet polynomials (16.4)	Part I's asymptotic constant A_m (2.10).
$B_0(m), B_4(\ell), B_z(k); B_i; B_q(u)$	local leading determinants (18.7), (17.5); blocks in Lemma 19.2; the numerator in (19.3)	Part I's $B_{\pm}(N)$ of Theorem 2.4.
$M_N; M$	the $D \times D$ Christoffel matrix (15.7); $\max_{\mathbf{k}} d_{\mathbf{k}}$ in Section 22	Part I's minors $M_{m,\ell}(N)$ (4.6).
E	forward shift, $(Ef)(N) = f(N + 1)$	Part I's shift E in Lemma 6.1 and its endpoint polynomial $E_m(N)$ (6.10).
Λ, Q_{λ}	set of distinct rates; grouped sector polynomial (20.1)	Part I's minor generating polynomial $Q_m(N; r)$ (5.7).
$(a_u, v_u)_{u \leq b}; (w_a, u_a)_{a \leq b}; b_t$	the distinct roots of q , their multiplicities and number b (Section 15); nodes and block sizes in Lemma 17.1; degree of the square-free factor f_t	Part I's parameters a, b , which Part II uses only in the first subsections of Sections 13 and 21.
p_n, μ	monic orthogonal polynomials (15.1) and the Catalan measure	they are Part I's $p_n^{(0)}$ and μ_0 ; Part I's $p_N^{(m)}$ and μ_m with $m \geq 1$ are different.

Symbol	Meaning in Part II	Not to be confused with
$\Delta_q, V, \mathcal{V}$	cross-root product (15.6); Vandermonde (matrix); confluent product (18.6)	Part I's Vandermonde $\Delta(x)$.
C (proof of Theorem 18.1)	a matrix of coefficient polynomials	the Catalan numbers C_n .

12.4 Where the merge had to choose

1. *Placement.* Part II is printed after Part I's Section 11 and before Part I's appendices, so that no section, equation, theorem or table number of Part I moves (checked against the `.aux` file of a build of the committed text). Part II's sections continue the numbering (Sections 12 to 27); the manuscript's Sections 1–13 are Sections 13–25, and its Appendices A and B are Sections 26 and 27.
2. *Labels.* Every label of Part II carries the prefix `rc::`; without it, the manuscript's `sec:main`, `thm:main`, `eq:R`, `eq:lambda`, `eq:measure`, `lem:Christoffel`, `eq:Christoffel` and several others would have collided with Part I's.
3. *Overlaps with Part I.* The manuscript re-derives, without assuming Part I correct, several things Part I already proves. Printed with a pointer and kept: the orthogonality of the p_n and $\det(C_{i+j}) = 1$ (a second route; Section 15); the Christoffel identity (Part I's Lemma 4.1 is its special case $\gamma = 1$, with Part I's normalization R_n , and Proposition 4.2 its confluent case at the root 0); the binomial determinant (17.3) (as in (5.3)); the minimal-annihilator lemma (Lemma 19.1 restates Lemma 6.1; the manuscript's proof is kept as a second proof); and the recovery of Part I's sharp order and exceptional orders (a second proof by a different route, Section 21). No statement of Part I was changed.
4. *Bibliography.* The manuscript's references to Barry and to Krattenthaler's Ramanujan Journal paper are Part I's [1, 3]; its versions and pointers are recorded in Section 12.2. Its references to Cigler and Krattenthaler and to Krattenthaler's Uvarov-formula preprint are added to the common bibliography. Its reference to the pinned repository report is replaced by Part I itself and the pin recorded in Section 12.1.
5. *File names.* Sections 23 and 27 give the shipped names beside the delivered ones. The manuscript's reproduction commands are printed as delivered but must not be run in place here; Section 27 gives a recipe on a scratch copy.
6. *Front matter.* The manuscript's title page, abstract, status paragraph and archive paragraph are replaced by Section 12.1, which prints the abstract and the status. The archive paragraph is not reprinted because the delivered source and PDF are not shipped; the shipped code and data are listed in Section 27.
7. *Notation.* The manuscript's symbols are kept, with the single rename $R_0 \rightarrow \mathcal{R}_0$ and the scoping convention and Table 1 of Section 12.3.

13 The research question and its provenance

13.1 A concrete extension of the ProveIt report

The starting point is the report *Shifted Catalan Hankel Polynomials: Exact coefficients, sharp recurrences, and a denominator conjecture*, dated 19 September 2026, in ProveIt, printed above as Part I (§12.1 records the pinned snapshot). For

$$D_N^{(m)}(a, b) = \det_{0 \leq i, j < N} (aC_{i+j+m} + bC_{i+j+m+1}), \quad C_n = \frac{1}{n+1} \binom{2n}{n},$$

it proves the sharp characteristic polynomial

$$(X^2 - (a + 2b)X + b^2)^{\binom{m}{2}+1} \quad (13.1)$$

when $b \neq 0$ and $a(a + 4b) \neq 0$ (this is (2.6) of Theorem 2.2). Its concluding discussion (Section 11) explicitly leaves multiple linear factors and their collision patterns for further work. That is the extension addressed here.

The repository was inspected at commit

fbba58593dc0622aa914972896150d4848f935b5.

The relevant source is under

SetTheory/Cardinals/docs/reports/hankel-determinants/
catalan-and-ballot/shifted-catalan-hankel-polynomials/article.tex.

The present proof does not assume that an unrefereed repository manuscript is correct. We rederive the determinant identity and the local degree calculus, then recover (13.1) as a specialization.

The question is not merely whether another rational generating function exists. It is this:

Given every root and its multiplicity in a polynomial multiplier, which exponential terms actually occur, what is the exact polynomial degree of each term, and precisely when does the minimal recurrence order drop?

A repeated root is not equivalent to several independently chosen roots. The associated determinant forces cancellations within a repeated block. Those cancellations are the main subject of this article.

13.2 What is classical, and what is being added

The Christoffel determinant formula and its confluent form are classical; Krattenthaler's treatment [3] gives several proofs and explicitly handles coalescing evaluation points. Its Section 8, Corollary 9, supplies a constant-coefficient recurrence of order at most 2^D for a multiplier of degree D , including Catalan moments. That account also credits Elouafi's earlier proof of the Catalan recurrence assertion. General rationality is therefore not a new conclusion of this article.

Barry's note [1] motivates the shifted linear combination and its quadratic-denominator exponent. The earlier work of Cigler and Krattenthaler [7] gives further context for polynomial linear combinations of moments. Our comparison is with these particular sources and the pinned repository report, not with an exhaustive classification of the literature.

The additions proved below are an exact degree law for every repeated-root block, its nonzero leading constant with endpoint factors included, the resulting sharp recurrence and a factorization detecting its exceptional locus. A root-free finite procedure handles resonances without assuming that a generic formula remains minimal after specialization. The underlying determinant techniques are familiar; the full multiplicity law and its assembled consequences are the proposed research contribution. An equivalent earlier formulation in the wider orthogonal-polynomial or symmetric-function literature has not been ruled out.

14 Definitions and principal results

14.1 The weighted determinant and its spectral parameters

For $q(x) = \sum_{r=0}^D q_r x^r \in \mathbb{C}[x] \setminus \{0\}$, define

$$H_N(q) = \det_{0 \leq i, j < N} \left(\sum_{r=0}^D q_r C_{i+j+r} \right), \quad N \in \mathbb{N}_0, \quad H_0(q) = 1. \quad (14.1)$$

Write its factorization as

$$q(x) = \gamma x^m (x - 4)^\ell \prod_{i=1}^s (x - c_i)^{t_i}, \quad (14.2)$$

where $\gamma \neq 0$, $m, \ell \in \mathbb{N}_0$, $t_i \geq 1$, and the c_i are pairwise distinct and avoid 0, 4. Thus

$$T = \sum_{i=1}^s t_i, \quad D = m + \ell + T.$$

An empty product is 1. Constant multipliers are included by taking $m = \ell = s = 0$. The identically zero multiplier is excluded; its sequence would simply be 1, 0, 0, ...

For each i choose a solution of

$$c_i = 2 + z_i + z_i^{-1}. \quad (14.3)$$

Then $z_i \in \mathbb{C}^\times \setminus \{1, -1\}$. Replacing z_i by z_i^{-1} only relabels an index below. No final recurrence depends on this choice. Define

$$\mathcal{K} = \prod_{i=1}^s \{0, 1, \dots, t_i\}, \quad (14.4)$$

$$\kappa = \binom{m}{2} + \binom{\ell + 1}{2}, \quad (14.5)$$

$$\lambda_{\mathbf{k}} = (-1)^{\ell+T} \gamma \prod_{i=1}^s z_i^{2k_i - t_i}, \quad (14.6)$$

$$d_{\mathbf{k}} = \kappa + \sum_{i=1}^s k_i(t_i - k_i), \quad r_{\mathbf{k}} = d_{\mathbf{k}} + 1. \quad (14.7)$$

When $s = 0$, the index set $\mathcal{K}$ has one element, the empty tuple.

Definition 14.1 (Resonance). The parameters are *nonresonant* if the values $\lambda_{\mathbf{k}}$, $\mathbf{k} \in \mathcal{K}$, are pairwise distinct. Otherwise there is a *resonance*. Equivalently, a resonance is an identity

$$\prod_i z_i^{2(k_i - j_i)} = 1 \quad \text{for some distinct } \mathbf{k}, \mathbf{j} \in \mathcal{K}. \quad (14.8)$$

This is different from a repeated root of q : repeated roots have already been recorded by the t_i .

14.2 The exact multiplicity law

Theorem 14.2 (Branch degrees and minimal recurrence). *For every multiplier (14.2), there are canonically branch-labelled polynomials $P_{\mathbf{k}} \in \mathbb{C}[N]$ such that*

$$H_N(q) = \sum_{\mathbf{k} \in \mathcal{K}} P_{\mathbf{k}}(N) \lambda_{\mathbf{k}}^N \quad (N \geq 0), \quad (14.9)$$

and

$$\deg P_{\mathbf{k}} = d_{\mathbf{k}} \quad \text{for every } \mathbf{k}. \quad (14.10)$$

Their leading coefficients are given by the nonzero product in Theorem 18.1. These individual assertions remain valid at resonances, when different branch labels can have the same numerical rate.

If the parameters are nonresonant, the minimal characteristic polynomial of the sequence is

$$\chi_q(X) = \prod_{\mathbf{k} \in \mathcal{K}} (X - \lambda_{\mathbf{k}})^{d_{\mathbf{k}} + 1}. \quad (14.11)$$

Its degree is

$$\mathcal{R}_0 = \prod_{i=1}^s (t_i + 1) \left(\kappa + 1 + \frac{1}{6} \sum_{i=1}^s t_i (t_i - 1) \right). \quad (14.12)$$

For arbitrary parameters, (14.11) is still an annihilating polynomial of degree $\mathcal{R}_0$. The true minimal order equals $\mathcal{R}_0$ if and only if the parameters are nonresonant.

A characteristic polynomial $g(X) = \sum_{j=0}^r g_j X^j$, with $g_r = 1$, means the recurrence

$$\sum_{j=0}^r g_j H_{N+j}(q) = 0 \quad \text{for every } N \geq 0. \quad (14.13)$$

Thus the assertions concern the entire sequence, including its initial values, not only an eventual recurrence. A polynomial of minimal degree with this property is unique.

Corollary 14.3 (One repeated nonendpoint root). *Let $q(x) = \gamma(x-c)^t$, where $c \notin \{0, 4\}$ and $c = 2+z+z^{-1}$. If $z^{2h} \neq 1$ for $1 \leq h \leq t$, then*

$$\text{ord}(H(q)) = t + 1 + \binom{t+1}{3}. \quad (14.14)$$

The exponential with rate $(-1)^t \gamma z^{2k-t}$ has polynomial degree $k(t-k)$. In particular the first orders, for $t = 1, \dots, 8$, are

$$2, 4, 8, 15, 26, 42, 64, 93.$$

The order in (14.14) is asymptotic to $t^3/6$, in contrast to the universal bound 2^t . The first strict improvement occurs at $t = 4$. This is a uniform family of exact minimality results, rather than an isolated fitted recurrence.

Corollary 14.4 (Simple factors and endpoints). *If every $t_i = 1$ and the parameters are nonresonant, the minimal order is*

$$2^s \left(\binom{m}{2} + \binom{\ell+1}{2} + 1 \right). \quad (14.15)$$

If $s = 0$, no nonresonance hypothesis is needed: $H_N(q)$ is $((-1)^\ell \gamma)^N$ times a polynomial of degree exactly κ . The minimal polynomial is $(X - (-1)^\ell \gamma)^{\kappa+1}$.

The asymmetry between 0 and 4 is essential. A multiplicity m at 0 contributes $\binom{m}{2}$ degrees, whereas a multiplicity ℓ at 4 contributes $\binom{\ell+1}{2}$. Treating both endpoints as a generic collision of two equally weighted branches gives the wrong answer.

14.3 An exact detector for exceptional parameters

Put $L_k = \text{lc}(P_k)$, and choose any ordering of $\mathcal{K}$.

Theorem 14.5 (Secondary Hankel factorization). *With $\mathcal{R}_0 = \sum_k r_k$, one has*

$$\begin{aligned} & \det_{0 \leq u, v < \mathcal{R}_0} (H_{u+v}(q)) \\ &= (-1)^{\sum_k \binom{r_k}{2}} \prod_k \left((d_k! L_k)^{r_k} \lambda_k^{r_k(r_k-1)} \right) \prod_{k < j} (\lambda_j - \lambda_k)^{2r_k r_j}. \end{aligned} \quad (14.16)$$

This identity holds at resonant as well as nonresonant parameters. All factors outside the final product are nonzero. Consequently this secondary determinant vanishes exactly on the resonance locus.

The formula is stronger than a statement that a generic matrix has full rank. It identifies every source of rank loss and its multiplicity in spectral coordinates. Sections 19 and 20 prove this assertion and make it algorithmic.

15 Catalan orthogonality and the fixed-size determinant

This section proves the classical infrastructure with our normalization. It is included to make the signs, initial index and confluent factorials fully auditable. The general identities are not claimed as new.

15.1 The orthogonal polynomials

Define monic polynomials by

$$p_0(x) = 1, \quad p_1(x) = x - 1, \quad p_{n+1}(x) = (x - 2)p_n(x) - p_{n-1}(x) \quad (n \geq 1). \quad (15.1)$$

The Catalan probability measure is

$$d\mu(x) = \frac{1}{2\pi} \sqrt{\frac{4-x}{x}} \mathbf{1}_{(0,4)}(x) dx. \quad (15.2)$$

Substitution $x = 4u$ and the beta integral give

$$\int x^n d\mu(x) = \frac{2 \cdot 4^n}{\pi} B\left(n + \frac{1}{2}, \frac{3}{2}\right) = \frac{(2n)!}{n!(n+1)!} = C_n.$$

There is also an elementary trigonometric verification of the orthogonality. For $x = 2 + 2 \cos \theta$, $0 < \theta < \pi$, the recurrence gives

$$p_n(x) = \frac{\sin((n + \frac{1}{2})\theta)}{\sin(\theta/2)}, \quad d\mu = \frac{1 - \cos \theta}{\pi} d\theta. \quad (15.3)$$

The product-to-sum formula then yields

$$\int p_n(x) p_j(x) d\mu(x) = \delta_{n,j}. \quad (15.4)$$

Indeed, after cancellation of $\sin^2(\theta/2)$ the integral is $\frac{2}{\pi} \int_0^\pi \sin((n + \frac{1}{2})\theta) \sin((j + \frac{1}{2})\theta) d\theta$. The change from monomials to monic p_n is triangular with determinant 1. Hence (15.4) also proves

$$\det_{0 \leq i, j < N} (C_{i+j}) = 1. \quad (15.5)$$

Relation to Part I. The measure μ is Part I's μ_0 of (3.1), and p_n is Part I's $p_n^{(0)}$, with the same recurrence (3.8). Part I proves (15.5), $H_N^{(0)} = 1$, from the constant squared norms of the $p_n^{(0)}$ (Section 3); the trigonometric orthogonality above is a second route, kept as the manuscript printed it.

15.2 Christoffel's determinant, including confluence

List the distinct roots in (14.2) in the order $0, 4, c_1, \dots, c_s$, omitting endpoints whose multiplicities are zero. Write this list as $(a_1, \nu_1), \dots, (a_b, \nu_b)$ and set

$$\Delta_q = \prod_{1 \leq u < v \leq b} (a_v - a_u)^{\nu_u \nu_v}. \quad (15.6)$$

It is nonzero. Let M_N have rows indexed by (a_u, r) , $0 \leq r < \nu_u$, and columns indexed by $j = 0, \dots, D - 1$, with entries

$$(M_N)_{(a_u, r), j} = \frac{p_{N+j}^{(r)}(a_u)}{r!}. \quad (15.7)$$

Lemma 15.1 (Confluent Christoffel identity). *For all $N \geq 0$,*

$$H_N(q) = (-1)^{ND} \gamma^N \frac{\det M_N}{\Delta_q}. \quad (15.8)$$

Proof. First suppose all roots $a_1, \dots, a_D$ are distinct. For an ordered list y , write $V(y) = \prod_{i < j} (y_j - y_i)$. Expanding two determinants under the integral and using the product measure proves the elementary integration identity

$$\int \det(f_i(x_j))_{i,j=1}^N \det(g_i(x_j))_{i,j=1}^N \prod_j d\mu(x_j) = N! \det \left(\int f_i(x) g_j(x) d\mu(x) \right)_{i,j=1}^N.$$

Taking monomials with the multiplier included gives

$$H_N(q) = \frac{1}{N!} \int V(x_1, \dots, x_N)^2 \prod_{i=1}^N q(x_i) d\mu(x_i).$$

Since

$$V(x, a) = V(x)V(a) \prod_{i,u} (a_u - x_i),$$

we obtain

$$H_N(q) = \frac{(-1)^{ND} \gamma^N}{N! V(a)} \int V(x) V(x, a) \prod_i d\mu(x_i). \quad (15.9)$$

Replace the monomials in both Vandermonde determinants by the monic polynomials p_j . Expand $V(x, a)$ along its first N rows. On integration against $V(x)$, orthogonality kills every selected set of columns except $0, 1, \dots, N-1$. That surviving integral is $N!$, and its complementary minor is $(p_{N+j}(a_u))_{u,j}$. The Laplace sign is positive because the selected columns are the first N . This proves (15.8) for distinct roots.

Now let ν distinct roots tend to a common value a . Taylor expansion shows that division by their internal Vandermonde replaces their successive evaluation rows by the rows $p_{N+j}^{(r)}(a)/r!$, $r = 0, \dots, \nu-1$. Cross-block differences tend to the product (15.6). Applying this separately to every block gives the stated formula. Equivalently, both sides are polynomial identities in the root variables after the removable Vandermonde factors are canceled. Thus the limiting argument does not assume that any weighted Hankel determinant is nonzero. At $N = 0$, both the empty moment determinant and the normalized confluent Vandermonde equal 1. $\square$

The important computational consequence is that the determinant on the right has size D , independent of N . The important theoretical consequence is different: all dependence on N has been localized in jets of a two-branch orthogonal-polynomial sequence.

Relation to Part I. Lemma 4.1 of Part I is the case $\gamma = 1$, roots $a_u = -y_u$, in the reversed normalization $R_n(y) = (-1)^n p_n(-y)$ of (4.1); its proof follows the same integration-and-Laplace route for distinct roots. Part I then takes the confluent limit only for the m -fold root at 0 (Proposition 4.2). Lemma 15.1 adds confluence at every root, both endpoints, and the leading coefficient γ with its sign, so it is printed in full.

16 Local jet structure

16.1 The two branches away from the endpoints

The characteristic equation of (15.1), at $x = 2 + z + z^{-1}$, has roots z, z^{-1} . Matching the first two values gives

$$p_n(2 + z + z^{-1}) = \frac{z^{n+1} - z^{-n}}{z - 1} = A_+(z)z^n + A_-(z)z^{-n}, \quad (16.1)$$

where

$$A_+(z) = \frac{z}{z-1}, \quad A_-(z) = -\frac{1}{z-1}, \quad s(z) = \frac{1}{z-z^{-1}}. \quad (16.2)$$

The letter $s(z)$ is a scalar function, not the number s of distinct nonendpoint roots. For clarity we often write $s_i = s(z_i)$. The inverse function theorem applies locally because $z \neq \pm 1$, and

$$\frac{d}{dx} \log z(x) = s(z), \quad \frac{d}{dx} = z s(z) \frac{d}{dz}. \quad (16.3)$$

Lemma 16.1 (Offendpoint jets). *For $c = 2 + z + z^{-1} \notin \{0, 4\}$ and $r \geq 0$,*

$$\frac{p_n^{(r)}(c)}{r!} = z^n A_{r,+}(n; z) + z^{-n} A_{r,-}(n; z), \quad (16.4)$$

where $A_{r,\pm}(n; z)$ are polynomials of degree exactly r in n . Their leading coefficients are

$$[n^r] A_{r,\pm}(n; z) = \frac{A_{\pm}(z)(\pm s(z))^r}{r!}. \quad (16.5)$$

They can be constructed recursively from $A_{0,\pm} = A_{\pm}$ by

$$A_{r+1,\pm}(n; z) = \frac{zs(z) \partial_z A_{r,\pm}(n; z) \pm ns(z) A_{r,\pm}(n; z)}{r+1}. \quad (16.6)$$

Proof. Differentiate (16.4) once and use (16.3). The normalization by $(r+1)!$ gives (16.6). The derivative term does not increase the degree in n , while multiplication by $\pm ns(z)$ increases it by one. Induction gives (16.5). All its factors are nonzero, so the degree is exact. $\square$

16.2 The two endpoint jets

From (15.1),

$$\sum_{n \geq 0} p_n(x) u^n = \frac{1+u}{1-(x-2)u+u^2}. \quad (16.7)$$

Differentiation in x , normalized by $r!$, gives

$$\sum_{n \geq 0} \frac{p_n^{(r)}(x)}{r!} u^n = \frac{(1+u)u^r}{(1-(x-2)u+u^2)^{r+1}}. \quad (16.8)$$

Lemma 16.2 (Endpoint degrees). *For all $n, r \geq 0$,*

$$\frac{p_n^{(r)}(0)}{r!} = (-1)^{n-r} \binom{n+r}{2r}, \quad (16.9)$$

$$\frac{p_n^{(r)}(4)}{r!} = \frac{2n+1}{2r+1} \binom{n+r}{2r}. \quad (16.10)$$

After the common factor $(-1)^n$ is removed at 0, the polynomial in n has degree $2r$ and leading coefficient $(-1)^r/(2r)!$. At 4 the degree is $2r+1$, with leading coefficient $2/(2r+1)!$.

Proof. At $x = 0$, (16.8) becomes $u^r/(1+u)^{2r+1}$, which gives (16.9). At $x = 4$ it becomes $(1+u)u^r/(1-u)^{2r+2}$. The coefficient is

$$\binom{n+r+1}{2r+1} + \binom{n+r}{2r+1} = \frac{2n+1}{2r+1} \binom{n+r}{2r}.$$

The assertions on leading coefficients follow immediately. For $n < r$ the binomial coefficient is zero, as it must be for a derivative whose order exceeds the polynomial degree. $\square$

The same quadratic recurrence has two different endpoint amplitudes: $p_n(0) = (-1)^n$ but $p_n(4) = 2n+1$. This single extra degree at 4 is what ultimately changes $\binom{\ell}{2}$ into $\binom{\ell+1}{2}$.

17 The determinant degree calculus

17.1 Polynomial shifts and confluent exponential rows

The next two identities isolate the cancellation responsible for the sharp order. If f is a polynomial of degree d with leading coefficient a , then

$$[j^h]f(N+j) \text{ has degree at most } d-h \text{ in } N, \quad [N^{d-h}j^h]f(N+j) = a \binom{d}{h}. \quad (17.1)$$

An expression with $h > d$ is interpreted as zero.

For distinct nonzero numbers $w_1, \dots, w_b$ and positive integers $u_1, \dots, u_b$ summing to D , form the matrix

$$V_{(a,h),j} = w_a^j j^h, \quad 0 \leq h < u_a, \quad 0 \leq j < D.$$

The convention $0^0 = 1$ is used only in the polynomial value j^0 .

Lemma 17.1 (Confluent exponential Vandermonde). *In the indicated block order,*

$$\det V = \prod_{a=1}^b \left(w_a^{\binom{u_a}{2}} \prod_{h=0}^{u_a-1} h! \right) \prod_{a < b} (w_b - w_a)^{u_a u_b}. \quad (17.2)$$

Proof. Start with the ordinary Vandermonde determinant and coalesce u_a variables to each w_a . The matrix of ordinary, unnormalized derivatives of w^j then has determinant $\prod_a \prod_{h < u_a} h! \prod_{a < b} (w_b - w_a)^{u_a u_b}$. Within a block replace ∂_w^h by $(w \partial_w)^h$. This is a triangular change of rows: the coefficient of ∂_w^h is w^h , and all other terms have smaller derivative order. The determinant is multiplied by $\prod_a w_a^{0+1+\dots+(u_a-1)}$. Since $(w \partial_w)^h w^j = j^h w^j$, the resulting matrix is V . $\square$

17.2 Three local leading determinants

For an endpoint block of size u , the leading matrices obtained from (17.1) have entries proportional to $\binom{2r}{h}$ or $\binom{2r+1}{h}$, $0 \leq r, h < u$. As a polynomial in its upper argument, $\binom{x}{h}$ has leading coefficient $1/h!$. Consequently

$$\det_{0 \leq r, h < u} \binom{2r}{h} = \det_{0 \leq r, h < u} \binom{2r+1}{h} = 2^{\binom{u}{2}}. \quad (17.3)$$

Here the Vandermonde in $0, 2, \dots, 2u-2$ or $1, 3, \dots, 2u-1$ contributes $2^{\binom{u}{2}} \prod_{h < u} h!$, canceling the column leading denominators. (The first equality with $u = m$ is the determinant evaluated in (5.3) of Part I, by the same argument.)

For the nonendpoint calculation let $t = k + h$, and write $A_+ = A_+(z)$, $A_- = A_-(z)$ and $s = s(z)$. Consider the $t \times t$ matrix with rows $r = 0, \dots, t-1$, followed first by k plus columns and then by h minus columns:

$$B_{r,(+,j)} = \frac{A_+ s^r}{r!} \binom{r}{j} \quad (0 \leq j < k), \quad B_{r,(-,j)} = \frac{A_- (-s)^r}{r!} \binom{r}{j} \quad (0 \leq j < h). \quad (17.4)$$

Lemma 17.2 (The mixed-block determinant). *The determinant in (17.4) equals*

$$B_z(k) = \frac{A_+^k A_-^h s^{\binom{k}{2}} (-s)^{\binom{h}{2}} (-2s)^{kh}}{\prod_{r=0}^{t-1} r!}, \quad h = t - k, \quad (17.5)$$

and is nonzero.

Proof. Remove the row factors $1/r!$ and the column factors $A_{\pm}$. Use

$$w^r \binom{r}{j} = w^j \frac{1}{j!} \left(\frac{d}{dw} \right)^j w^r.$$

The remaining normalized-derivative matrix is the transpose of a confluent Vandermonde at $w = s$ and $w = -s$. Its determinant is $(-2s)^{kh}$. The column powers supply $s^{\binom{k}{2}}(-s)^{\binom{h}{2}}$, proving (17.5). Every factor is nonzero because $z \notin \{0, 1, -1\}$. $\square$

This lemma is the point at which a mere degree bound becomes an exact degree theorem. It sums all competing assignments of the derivative rows to the two branches, including their determinant signs. Ignoring those signs, or arguing that one assignment has a nonzero leading term, would not establish noncancellation of the whole sector.

18 Canonical sectors and the proof of sharpness

18.1 A definition that still makes sense at resonance

Set

$$f_{0,r}(n) = (-1)^r \binom{n+r}{2r}, \quad f_{4,r}(n) = \frac{2n+1}{2r+1} \binom{n+r}{2r}.$$

Introduce independent marker variables $y_1, \dots, y_s$. Let $\tilde{M}(N; \mathbf{y})$ be the $D \times D$ matrix whose column j contains the following rows, in the same root order as before:

$$(-1)^j f_{0,r}(N+j), \quad 0 \leq r < m, \quad (18.1)$$

$$f_{4,r}(N+j), \quad 0 \leq r < \ell, \quad (18.2)$$

$$y_i z_i^j A_{r,+}(N+j; z_i) + z_i^{-j} A_{r,-}(N+j; z_i), \quad 0 \leq r < t_i. \quad (18.3)$$

Its determinant is a polynomial in N and the markers. Define

$$P_{\mathbf{k}}(N) = \frac{1}{\Delta_q} [y_1^{k_1} \cdots y_s^{k_s}] \det \tilde{M}(N; \mathbf{y}). \quad (18.4)$$

This definition does not require distinct numerical rates. It is the meaning of “canonically branch-labelled” in Theorem 14.2.

In the original matrix M_N , remove $(-1)^N$ from each of its m zero-endpoint rows and z_i^{-N} from each of the t_i rows at c_i . The remaining matrix is (18.1)–(18.3) with $y_i = z_i^{2N}$. Therefore (15.8) gives

$$H_N(q) = \left((-1)^{D+m} \gamma \prod_i z_i^{-t_i} \right)^N \sum_{\mathbf{k}} P_{\mathbf{k}}(N) \prod_i z_i^{2N k_i}.$$

Since $D+m$ has the parity of $\ell + T$, this proves (14.9), with the rates in (14.6).

18.2 The leading coefficient in product form

For each label $\mathbf{k}$, form an ordered list of spectral blocks

$$\mathcal{W}_{\mathbf{k}} = ((-1, m), (1, \ell), (z_1, k_1), (z_1^{-1}, t_1 - k_1), \dots, (z_s, k_s), (z_s^{-1}, t_s - k_s)), \quad (18.5)$$

omitting all blocks of size zero. Its nodes are pairwise distinct: an equality between a node from c_i and a node from c_j would imply $c_i = c_j$, and none is an endpoint node ± 1 . For any such ordered list write

$$\mathcal{V}((w_a, u_a)_{a=1}^b) = \prod_a \left(w_a^{\binom{u_a}{2}} \prod_{h=0}^{u_a-1} h! \right) \prod_{a < b} (w_b - w_a)^{u_a u_b}. \quad (18.6)$$

Finally define

$$B_0(m) = \frac{(-2)^{\binom{m}{2}}}{\prod_{r=0}^{m-1} (2r)!}, \quad B_4(\ell) = \frac{2^{\binom{\ell+1}{2}}}{\prod_{r=0}^{\ell-1} (2r+1)!}. \quad (18.7)$$

Both equal 1 at multiplicity zero.

Theorem 18.1 (Exact leading coefficient). *For P_k defined by (18.4),*

$$[N^{d_k}]P_k(N) = \frac{B_0(m)B_4(\ell) \prod_{i=1}^s B_{z_i}(k_i)}{\Delta_q} \mathcal{V}(\mathcal{W}_k) \neq 0. \quad (18.8)$$

Here $B_{z_i}(k_i)$ is (17.5) with $t = t_i$. Moreover P_k has no term of degree greater than d_k .

Proof. Expand each row of $\tilde{M}$ in the finite collection of basis rows

$$(w^j j^h)_{j=0}^{D-1}. \quad (18.9)$$

Only the finitely many h up to the largest relevant polynomial degree are needed. This factors $\tilde{M} = C(N; \mathbf{y})V$, with C a matrix of coefficient polynomials and V the matrix of basis rows. Cauchy–Binet expresses its determinant as a sum over selections of D basis rows.

A nonzero coefficient minor must select exactly m rows associated with node -1 , exactly ℓ with node 1 , and exactly t_i rows from the two nodes z_i, z_i^{-1} for each i . Otherwise a root block of C , whose support is confined to its own nodes, cannot have full rank. The coefficient of $\mathbf{y}^k$ selects precisely k_i rows at z_i and $t_i - k_i$ rows at z_i^{-1} .

For a fixed node with u selected basis rows, their h indices must be distinct. Their sum is therefore at least $0 + 1 + \dots + (u-1) = \binom{u}{2}$, with equality only for the set $\{0, \dots, u-1\}$. By (17.1), selecting such an index h subtracts h from the maximum possible degree in N of its coefficient. Thus the zero-endpoint block contributes at most

$$\sum_{r=0}^{m-1} 2r - \binom{m}{2} = \binom{m}{2},$$

the four-endpoint block contributes at most

$$\sum_{r=0}^{\ell-1} (2r+1) - \binom{\ell}{2} = \binom{\ell+1}{2},$$

and the i th nonendpoint block contributes at most

$$\sum_{r=0}^{t_i-1} r - \binom{k_i}{2} - \binom{t_i - k_i}{2} = k_i(t_i - k_i).$$

The sum of these bounds is d_k .

There is exactly one selection of basis rows that can attain this total degree: at every node take $h = 0, \dots, u-1$. The associated basis determinant is (18.6) for the list $\mathcal{W}_k$, by Lemma 17.1. In its coefficient minor, the root blocks separate. For the endpoint 0, the leading row factors from Lemma 16.2, followed by (17.3), give $B_0(m)$. At 4 they give $B_4(\ell)$. At c_i the leading coefficient matrix is exactly (17.4), so Lemma 17.2 gives $B_{z_i}(k_i)$. Dividing by Δ_q proves (18.8).

Every factor in (18.8) is nonzero. In particular, the unique maximal-degree basis selection cannot vanish or be canceled by another selection, because every other selection has strictly smaller degree. This proves both exactness and noncancellation. $\square$

Remark 18.2 (Why the naive degree is too large). For a block of multiplicity t , multiplying row degrees without taking the determinant into account suggests $\binom{t}{2}$ degrees. The k plus-branch rows consume $\binom{k}{2}$ degrees, and the $t - k$ minus rows consume $\binom{t-k}{2}$. The surviving degree is their difference, $k(t - k)$. The consumption is forced by linear dependence of equal low-degree basis rows; it is not an accidental cancellation at special numerical parameters.

19 Minimal recurrences, order, and the secondary factorization

19.1 Exponential polynomials and minimality

Let $(Ef)(N) = f(N + 1)$ denote forward shift.

Lemma 19.1 (Minimal annihilators). *Suppose $\lambda_1, \dots, \lambda_b$ are distinct nonzero complex numbers and Q_i are nonzero polynomials of degrees d_i . Then $f(N) = \sum_i Q_i(N)\lambda_i^N$ has minimal characteristic polynomial*

$$\prod_{i=1}^b (X - \lambda_i)^{d_i+1}. \quad (19.1)$$

Its ordinary generating function has reduced denominator $\prod_i (1 - \lambda_i u)^{d_i+1}$.

This is Lemma 6.1 of Part I, restated in Part II's notation (Part I writes E for the shift and t for the generating-function variable). The manuscript proves it again; its proof isolates each block by annihilating the others instead of using Chinese-remainder projectors, and is kept as a second route.

Second proof (manuscript 03). On a term $Q(N)\lambda^N$, the operator $E - \lambda$ gives $\lambda^{N+1}(Q(N+1) - Q(N))$. It lowers the degree by exactly one until the polynomial is zero. In contrast, $E - \mu$, for $\mu \neq \lambda$, preserves the degree and multiplies the leading coefficient by $\lambda - \mu$.

These observations also prove independence of distinct exponential blocks. To isolate one block in a purported zero sum, annihilate all other blocks by their respective powers of $E - \lambda_i$. The selected block remains nonzero unless it was zero initially. If a polynomial g has a zero of multiplicity e at λ_i , then its action on that block is nonzero for $e \leq d_i$ and zero for $e > d_i$. Hence each factor in (19.1) is necessary and their product suffices.

For the generating function, apply powers of $u d/du$ to $(1 - \lambda_i u)^{-1}$. A degree- d_i polynomial multiplier yields a pole of order exactly $d_i + 1$ at $u = \lambda_i^{-1}$. Distinct poles cannot cancel. This gives the stated reduced denominator. $\square$

Applying the lemma to Theorem 18.1 proves (14.11) under nonresonance. Without that hypothesis its labelled product still annihilates every term. Summing its exponents uses

$$\sum_{k=0}^t k(t-k) = \frac{t(t+1)(t-1)}{6}, \quad (19.2)$$

and gives (14.12). If two or more labels have the same rate, their sum has degree at most their largest degree, strictly less in order contribution than the sum of their separate $d + 1$ values. Thus any resonance strictly lowers the order below $\mathcal{R}_0$. This completes the proof of Theorem 14.2 and its corollaries.

In the nonresonant case the ordinary generating function is therefore

$$\sum_{N \geq 0} H_N(q)u^N = \frac{B_q(u)}{\prod_k (1 - \lambda_k u)^{d_k+1}}, \quad \deg B_q < \mathcal{R}_0, \quad B_q(0) = 1, \quad (19.3)$$

and the displayed fraction is reduced. A limiting specialization of this numerator and denominator need not remain reduced.

19.2 Comparison with the classical exponential bound

There is a useful elementary consistency check on the sharp formula. For $0 \leq k \leq t$,

$$1 + k(t-k) \leq \binom{t}{k}. \quad (19.4)$$

To see this, identify a k -subset of $\{1, \dots, t\}$ with a partition inside a $k \times (t - k)$ rectangle. Every size from 0 to $k(t - k)$ is attained: fill complete rows and then at most one partial row. There are therefore at least $k(t - k) + 1$ different partitions. Also

$$1 + \binom{m}{2} \leq 2^m, \quad 1 + \binom{\ell + 1}{2} = 1 + \ell + \binom{\ell}{2} \leq 2^\ell.$$

For nonnegative numbers a_j , $1 + \sum_j a_j \leq \prod_j (1 + a_j)$. Combining these inequalities, label by label and then summing, proves

$$\mathcal{R}_0 \leq 2^{m+\ell} \prod_i \sum_{k_i=0}^{t_i} \binom{t_i}{k_i} = 2^D. \quad (19.5)$$

Thus the order bound here is compatible with the classical one, while retaining the determinant cancellations that the larger bound ignores. For a single repeated root, summing $1 + k(t - k)$ gives $t + 1 + \binom{t+1}{3}$. Strictness for $t \geq 4$ already follows from $\binom{t}{2} > 1 + 2(t - 2)$ at $k = 2$.

19.3 Proof of the secondary Hankel factorization

We prove a general identity, then substitute the explicit leading coefficients of our determinant sequence.

Lemma 19.2 (A finite Hankel determinant of exponential polynomials). *Suppose $f(N) = \sum_{i=1}^b Q_i(N) \lambda_i^N$, with $\deg Q_i = d_i$, leading coefficient a_i , and $r_i = d_i + 1$. Put $R = \sum_i r_i$. The rates may coincide. Then*

$$\det_{0 \leq u, v < R} f(u + v) = (-1)^{\sum_i \binom{r_i}{2}} \prod_i (d_i! a_i)^{r_i} \lambda_i^{r_i(r_i-1)} \prod_{i < j} (\lambda_j - \lambda_i)^{2r_i r_j}. \quad (19.6)$$

Proof. Write $Q_i(N) = \sum_{h=0}^{d_i} a_{i,h} N^h$. Let $U_{u,(i,h)} = \lambda_i^u u^h$, $0 \leq h \leq d_i$, and let B be block diagonal, with i th block

$$(B_i)_{h,j} = \begin{cases} \binom{h+j}{h} a_{i,h+j}, & h + j \leq d_i, \\ 0, & h + j > d_i. \end{cases}$$

The binomial expansion of $Q_i(u + v)$ gives the exact factorization $(f(u + v)) = U B U^T$. Each B_i is anti-triangular, so

$$\det B_i = (-1)^{\binom{r_i}{2}} a_i^{r_i} \prod_{h=0}^{d_i} \binom{d_i}{h}.$$

Lemma 17.1 gives $\det U$. The factorials simplify using

$$\left(\prod_{h=0}^{d_i} h! \right)^2 \prod_{h=0}^{d_i} \binom{d_i}{h} = (d_i!)^{r_i}.$$

Taking determinants proves (19.6) for distinct rates. The same identity holds for coincident rates by polynomial continuation in the λ_i , or directly because U then has repeated columns and both sides vanish. $\square$

Theorem 14.5 follows on taking the canonical labels $\mathbf{k}$ as the blocks of Lemma 19.2 and inserting (18.8). All its non-resonance factors are nonzero throughout the admissible root-multiplicity stratum. In particular, no hidden amplitude-zero locus can lower the maximal recurrence order.

Suppose, locally, exactly two rates meet, their difference vanishes to first order along a parameter curve, and no root itself reaches another root or an endpoint. Formula (14.16) then shows that the secondary determinant vanishes to exactly $2r_k r_j$ orders along that curve. This is a quantitative statement about rank loss, not just a test that some exceptional parameter exists.

20 Resonance and a root-free exact algorithm

20.1 The exact cancellation rule

Let Λ be the set of distinct numerical values among the rates. For $\lambda \in \Lambda$ define

$$Q_\lambda(N) = \sum_{k:\lambda_k=\lambda} P_k(N). \quad (20.1)$$

Some of these sums may be identically zero even though every summand has a nonzero leading coefficient.

Theorem 20.1 (All-parameter minimality rule). *At every admissible parameter choice, the minimal characteristic polynomial is*

$$\chi_q(X) = \prod_{\substack{\lambda \in \Lambda \\ Q_\lambda \neq 0}} (X - \lambda)^{1+\deg Q_\lambda}. \quad (20.2)$$

Before computing any cancellations, a valid smaller annihilator than the labelled product is

$$\Psi_q(X) = \prod_{\lambda \in \Lambda} (X - \lambda)^{1+\max\{d_k:\lambda_k=\lambda\}}. \quad (20.3)$$

Proof. Group (14.9) by numerical rate, discard zero polynomials, and apply Lemma 19.1. The degree of a sum is at most the maximum degree of its summands, proving (20.3). $\square$

Equation (20.2) is exact, but it is not a short classification of every possible resonant cancellation. Such a classification would require a more detailed description of the polynomials Q_λ . The next theorem instead gives a terminating algebraic procedure for each exact input, without constructing any spectral roots.

[Added 29 September 2026, batch 42: for a single repeated root $q = \gamma(x - c)^t$, $c \notin \{0, 4\}$, Part III describes the grouped polynomials exactly (Section 34): after grouping, at most one group loses further multiplicity, and exactly one, except at $c = 2$ (Theorems 30.1 and 30.4).]

20.2 Why a finite prefix can be a proof

Theorem 20.2 (Certified finite-prefix recovery). *Let $h = (h_N)_{N \geq 0}$ be a nonzero sequence over a characteristic-zero field, known to satisfy a monic constant-coefficient recurrence of order at most R from $N = 0$ onward. Its minimal recurrence can be recovered from $h_0, \dots, h_{2R-1}$ as follows. For $r = 1, \dots, R$, solve*

$$h_{n+r} + \sum_{j=0}^{r-1} g_j h_{n+j} = 0, \quad 0 \leq n < R. \quad (20.4)$$

The first soluble system gives the unique minimal monic characteristic polynomial $X^r + \sum_{j < r} g_j X^j$.

Proof. Let $A(E)h = 0$ be a known-to-exist monic annihilator of order R , padding a lower-order one by a power of X if necessary. For any candidate g the residual sequence $u = g(E)h$ also satisfies $A(E)u = 0$, because polynomials in shift commute. If (20.4) holds, its first R values are zero. The monic recurrence then forces every subsequent value to be zero by induction. Thus every soluble system gives a genuine all-index annihilator, not merely a fitted prefix.

The actual minimal annihilator gives a solution for its order, so the first successful order is minimal. Two different monic annihilators of that degree would differ by a nonzero annihilator of smaller degree, which is impossible. The largest index used by the algorithm is $R - 1 + R = 2R - 1$. $\square$

The prior order bound is indispensable. Without it, an arbitrarily long matching prefix does not certify a recurrence for an arbitrary sequence. For (14.1), Theorem 14.2 supplies exactly the bound needed.

20.3 No factorization into individual roots is required

Suppose $q \in \mathbb{Q}[x] \setminus \{0\}$. Strip its powers of x and $x - 4$, and perform square-free decomposition on the remainder:

$$q(x) = \gamma x^m (x - 4)^\ell \prod_{t \geq 1} f_t(x)^t, \quad (20.5)$$

where the nonconstant f_t are monic, square-free and pairwise coprime. Let $b_t = \deg f_t$. Ordinary polynomial gcds with the derivative produce this decomposition; irreducible factorization and numerical root finding are unnecessary. The bound becomes

$$R = \prod_{t \geq 1} (t + 1)^{b_t} \left(\kappa + 1 + \frac{1}{6} \sum_{t \geq 1} b_t t(t - 1) \right). \quad (20.6)$$

Compute the first $2R$ determinants directly from their rational moment matrices and apply Theorem 20.2. This determines the true minimal polynomial even when the roots are complex, multiplicatively dependent, or not conveniently expressible by radicals.

All recovered coefficients lie in $\mathbb{Q}$. More generally, if the input coefficients lie in an exactly computable algebraic number field, the same linear systems work over that field. Solvability over its algebraic closure is equivalent to solvability over the field itself, by Gaussian elimination; uniqueness of the minimal monic polynomial gives descent. For arbitrary unspecified complex numbers, “algorithm” presupposes an exact representation with equality and field operations available. We do not assert an effective equality test for arbitrary complex oracles.

The transparent root-free implementation uses $O(R^4)$ field operations for generating the determinants and searching all orders with naive Gaussian elimination, apart from polynomial preprocessing and moment generation. This is an arithmetic-operation estimate, not a bit-complexity claim. The included fixed-size Christoffel path is much faster when the roots are already given exactly.

20.4 Algebraic rank-drop loci

The finite-prefix proof also turns exceptional orders into explicit polynomial conditions. Fix a degree $D \geq 0$ and consider nonzero polynomial multipliers of exactly that degree. The uniform all-index bound $S = 2^D$ follows from (19.5). For $1 \leq r < S$, form

$$\mathcal{H}_{S,r}(q) = (H_{n+j}(q))_{0 \leq n < S, 0 \leq j \leq r}. \quad (20.7)$$

Then

$$\text{ord}(H(q)) \leq r \iff \text{rank } \mathcal{H}_{S,r}(q) \leq r. \quad (20.8)$$

For the forward direction, the recurrence gives a column dependence. For the reverse direction, take a nonzero dependence, normalize its highest nonzero coefficient, and apply the residual argument of Theorem 20.2. Its degree is at most r .

Every entry $H_n(q)$ is a polynomial in the coefficients of q . Consequently the order-at-most- r locus is defined, within the degree- D coefficient space, by the vanishing of all $(r + 1)$ -minors of (20.7). It is relatively Zariski closed. On a fixed root-multiplicity stratum, S may be replaced by the sharper $\mathcal{R}_0$. Theorem 14.5 is an explicit factorization of the maximal-rank test on that stratum; lower-rank tests can involve further cancellation conditions after rates have already collided.

21 Worked examples and checks against the repository

21.1 Recovering the shifted linear multiplier

Take $q(x) = x^m(a + bx)$, with $b \neq 0$ and $a(a + 4b) \neq 0$. Then $\gamma = b$, $\ell = 0$, $s = 1$, $t_1 = 1$ and $c_1 = -a/b$. The two rates are $-bz$ and $-bz^{-1}$, so their sum and product are

$$-b(z + z^{-1}) = -b(c_1 - 2) = a + 2b, \quad (-bz)(-bz^{-1}) = b^2.$$

Their two degrees are both $\binom{m}{2}$. Theorem 14.2 gives exactly (13.1). At $a = 0$ the multiplier is bx^{m+1} , whose order is $\binom{m+1}{2} + 1$. At $a = -4b$ it is $bx^m(x - 4)$, whose order is $\binom{m}{2} + 2$. These endpoint results follow from the same theorem, without taking a singular limit of the two nonendpoint amplitudes.

Relation to Part I. In this subsection a and b are Part I's parameters, $q(x) = x^m(a + bx)$ is the multiplier of $D_N^{(m)}(a, b) = H_N(q)$, and $-z^{\pm 1}$ are Part I's roots $r_{\pm}$ of (6.1), so the two rates are Part I's $\lambda_{\pm}$ of (6.3). The paragraph is thus a second proof, by a different route, of the minimality in Theorem 2.2 and of the rows $a = 0$, $a = -4b$ and $a(a + 4b) \neq 0$ of Theorem 2.3; the row $b = 0$, $a \neq 0$ is the case $s = 0$ of Corollary 14.4 (an observation of this merge). It does not reprove Part I's coefficient formula, numerator degree, reciprocity or asymptotic constant. One further consistency check (an observation of this merge, not of the manuscript): for $q = x^m$, Theorem 18.1 gives the leading coefficient $B_0(m) (-1)^{\binom{m}{2}} \prod_{h < m} h! = 2^{\binom{m}{2}} \prod_{h < m} h! / \prod_{r < m} (2r)!$, which is Part I's h_m of (1.6), the leading coefficient of $H_N^{(m)}$.

21.2 A repeated fourth power with order fifteen

For

$$q(x) = (2x - 9)^4 = 16(x - \frac{9}{2})^4, \quad (21.1)$$

choose $z = 2$. The five rates, their exact degrees and their leading coefficients are as follows.

k	λ_k	d_k	$\text{lc}(P_k)$
0	1	0	1/729
1	4	3	8/81
2	16	4	-64/27
3	64	3	-2048/81
4	256	0	65536/729

The exact minimal characteristic polynomial is

$$\chi_q(X) = (X - 1)(X - 4)^4(X - 16)^5(X - 64)^4(X - 256). \quad (21.2)$$

The full expansion $H_N = \sum_{k=0}^4 P_k(N) \lambda_k^N$ is

$$\begin{aligned} P_0(N) &= \frac{1}{729}, \\ P_1(N) &= \frac{24N^3 + 204N^2 + 572N + 520}{243}, \\ P_2(N) &= -\frac{1728N^4 + 13824N^3 + 43344N^2 + 62784N + 35648}{729}, \\ P_3(N) &= -\frac{6144N^3 + 21504N^2 + 23552N + 10240}{243}, \\ P_4(N) &= \frac{65536}{729}. \end{aligned} \quad (21.3)$$

These polynomials may be obtained by (18.4), or by exact interpolation after the theorem has established their degrees and nonvanishing. The first determinant values are

$$1, \quad 3401, \quad 2480745, \quad 1026904425, \quad 327789036905, \quad 92503701028201.$$

No polynomial of degree 14 can annihilate this sequence: all five rates are distinct and every displayed top coefficient is nonzero. This nonexistence assertion is proved for the entire sequence, not inferred from the displayed six values.

21.3 Resonance can cancel an entire branch

For $q(x) = (x - 2)^2$, choose $z = i$. The three canonical sectors are

$$P_0(N) = \frac{i}{4}, \quad P_1(N) = N + 1, \quad P_2(N) = -\frac{i}{4}, \quad (21.4)$$

with rates $-1, 1, -1$. The two nonzero constant sectors cancel exactly, so

$$H_N((x - 2)^2) = N + 1, \quad \chi_q(X) = (X - 1)^2.$$

For completeness, the leading-coefficient formula gives the two constants and the coefficient 1 of N in P_1 . The condition $H_0 = 1$ then fixes its constant term. This already proves (21.4). Taking the maximum degree separately for each collided rate, without checking cancellation, would retain an unnecessary $X + 1$ factor.

A similar phenomenon need not involve repeated roots of q . Take

$$q(x) = (x - \frac{9}{2})(x + \frac{1}{2}), \quad z_1 = 2, \quad z_2 = -2. \quad (21.5)$$

The four canonical constants, in order $(0, 0)$, $(0, 1)$, $(1, 0)$, $(1, 1)$, are $-1/15, -1/3, 1/3, 16/15$; their rates are $-1/4, -1, -1, -4$. Hence

$$H_N(q) = \frac{16}{15}(-4)^N - \frac{1}{15}(-\frac{1}{4})^N, \quad \chi_q(X) = X^2 + \frac{17}{4}X + 1. \quad (21.6)$$

The multiplicity profile permits generic order 4, the grouped annihilator has order 3, and exact cancellation lowers the true order to 2. Thus root multiplicities alone cannot determine the order on the resonance locus.

21.4 A nine-value proof of a resonant fourth-power formula

For $q = (x - 2)^4$ and $z = i$, the degree law and grouped annihilator give

$$(E - 1)^5(E + 1)^4H = 0. \quad (21.7)$$

The exact closed formula is

$$H_N((x - 2)^4) = \frac{(N + 1)(N + 2)^2(N + 3)}{12}. \quad (21.8)$$

Here is a finite proof using the already established all-index bound, rather than an extrapolation from data. The right side is a polynomial of degree 4, so it satisfies (21.7). The first nine values of both sides are

N	0	1	2	3	4	5	6	7	8
H_N	1	6	20	50	105	196	336	540	825

These determinant values follow from the 4×4 formula (15.8). They can be reproduced using only integers: write $u_{n,r} = p_n^{(r)}(2)/r!$, start with

$$(u_{0,r})_{r=0}^3 = (1, 0, 0, 0), \quad (u_{1,r})_{r=0}^3 = (1, 1, 0, 0),$$

and apply

$$u_{n+1,r} = u_{n,r-1} - u_{n-1,r}, \quad u_{n,-1} = 0.$$

Then $H_N = \det(u_{N+j,r})_{0 \leq r, j < 4}$. The difference between the two sides of (21.8) satisfies a monic recurrence of order 9 and has its first nine values zero. Induction proves that it is identically zero. Since the product polynomial has degree 4 and nonzero leading coefficient $1/12$, its minimal polynomial is $(X - 1)^5$. The generic order 15 has dropped to 5.

[Added 29 September 2026, batch 42: (21.4) and (21.8) are the cases $t = 2, 4$ of a known product of Cigler for every t , printed with a condensation proof as Theorem 30.4 in Part III.]

21.5 An irreducible rational input

No spectral representation is needed to run the exact algorithm. For $q = x^2 + 1$, the square-free profile is $b_1 = 2$, so $R = 4$. The root-free calculation gives

$$\chi_q(X) = X^4 - 5X^3 + 4X^2 - 5X + 1. \quad (21.9)$$

For $q = (x^2 + 1)^2$ the profile is $b_2 = 2$, and the order is 15. The archive records its full characteristic polynomial and exact input coefficients. These examples illustrate the intended use of the algorithm: its input is a rational coefficient list, not approximate complex roots or guessed multiplicative relations among them.

22 Exact asymptotics and stability of the leading sector

The expansion (14.9) is finite and exact. Consequently it contains more information than a leading asymptotic equivalence. Suppose a label $\mathbf{k}_*$ has strictly larger rate modulus than every other label. Put $\lambda_* = \lambda_{\mathbf{k}_*}$, $d_* = d_{\mathbf{k}_*}$, and let $M = \max_{\mathbf{k}} d_{\mathbf{k}}$. Choose $\rho < |\lambda_*|$ no smaller than the other rate moduli. Then

$$H_N(q) = \lambda_*^N P_{\mathbf{k}_*}(N) + O(\rho^N N^M). \quad (22.1)$$

When there are no other labels the error term is zero. In particular,

$$H_N(q) \sim L_{\mathbf{k}_*} N^{d_*} \lambda_*^N.$$

Every algebraic correction in inverse powers of N can be read from the finite polynomial $P_{\mathbf{k}_*}$; no infinite asymptotic series is needed for fixed parameters.

Corollary 22.1 (Repeated roots outside the interval). *Suppose all nonendpoint roots satisfy $c_i > 4$, and choose $z_i > 1$. Then the unique largest-modulus sector is $k_i = t_i$ for every i . Thus*

$$H_N(q) \sim L_{(t_1, \dots, t_s)} N^\kappa \left((-1)^{\ell+T} \gamma \prod_i z_i^{t_i} \right)^N. \quad (22.2)$$

No multiplicative-independence assumption is needed for this conclusion.

Proof. Reducing any k_i multiplies the modulus by a factor $z_i^{-2} < 1$. Hence the maximal sector is unique even if some of the smaller sectors resonate. At $k_i = t_i$ all $k_i(t_i - k_i)$ vanish, leaving degree κ . Its leading coefficient is nonzero by Theorem 18.1. $\square$

This separates two effects that are easy to conflate. Repeating a root outside the support changes the exponential growth rate and can greatly increase the recurrence order. It does *not* increase the power of N in the leading asymptotic term; that power comes entirely from the endpoint multiplicities. The repeated offendpoint roots instead create higher polynomial degrees in intermediate, subdominant sectors. For (21.1), for instance,

$$H_N((2x - 9)^4) \sim \frac{65536}{729} 256^N,$$

even though its 16^N sector has degree 4 and its recurrence has order 15.

The estimate is for fixed parameters. As roots approach 0, 4 or each other, the spectral gaps and the constants need not be uniform. A double-scaling limit cannot be justified by simply using (22.1) with moving parameters. That is one of the separate research directions listed below.

[Added 29 September 2026, batch 53: for roots approaching 0 or 4 on the scale N^{-2} , Part IV proves the uniform double-scaling limit directly (Theorems 47.1 and 47.4), without using (22.1); roots approaching each other away from the endpoints are not covered.]

23 Verification, implementation, and formalization status

23.1 Independent exact calculations

The archive contains a reference implementation in Python 3.10 or later, using only the standard library. Every numerical calculation uses integers or exact rational numbers. Gaussian elimination uses pivoting; there is no assumption that a signed or resonant moment matrix is positive definite.

The implementation has two determinant paths. The first constructs the original matrix in (14.1) from Catalan numbers. The second computes jets from the polynomial recurrence (15.1) and uses the fixed-size Christoffel determinant. The paths share the low-level elimination routine, but neither uses the explicit sector leading constants. A third calculation interpolates sector polynomials at distinct rational spectral parameters and checks their degrees, leading constants and extra values. The secondary-Hankel factorization is tested by yet another determinant, built from the already computed sequence values.

The following are the actual results recorded by `code/verify.py` (shipped here as `code/02-root-collisions-verify.py`; its recorded run is `data/02-root-collisions-verification.json`, with console text `data/02-root-collisions-verification.txt`). Counts include simple and boundary instances; they measure executed identities, not independent universal theorems.

Executed check	Count
Endpoint-jet identities	304
Original moment determinant versus Christoffel determinant	1,554
Summed sector orders versus the multiplicity formula	12
Secondary-Hankel product factorizations, including resonances	15
Exact sector degrees and leading coefficients	48
Full exponential-polynomial evaluations	328
Nonresonant recurrence residuals	200
Nonresonant minimal-recurrence recoveries	9
Resonant recurrence residuals	204
Resonant minimal-recurrence recoveries	12
Inter-root resonance recoveries	3
Root-free multiplicity profiles	111
Root-free recurrence recoveries	3
Central-root fourth-power product evaluations	40
Total; all passed	2,843

The original-matrix comparisons cover 222 multiplier/scale choices at sizes $N = 0, \dots, 6$. They include roots below the support, in its interior, at both endpoints and above the support; multiplicities up to five occur. The sector checks use twelve nonresonant patterns, including mixed endpoint and repeated-root blocks, with recurrence orders up to 26. The central fourth-power formula is checked through $N = 39$, although its finite proof in Section 21 needs only the first nine values.

The distinction between evidence and proof matters. A successful interpolation for twelve parameter choices does not prove Theorem 18.1. Its proof is the unique maximal-degree Cauchy–Binet selection and the three nonzero local determinants. Conversely, once a universal order bound is available, the finite certificate for a *particular* specialization is a rigorous proof by Theorem 20.2. The article uses both kinds of computation but does not conflate them.

23.2 The implementation interface

The module `code/catalan_collisions.py` provides the following principal entry points. (It is shipped here as `code/02-root-collisions-catalan_collisions.py`; §27 explains how to import it.) Polynomial arrays always use ascending powers.

Routine	Purpose
<code>direct_hankel</code>	Original moment determinant for rational coefficients.
<code>christoffel_values</code>	Fixed-size evaluation with rational roots and multiplicities.
<code>sectors</code>	Rates, exact degrees and leading constants for rational z_i .
<code>secondary_hankel_formula</code>	The product in (14.16).
<code>fit_sectors</code>	Exact polynomial interpolation when rates are distinct.
<code>multiplicity_profile</code>	Endpoint multiplicities and square-free root counts.
<code>minimal_recurrence</code>	Recovery from a prefix, given a proved order bound.
<code>recurrence_from_coefficients</code>	Root-free minimal recurrence for rational q .

The last routine is intentionally transparent, not optimized for large bounds. Its optional `max_bound` argument prevents an unexpected large computation. The mathematical theorem permits all complex roots; the reference implementation uses rational inputs, and does not purport to implement a general algebraic-number package.

23.3 A feasible formalization route

No Lean source has been compiled or checked for this article. The repository location does not confer formal status on the present proofs. A formalization should separate the argument into small algebraic layers.

First formalize normalized jets of (15.1) over a characteristic-zero field and their generating functions. Next prove the confluent Vandermonde identity, including the Euler-derivative row change, and formalize the polynomial-shift coefficient bound (17.1). The decisive reusable lemma is the maximal-degree Cauchy–Binet argument: it tracks one distinct list of derivative orders per spectral node and identifies the unique smallest such list.

The orthogonality and integral proof of Christoffel may be formalized analytically, but an algebraic alternative is attractive. One may define the moment functional by $L(x^n) = C_n$, prove its Gram identities for p_n directly from generating functions, and prove Christoffel by determinant multilinearity. This would avoid importing measure theory into the core recurrence theorem.

Finally, the shift-operator minimality lemma, the UBU^T secondary factorization and the finite-prefix recovery theorem are pure finite linear algebra over a field. They should be reusable outside Catalan determinants. An audit must distinguish imported classical facts from newly checked lemmas and must not replace a missing confluence proof with an admitted theorem hidden inside a computation.

24 Further research questions and proposed next steps

The questions below are precise continuations of this article. They are not all asserted to be unsolved in the published literature; a broader literature check is part of each proposed project.

24.1 Closed classification of cyclotomic resonances

For a single repeated root, classify the minimal recurrence as an explicit function of t and the order of z^2 when z is a root of unity other than ± 1 . The degree law reduces the task to finitely many grouped polynomials, but the cancellations can be substantial. The examples at $z = i$ show that counting distinct rates is not enough. A first target is a uniform closed form for $H_N((x-2)^t)$ for every t , together with a direct explanation of the parity effects, rather than a separate finite certificate for each t .

[Added 29 September 2026, batch 42: answered by Part III for one repeated root away from the endpoints, with a correction to the question itself: t and the order L of z^2 do not determine the minimal recurrence, and the sign of z^L is also needed (for $t = 3$, $c = 1$ and $c = 3$ both have $L = 3$ but orders 7 and 6). Theorem 30.1 gives every characteristic multiplicity for $c \neq 2$ in terms of t , L and z^L , and Theorem 37.1 computes it from c without roots. The uniform central formula is a known product of Cigler, credited and re-proved in Theorem 30.4; the parity effects come from the single tied pair of Lemma 34.2. The endpoint cases $c = 0, 4$ are outside Part III.]

24.2 Pairwise resonance and cancellation order

Suppose exactly two labelled rates agree and all other rates are separate. Determine when their leading coefficients cancel, and how many further coefficients can cancel. Theorem 18.1 makes the first test explicit, while (16.6) determines all later tests. The aim is a structural criterion in the root parameters, not just the output of Gaussian elimination. The distinct-root example (21.5) is a small initial case.

[Added 29 September 2026, batch 42: partly answered. Within the rate groups of one repeated root, Part III's Section 34 gives the structural criterion: only the tied pair $k_{\pm} = (t \mp L)/2$ can cancel at the leading order, it cancels exactly when the sign ρ of (34.5) is -1 , and then exactly one further degree is lost unless $c = 2$. Collisions between distinct roots, including the example above, remain open.]

24.3 Sector polynomials and symmetric functions

The degree $k(t-k)$ is the area of a $k \times (t-k)$ rectangle, and the proof uses a mixed confluent Vandermonde. Seek a Schur-function or factorial-Schur description of the entire P_k , not only its leading coefficient. Useful outputs would be exact factorizations, coefficient sign rules for real roots outside $[0, 4]$, and formulas that remain stable as two parameters approach one another. Such a description could also identify an earlier equivalent formulation of the multiplicity law.

24.4 Uniform endpoint and root-collision limits

A concrete endpoint scaling is

$$c_N = 2 + 2 \cosh(\tau/N), \quad q_N(x) = (x - c_N)^t.$$

Determine the limit, locally uniformly in τ , of

$$\frac{H_N(q_N)}{(-1)^{tN} N^{t(t+1)/2}},$$

including its value at $\tau = 0$. The analogous left-endpoint scaling $c_N = 2 - 2 \cosh(\tau/N)$ should use the exponent $t(t-1)/2$. For $t = 1$ the branch formula already reduces these limits to elementary hyperbolic

functions. The general problem asks for a uniform confluent determinant limit and error bounds. A second version lets two distinct nonendpoint roots approach at a rate of order N^{-1} .

[Added 29 September 2026, batch 53: the endpoint half is answered by Part IV for any fixed numbers m and ℓ of roots approaching 0 and 4 together, with arbitrary internal collisions: with $\nu = N + (m + \ell)/2$ the normalized determinant is a convergent series in ν^{-2} , locally uniform in the scaled coordinates (Theorem 47.1); its leading term factorizes into confluent cosine and sine determinants, its first correction is explicit (Theorem 47.2), and in the scaling posed here the limits are $F_t^+(\tau)$ at the right endpoint and $F_t^-(\tau)$ at the left, with the stated values at $\tau = 0$ and an N^{-1} term that is only the centring $N \mapsto N + t/2$ (Corollary 54.1); an explicit error bound is Theorem 50.2. Relative to the fixed-endpoint determinant the ratio tends to 1 exactly when every outward displacement is $o(N^{-2})$ (Theorem 47.4). The nonendpoint collisions of the second version remain open (Section 57.1).]

24.5 Other eventually constant Jacobi recurrences

Krattenthaler's recurrence theorem covers many moment families beyond Catalan numbers [3]. Extend the exact degree analysis to a three-term recurrence whose coefficients are eventually constant. Away from branch points, the main new issue is a zero of one branch amplitude caused by the finite initial perturbation. Classify how its vanishing order changes the jet degrees and the noncancellation matrix. This would distinguish universal collision behavior from behavior tied to the Catalan boundary condition $p_1 = x - 1$.

24.6 Rational multipliers and finite poles

Replace $q(x)$ by a rational function whose poles avoid the support. The determinant identities underlying Uvarov transformations provide a starting point [8]. The target is an exact analogue of the sector-degree formula when evaluation and Cauchy-transform blocks occur together. One must first specify which normalizing factors are removed and which recurrences hold from the initial index rather than only eventually.

24.7 Coefficient-space equations and arithmetic reductions

Eliminate the spectral variables from (14.16) to obtain small defining equations for maximal-order failure directly in the coefficients of q . The finite minors in (20.7) are correct but can be very large. Seek sharper degrees, irreducible components and efficient membership tests. For monic integral multipliers, a related problem is to classify primes at which the minimal recurrence drops after reduction. The factorials in the leading constants show why characteristic-zero noncancellation cannot simply be reused unchanged in small characteristic.

24.8 Fast certified algorithms and formal verification

Combine the proved order bound with fast minimal-recurrence algorithms, but reduce the cost of obtaining the initial determinants as well. A useful target is an output-sensitive exact algorithm with bit-complexity bounds in D , the coefficient heights and the final recurrence order. Separately, formalize the maximal-degree determinant lemma and the finite-prefix certification theorem first: these are the shortest reusable path from the present article to a machine-checked result.

25 Conclusion

The selected repository extension has a uniform answer at the level of exponential sectors. A root of multiplicity t_i away from 0, 4 contributes the exact degree $k_i(t_i - k_i)$ to its k_i th branch sector. The two endpoints contribute different triangular numbers. The leading coefficient is a nonzero product, so distinct

sectors cannot disappear at nonresonant parameters. This proves the sharp order formula (14.12), with cubic rather than exponential complexity for a single repeated root.

The secondary-Hankel factorization identifies the maximal-order exceptional locus exactly. Resonances may cause further cancellations, but they do not obstruct an exact answer for a specified rational or algebraic multiplier: a proved order bound turns finitely many determinant values into a complete minimal-recurrence certificate. The remaining research questions concern structural descriptions of those cancellations, uniform collision limits, broader moment families and formalization, not gaps in the stated multiplicity proof.

26 Normalization and sign audit

There are four easy ways to obtain an incorrect statement in this problem. This section (Appendix A of the manuscript) records the conventions that prevent them.

Size versus last index. Throughout, N is the determinant size. In a convention that indexes rows by $0 \leq i, j \leq n$, one must substitute $N = n + 1$. The initial term $H_0 = 1$ is part of the recurrence sequence used here.

The Christoffel sign. The product in the multiplier is $\prod(x - a_u)$, whereas the mixed Vandermonde supplies $\prod(a_u - x)$. This contributes $(-1)^{ND}$. The m zero-endpoint rows add $(-1)^{mN}$ when their factors are removed. The remaining sign is therefore $(-1)^{(\ell+T)N}$, exactly the sign in (14.6).

Normalized versus unnormalized jets. The rows of M_N are $p_{N+j}^{(r)}(a)/r!$. Thus Δ_q contains only cross-root differences. Extra products of $r!$ in its denominator would double-count confluence. The factorials appearing later in B_0, B_4, B_z and $\mathcal{V}$ have separate origins and must not be removed by changing the jet convention halfway through the proof.

Two different kinds of collision. A collision of roots of q changes the factorization profile and must be treated by normalized jets. A collision of rates λ_k can occur even when all roots of q are distinct, and must be treated by adding their polynomial coefficients. These are logically different operations. The first creates the degree law; the second can reduce the actual recurrence order.

As a final check, $H_N(uq) = u^N H_N(q)$. The determinant on the left of (14.16) therefore scales by $u^{\mathcal{R}_0(\mathcal{R}_0-1)}$. On its right, the rate powers and rate differences contribute

$$\sum_k r_k(r_k - 1) + 2 \sum_{k < j} r_k r_j = \mathcal{R}_0(\mathcal{R}_0 - 1),$$

while the sector leading coefficients do not depend on the overall scale γ . The scaling agrees exactly.

27 Reproducing the package

This section is Appendix B of the manuscript, adapted to the shipped file names. As delivered, the package was reproduced from its top directory by

```
python3 code/verify.py --out data
latexmk -pdf -interaction=nonstopmode -halt-on-error article.tex
```

The first command regenerates the recorded numerical outputs; the second builds the PDF. There are no third-party Python dependencies. *In this report these commands must not be run in place.* Here `code/verify.py` and `data/` belong to Part I, so the first command would run Part I's suite and rewrite Part I's data; the shipped

verifier code/02-root-collisions-verify.py still imports its sibling module under the delivered name `catalan_collisions`, which is not importable under the shipped name, and its default output directory `data` would overwrite Part I's `data/verification.json` and `data/verification.txt`. The delivered Makefile, shipped as `code/02-root-collisions-Makefile`, has the same defect: its `check target` runs `code/verify.py --outdata`, which here is Part I's verifier. Restore the delivered names in a scratch copy instead (POSIX shell, run from the report directory):

```
W=/path/to/scratch/root-collisions
mkdir -p "$W/code" "$W/recorded"
for f in code/02-root-collisions-*.py; do
  cp "$f" "$W/code/${f#code/02-root-collisions-}"; done
for f in data/02-root-collisions-*.json; do
  cp "$f" "$W/recorded/${f#data/02-root-collisions-}"; done
cd "$W" && python3 code/verify.py --out data
```

The run writes seven files to `$W/data`, which can be compared with the recorded copies in `$W/recorded`; on 28 September 2026 this recipe passed all 2,843 checks, and the seven files agreed with the recorded ones apart from Windows line endings. The following root-free example, run in the same scratch directory, needs no spectral factorization:

```
import sys
sys.path.insert(0, "code")
from catalan_collisions import recurrence_from_coefficients

chi, bound = recurrence_from_coefficients([1, 0, 1], max_bound=16)
print(bound) # 4
print(chi) # ascending coefficients: 1, -5, 4, -5, 1
```

The main data files are shipped in `data/` with the prefix `02-root-collisions-` (delivered without it):

- `verification.json`, `verification.txt`: the recorded run of Section 23;
- `sectors.json`: the twelve nonresonant patterns, with sector polynomials, leading constants and characteristic polynomials;
- `resonances.json`: exact minimal recurrences at resonant parameters;
- `rational_input_examples.json`: root-free rational-coefficient examples;
- `central_root_certificate.json`: the nine-value certificate of (21.8);
- `fourth_power_sequence.csv`: exact values of $H_N((2x - 9)^4)$.

The code is shipped in `code/` with the same prefix (`verify.py`, `catalan_collisions.py` and the delivered root Makefile), and the provenance notes, delivered as `notes/provenance.md`, as `02-root-collisions-provenance.md` at the report root. Rational values are stored as exact strings, for example `"8/81"`. Arrays representing polynomials use increasing powers. Sector polynomials are polynomials in the determinant size N , not generating-function numerators.

Part III. Cyclotomic resonances of one repeated root

Part III was added on 29 September 2026 in batch 42 of ProveIt's incoming-report intake, from one manuscript written against the pinned state of Parts I and II. It answers Part II's research question "Closed classification of cyclotomic resonances" (Section 24.1) for a single repeated root away from the endpoints: at every parameter it determines the minimal recurrence, every characteristic multiplicity, and a root-free factorization, and it shows that the question's proposed parameters, the multiplicity t and the order of z^2 , do not determine the answer. Parts I and II are unchanged except for dated pointers to this addition (Sections 11, 20, 21 and 24), a line on the title page (whose top space was reduced by 3 mm so that it still fits on one page) and a paragraph of the abstract, a contents entry, one bibliography entry used only by Part III and a clause in the closing note on source dates, and two new labels on Part II's research subsections. Section 28 records the provenance and fixes the notation of the addition.

28 Provenance and conventions of Part III

28.1 One manuscript, one addition

Part III is batch-42 manuscript 01, *A One-Defect Law for Cyclotomic Catalan Hankel Recurrences*, subtitled *Complete single-root classification, a necessary sign parameter, and the central exception*, a research article prepared with OpenAI ChatGPT for Vladimir Reshetnikov's ProveIt research program and dated 29 September 2026 (22 A4 pages as delivered). It arrived as the archive `ProveIt_Cyclotomic_Catalan_Recurrences.zip` (inner directory `cyclotomic_catalan/`, main file `article.tex`) in ProveIt commit 8315d24e3 and was placed in 3609d0473. It was written against ProveIt commit 9754e83603e223811b7515eb892f22c847144593, reading this report's combined `article.tex` (Git blob b29ea42fc5bba69d23c3b036bee25357e202f356), its provenance notes and the directory listing at that commit; its source notes add that the standalone Part II manuscript was read as well. The report's files are unchanged between the pin and the placement commit (the same blob), so every statement the manuscript makes about "the repository report" or "the prior report" refers to Parts I and II as printed above.

The manuscript's code and data are shipped beside the earlier files with the prefix `03-cyclotomic-` (Part I's unprefixed files count as 01, Part II's as 02), and its source and comparison notes, delivered as `SOURCES.md`, as `03-cyclotomic-SOURCES.md` at the report root. Its manuscript, delivery README and PDF are not shipped; they survive in the arrival commit. Section 39 lists the shipped files.

The manuscript contributes, in the order printed here: the repository question, what is already known, and why grouping equal rates is not enough (Section 29); the definitions, the complete noncentral classification with its elementary sign test, and the central theorem (Section 30); a self-contained derivation of the confluent Christoffel identity in its normalization (Section 31); an exact subset expansion of every canonical sector (Section 32); the leading coefficient, the first centred correction and an exact reflection of the sector polynomials (Section 33); the one-defect proof (Section 34); closed orders, the exceptional locus and its count, the reflection $c \mapsto 4 - c$ and quadratic growth in the multiplicity (Section 35); the central exception via Cigler's known product, with a self-contained condensation proof (Section 36); a root-free factorization (Section 37); worked examples and the classification of all rational roots (Section 38); its verification record and rebuild instructions (Section 39); eight research directions (Section 40); a conclusion; a normalization and proof audit (Section 42); and a source and package concordance (Section 43). Every result, proof, example, remark, limitation and question of the manuscript is printed. Where it re-derives something Parts I or II already prove, the overlap is marked (Section 28.5).

The manuscript's own abstract reads:

Let $H_N^{(t)}(c)$ be the size- N Hankel determinant of the Catalan moments multiplied by $(x - c)^t$, with $c \neq 0, 4$. Write $c = 2 + z + z^{-1}$ and let L be the order of z^2 , possibly infinite. The nonresonant minimal order $t + 1 + \binom{t+1}{3}$ is already established in the ProveIt report that motivates this work. We determine the minimal recurrence at every remaining specialization.

For $3 \leq L \leq t$, put $\eta = \mathbf{1}_{t \equiv L \pmod{2}}$. The exact order is

$$R(t, c) = \frac{L(3t^2 - L^2 + 13 - 3\eta)}{12} - \delta, \quad \delta \in \{0, 1\}.$$

The defect occurs precisely when $\eta = 1$ and $(-1)^{L(t+1)/2} z^{Lt} = -1$. We determine the multiplicity of every characteristic root, not only their total. The proof isolates the first two coefficients of each canonical exponential sector using an exact signed-Vandermonde identity. A collided sector can lose just one degree away from $c = 2$; at $c = 2$ an entire group can disappear. The latter case is governed by a product formula already proved by Cigler for middle-binomial Hankel determinants, which we credit and rederive with the required normalization.

The answer cannot depend on t and L alone: for $t = 3$, $c = 1$ and $c = 3$ both have $L = 3$, but their minimal orders are 7 and 6. Consequences include a complete rational-parameter classification, an exact count of exceptional parameters, an eventual quadratic quasipolynomial law in t , and a root-free factorization computed with $O(t)$ scalar field operations. A standard-library Python audit supplies 22,124 successful exact checks, including independent determinant computations in cyclotomic integer rings.

Status (manuscript 01). This is an unrefereed, AI-assisted manuscript with ordinary mathematical proofs, not a Lean formalization. The proposed extension is the noncentral cancellation classification and its coefficient mechanism: the explicit first centred sector coefficient, the noncentral one-defect theorem, the minimal characteristic polynomial at every cyclotomic specialization and the resulting sign-sensitive order laws, with the exceptional-parameter count, the reflection law and the rational-input classification as corollaries. Classical determinant identities (the Christoffel formula and its confluent form, Andréief's identity, Desnanot–Jacobi condensation), general rationality and the order-at-most- 2^t construction [3], Part II's generic degree law and finite recurrence-recovery algorithm, and Cigler's central product [9] are not presented as new; the signed Vandermonde evaluation and the shift-operator minimality lemma are elementary uses of established techniques, not standalone priority claims. First-discovery priority has not been established: a specialized equivalent result may exist in Wronskian, orthogonal-polynomial or symmetric-function notation, the targeted searches that returned unrelated results do not establish absence from the literature, and independent expert review and a broader priority audit are still appropriate. The comparison with this report is targeted, not a proof audit of every report in ProveIt. The manuscript does not solve multi-root resonance, does not claim uniform collision asymptotics, and does not transfer its characteristic-zero minimality theorem unchanged to positive characteristic; the endpoints $c = 0, 4$ are outside its theorems. Its 22,124 exact assertions are finite supporting checks, not a universal proof. No proof assistant was run, nothing in Part III is formalized, and its placement beside a Lean project confers no formal status. (Sources: the manuscript, its delivery README and its source notes, 03-cyclotomic-SOURCES.md.)

28.2 Reading conventions

Inside Sections 29–43, “this article” and “the present paper” mean manuscript 01 as printed in Part III; “the repository report”, “the ProveIt report”, “the prior report” and “the preceding repository work” mean Parts I and II at the pin 9754e8360; and “the older root-collisions manuscript” is Part II. The manuscript cites Part II's standalone section numbers: its Section 12.1 and Section 12.2 are Sections 24.1 and 24.2 here (“Closed classification of cyclotomic resonances” and “Pairwise resonance and cancellation order”). “The archive” or “the package” means the delivered package, whose code and data are shipped under the prefixed names listed in Section 39. The manuscript's two appendices are printed as Sections 42 and 43, so that Part I's appendices keep their letters. The manuscript cites Krattenthaler's Ramanujan

Journal paper through arXiv:2101.04225, version 5 of 10 September 2021, and points to its Theorem 1, its confluence discussion and Section 8; that is Part I’s reference [3]. Its Barry reference is Part I’s [1], cited as background only. Bracketed paragraphs beginning “Merge note” are this merge’s, not the manuscript’s.

28.3 Notation of Part III

Part III keeps the manuscript’s symbols. Within Sections 29–43 every symbol has its Part III meaning, except in merge notes and where a sentence names Part I or II or cites one of their equations. The single multiplier is $(x - c)^t$ with one root $c \notin \{0, 4\}$; in Part II’s notation this is $q = \gamma x^m (x - 4)^\ell \prod_i (x - c_i)^{t_i}$ with $\gamma = 1, m = \ell = 0, s = 1, t_1 = t, c_1 = c, z_1 = z$, and the manuscript’s $H_N^{(t)}(c)$ is Part II’s $H_N((x - c)^t)$ (14.1). Table 2 lists the symbols that a reader of Parts I and II is likely to misread, and the notions that need care. No symbol was renamed and no normalization changed: the manuscript’s rates λ_k are Part II’s λ_k for one root, and its canonical sector polynomials $P_{t,k}$ have Part II’s normalization (Section 28.5).

Table 2: Symbols and notions of Part III. The last column gives the meaning of the same letter in Part I or II, or the tempting false reading.

Symbol or notion	Meaning in Part III	Not to be confused with
$H_N^{(t)}(c)$	$H_N((x - c)^t)$, the superscript being the multiplicity t (30.1)	Part I’s $H_N^{(m)} = \det(C_{i+j+m})$, whose superscript is the shift; formally $H_N^{(t)}(0) = H_N^{(t)}$, but $c = 0$ is outside the theorems.
<i>resonance</i> ; <i>exceptional</i> ; the “resonant range”	for one root, Part II’s resonance (two equal rates) happens exactly when $L \leq t$, and Corollary 35.1 shows that this is exactly when $R(t, c) < R_{\text{gen}}(t)$ (“exceptional”); the manuscript’s “resonant range” $3 \leq L \leq t$ omits the central case $L = 2, c = 2$	reading “resonant range” as all resonances: the central root is resonant for $t \geq 2$ but is treated separately (Theorem 30.4).
<i>one defect</i> ; δ	further loss <i>after</i> grouping equal rates, 0 or 1; the baseline $R_{\text{grp}} = \sum_r (d_r + 1)$ is the degree of Part II’s grouped annihilator Ψ_q (20.3)	a loss of at most one from the generic order $R_{\text{gen}}(t)$, which is false (Remark 30.2).
t, c, z, γ	multiplicity, root, spectral parameter $c = 2 + z + z^{-1}$, scale	Part II’s t_i, c_i, z_i, γ (agreeing for one root); Part I’s generating-function variable t and coefficients $c_{N,m,k}$.
$L = \text{ord}(z^2)$ $\lambda_k = (-1)^t z^{2k-t}$	the order of z^2 , possibly infinite rate of the k th sector (30.4)	Part II’s sector leading coefficients L_k . equal to Part II’s λ_k for one root with $\gamma = 1$; Part I’s $\lambda_\pm$.
$R(t, c), \chi_{t,c}, R_{\text{gen}}(t), R_{\text{grp}}$	minimal order, minimal polynomial, generic order (29.1) (Part II’s $\mathcal{R}_0$ for one root, (14.14)), grouped order (35.2)	Part I’s polynomials $R_n(y)$ (4.1); Part II’s plain R (a bound or a size).
$I_r, d_r, e_r, G_r(N)$	residue classes of sector indices, their maximal degree, exponents, and grouped sector polynomials (34.2)	Part I’s $d = \binom{m}{2}$ and $e = d + 1$; Part II’s d_k ; G_r is Part II’s Q_λ (20.1).
$\eta, k_\pm, r_*, d_*, \rho, \lambda_*$	parity indicator, tied indices, their class, tied degree, leading ratio, defect root	—
$T = \binom{t}{2}$ (Section 32 onward)	a triangular number	Part II’s $T = \sum_i t_i$ (which is t here); Part I’s $T_{n,k,s}$.
$A_u = \prod_{j < u} j!, K_t(z)$	superfactorial and the common sector factor (32.1)	Part II’s amplitudes $A_\pm(z)$; Part I’s constant A_m and $K = (m - 1)^2$; Part II’s index set $\mathcal{K}$.

Symbol or notion	Meaning in Part III	Not to be confused with
$J, S, S^c, \sigma(S), \Delta(S)$	$\{0, \dots, t-1\}$, a subset, its complement, its sum, its Vandermonde magnitude (32.2)	$\sigma = z^L$ without argument (Section 37); S_a in (37.4); Part II's $S = 2^D$ and Δ_q ; $\Delta_N(x)$ in Section 31 is a Vandermonde.
$F_{t,k}, P_{t,k}(N; z)$	raw and canonical sector polynomials (32.3), (32.4)	$P_{t,k}$ is Part II's P_k for one root; Part I's numerator $P_m(t)$.
q	a multiplier in (31.5), but an indeterminate in Lemma 33.1 and $q = z^2$ in the proof of Theorem 33.2	Part II's multiplier $q(x)$; Part I's denominator $q(t)$.
$h = t - k, d = kh, w = N + t/2$	complementary index, sector degree, centred size variable	Part I's h_m and $d = \binom{m}{2}$; Part II's nodes w_a .
$\ell_{t,k}(z), b_{t,k}(z), Q_{t,k}(w)$	leading coefficient (33.3), first centred correction (33.7), remainder of degree at most $d - 2$	Part II's multiplicity ℓ of the root 4; Part I's parameter b ; $Q_a(X)$ below; Part II's Q_λ and Part I's $Q_m(N; r)$.
$B(S), \bar{B}; B_{a,b}(N)$	offset sum and its weighted mean (33.5); the rectangular product (30.12) with integers a, b	Part I's $B_\pm(N)$ and parameters a, b ; Part II's B_0, B_4, B_z, B_i and $B_q(u)$.
$\mu; \mu_n; \alpha$	the Catalan measure; centred moments (36.2); the arcsine measure	Part I's shifted measures μ_m (μ is Part I's μ_0 and Part II's μ); Part III's μ_n is a number, not a measure.
$D_t(N); h_N^{(t)}$	Cigler's middle-binomial Hankel determinant; $\det(\mu_{i+j+t})$ (Section 36)	Part I's $D_N^{(m)}(a, b)$ and h_m ; Part II's $D = \deg q$.
$s_N = (-1)^{N(N+1)/2}$	a period-four sign	Part I's sign s_m ; Part II's s and $s(z)$.
$v, U_a, S_a, \sigma, Q_a(X), E_a$	$c - 2$; Chebyshev values; $z^a + z^{-a}$; z^L ; quadratic factors and their exponents (Section 37)	Part II's matrix U in UBU^T ; the shift operator E (as in Part II); Part I's $E_m(N)$.
$M, \Phi_M, \varphi, \mathcal{E}_t$	primitive order of z in the audit ring (so $L = M/\gcd(M, 2)$), cyclotomic polynomial, Euler's function, the exceptional set (35.3)	Part II's M_N and M ; Part I's $M_{m,\ell}$; E the shift.

28.4 What Part III answers, and how it corrects the question

Part II's first research question (Section 24.1) asked, for a single repeated root, for the minimal recurrence “as an explicit function of t and the order of z^2 ”, with a uniform closed form for $H_N((x-2)^t)$ and an explanation of the parity effects. Part III answers it for one repeated root away from the endpoints: Theorem 30.1 gives every characteristic multiplicity for $c \neq 2$, Theorem 30.4 the central root (a known product of Cigler, credited), and Theorem 37.1 a root-free factorization. It also shows that the question's framing was incomplete: t and $L = \text{ord}(z^2)$ do *not* determine the answer in odd parity, and the sign of z^L is a necessary further datum. The smallest instance is $t = L = 3$, where $c = 1$ and $c = 3$ have minimal orders 7 and 6 (Corollary 35.2 and (38.1)–(38.2)). The corrected statement of the answer is therefore a function of t, L and $z^L \in \{1, -1\}$, and the parity effects are explained by the tied pair of Lemma 34.2 and the sign ρ of (34.5). Part II's second question (“Pairwise resonance and cancellation order”) is answered only for the tied groups of one repeated root (Section 34); collisions between distinct roots, and endpoint factors, remain open (Section 40).

Consistency with Part II's data (an observation of this merge, checked by a separate exact computation not shipped with the report). Part II's `data/02-root-collisions-resonances.json` records twelve single-root cases, $c = 1, 2, 3$ with $t = 1, \dots, 4$, each with its minimal order, full characteristic polynomial and initial values. All twelve characteristic polynomials, not only the orders, coincide with those given by Theorems 30.1, 30.4 and 37.1, and the orders coincide with the rows $t \leq 4$ of `data/03-cyclotomic-orders_c1_c2_c3.csv`: 0 mismatches. Seven of the twelve lie below the generic order ($c = 1, 3$ at $t = 3, 4$ and $c = 2$ at $t = 2, 3, 4$); the other five ($t = 1$ for all three roots, and

$t = 2$ for $c = 1, 3$) have $L > t$, so they are nonresonant by Corollary 35.1 and have the generic order, although that file is described as recording resonant parameters. The case $(x - 2)^3$ of `data/02-root-collisions-rational_input_examples.json`, with characteristic polynomial $(X^2 + 1)^3$, is the case $m = 1$ of (30.16). The central products at $t = 2, 4$ are $B_{1,1}(N) = N + 1$ and $B_{2,2}(N) = (N + 1)(N + 2)^2(N + 3)/12$, which are (21.4) and (21.8); the manuscript's residue-class formulas (38.3) and the table for $c = 1$ in Section 38 agree with direct determinant evaluation.

28.5 Where the merge had to choose

1. *Placement.* Part III is printed after Part II and before Part I's appendices, so that no section, equation, theorem or table number of Parts I and II moves (checked against the `.aux` file of a build of the committed text). Its sections continue the numbering (Sections 28 to 43); the manuscript's Sections 1–13 are Sections 29–41, and its Appendices A and B are Sections 42 and 43.
2. *Labels.* Every label of Part III carries the prefix `cyc:`; without it, eight of the manuscript's labels, `sec:main`, `sec:examples`, `thm:main`, `lem:Christoffel`, `lem:minimal`, `eq:Christoffel`, `eq:lambda` and `eq:measure`, would have collided with Part I's bare labels. Two labels were added to Part II's research subsections so that they can be cited; no existing label was changed.
3. *Overlaps with Parts I and II.* The manuscript re-derives, without assuming the repository report correct, several things already proved there. Printed with a pointer and kept: the Catalan measure, orthogonality and $\det(C_{i+j}) = 1$ (a third route, after Part I's Section 3 and Part II's Section 15); the confluent Christoffel identity (Lemma 31.1 is Part II's (15.8) for $q = (x - c)^t$, and Part I's Lemma 4.1 and Proposition 4.2 are other special cases); the minimal-annihilator lemma (Lemma 34.1 restates Part I's Lemma 6.1 and Part II's Lemma 19.1, with a third proof and the generating-function statement); the exact sector degree $k(t - k)$ and the generic order (Part II's Theorem 14.2 and Corollary 14.3, re-proved here by the subset expansion and the explicit leading coefficient (33.3), a different route from Part II's Cauchy–Binet argument); and the central examples at $t = 2, 4$, which the central product of Theorem 30.4 contains. As an observation of this merge, the leading coefficients (33.3) reproduce all five leading coefficients of Part II's order-fifteen example (the table before (21.2), at $z = 2$) and the two constants of (21.4): the two sector normalizations agree. No statement of Parts I or II was changed.
4. *A third route to half of the central product* (an observation of this merge). The neighbouring report `cigler-conjecture-8-schur` (same directory) proves, for Cigler's polynomials $b_m(t)$ with $b_m(1) = \binom{m}{\lfloor m/2 \rfloor}$, an even-shift Hankel–Schur identity (its Theorem 3.1) and, at $t = 1$, the dimension product $\prod_{i,j \leq k} (n + i + j - 1)/(i + j - 1)$ (its Proposition 7.2). Together they give $D_{2k}(n) = B_{k,k}(n)$, the even case of Cigler's product used in Section 36, by a Schur-function route; the odd case is not covered there. That report and its two siblings concern conjectures of the same Cigler paper [9]; Part III uses its proved Theorem 1, not a conjecture.
5. *Bibliography.* The manuscript's references to Krattenthaler and Barry are Part I's [3, 1]; its reference to Cigler's middle-binomial paper is added to the common bibliography as [9]. Its reference to the pinned repository report is replaced by Parts I and II themselves and the pin recorded in Section 28.1; the sentence that cited it now points to Section 28.1.
6. *File names.* Section 39 gives the shipped names beside the delivered ones. The manuscript's rebuild commands are printed as delivered but must not be run in place here; a recipe on a scratch copy follows them.
7. *Front matter.* The manuscript's title page, abstract, status paragraph and package paragraph are replaced by Section 28.1, which prints the abstract and the status. The package paragraph is not reprinted because the delivered source and PDF are not shipped; the shipped code and data are listed in Section 39.
8. *Notation.* The manuscript's symbols are kept, with the scoping convention and Table 2 of Section 28.3; nothing was renamed. One evident index slip in (31.7) is corrected, with a merge note after Lemma 31.1.

29 The repository question and the scope of the answer

The relevant ProveIt package is *Shifted Catalan Hankel Polynomials*, whose Part II incorporates *Root Collisions and Sharp Recurrences for Polynomially Weighted Catalan Hankel Determinants* (Parts I and II above; the manuscript’s bibliography entry for this repository report is replaced by Section 28.1). The selected source was checked against repository commit

9754e83603e223811b7515eb892f22c847144593.

Its directory is

SetTheory/Cardinals/docs/reports/hankel-determinants/catalan-and-ballot/
shifted-catalan-hankel-polynomials/.

The combined `article.tex` has Git blob identifier `b29ea42fc5bba69d23c3b036bee25357e202f356`.

For a single nonendpoint root of multiplicity t , that report gives a canonical expansion into $t + 1$ exponential sectors, of respective polynomial degrees $k(t - k)$ for $0 \leq k \leq t$. Distinct exponential rates therefore give the exact order

$$R_{\text{gen}}(t) = \sum_{k=0}^t (k(t - k) + 1) = t + 1 + \binom{t + 1}{3}. \quad (29.1)$$

At resonances it provides a finite exact algorithm, but its research agenda asks for a structural classification when z is a root of unity. It also asks for a uniform central-root formula and an explanation of parity, beyond its separately certified examples at multiplicities two and four. These are the local research questions addressed here.

[Merge note: in this report the canonical expansion and its degrees are Part II’s Theorem 14.2, the generic order (29.1) is its Corollary 14.3, the finite exact algorithm is Theorems 20.1 and 20.2, the research agenda is Section 24, and the certified central examples at multiplicities two and four are (21.4) and (21.8).]

There are two distinct conclusions. First, the noncentral cyclotomic problem admits a closed answer: at most one collided rate has a leading cancellation, and that cancellation loses exactly one multiplicity. Second, the suggested input “multiplicity and order of z^2 ” is insufficient in odd parity. The sign of z^L is necessary. Thus the result both completes a classification and corrects the parameter list needed to state it.

29.1 What is already known

The Christoffel determinant formula, its confluent version, and general constant-coefficient recurrences for polynomially modified Catalan moments are classical. Krattenthaler [3] treats these identities and, in Section 8, gives the familiar order-at-most- 2^t recurrence construction. Barry’s note [1] is part of the background for the repository’s shifted-linear-multiplier work. Neither rationality alone nor the existence of an exponential expansion is a novelty claim here.

The central product used in Section 36 is also not new. Cigler’s Theorem 1, equation (23), in [9] evaluates the shifted Hankel determinants of $\binom{n}{\lfloor n/2 \rfloor}$. A translation and a row–column sign change give exactly the $c = 2$ product required here. His proof also uses condensation. Recognizing this prior result closes the repository’s central subproblem without assigning it new priority.

The additional content is the explicit first centered sector coefficient, the noncentral one-defect theorem, the minimal characteristic polynomial at every cyclotomic specialization, and the resulting sign-sensitive order laws. These statements are not present in the selected repository source. A broader literature equivalence in different orthogonal-polynomial or symmetric-function notation has not been ruled out. The comparison is not a proof audit of every report in ProveIt.

29.2 Why the problem is not solved by identifying equal rates

An exponential polynomial $\sum_k P_k(N) \lambda_k^N$ must first be grouped by equal numerical values of λ_k . The largest degree within one group is only an upper bound for the degree of its sum. Equal leading coefficients can cancel, and symmetry can make the whole sum zero. The prior report's central square already illustrates the latter phenomenon [(21.4)].

The main proof computes one more coefficient than the degree law. That additional coefficient is sufficient everywhere except at the single central value $c = 2$. This is what turns a finite procedure for each example into a uniform classification.

30 Definitions and the complete classification

Write $\mathbb{N}_0 = \{0, 1, 2, \dots\}$ and

$$C_n = \frac{1}{n+1} \binom{2n}{n}, \quad H_N^{(t)}(c) = \det_{0 \leq i, j < N} \left(\sum_{a=0}^t \binom{t}{a} (-c)^{t-a} C_{i+j+a} \right), \quad H_0^{(t)}(c) = 1. \quad (30.1)$$

Throughout the main classification, $t \geq 1$ and $c \in \mathbb{C} \setminus \{0, 4\}$. Choose a root z of

$$z + z^{-1} = c - 2. \quad (30.2)$$

Then $z \neq 0, 1, -1$. The two choices are reciprocal and give the same answers after relabeling.

A monic polynomial $g(X) = \sum_{j=0}^R g_j X^j$ is an annihilator when

$$\sum_{j=0}^R g_j H_{N+j}^{(t)}(c) = 0 \quad \text{for every } N \geq 0. \quad (30.3)$$

The least possible degree is denoted $R(t, c)$, and the monic polynomial of that degree by $\chi_{t,c}$. All characteristic roots occurring below are nonzero; consequently the same minimal polynomial is necessary even for recurrences required only after a finite initial segment.

Let $L = \text{ord}(z^2) \in \{2, 3, \dots\} \cup \{\infty\}$. Define

$$\lambda_k = (-1)^t z^{2k-t} \quad (0 \leq k \leq t). \quad (30.4)$$

If $L < \infty$, equality of these rates means exactly equality of the indices modulo L .

Theorem 30.1 (Minimal recurrence at every noncentral parameter). *Assume $c \notin \{0, 2, 4\}$.*

If $L > t$, including $L = \infty$, then

$$\chi_{t,c}(X) = \prod_{k=0}^t (X - \lambda_k)^{k(t-k)+1}, \quad R(t, c) = R_{\text{gen}}(t). \quad (30.5)$$

If $3 \leq L \leq t$, define, for $0 \leq r < L$,

$$I_r = \{k : 0 \leq k \leq t, k \equiv r \pmod{L}\}, \quad d_r = \max_{k \in I_r} k(t-k), \quad \eta = \mathbf{1}_{t \equiv L \pmod{2}}. \quad (30.6)$$

When $\eta = 1$, also put

$$k_- = (t-L)/2, \quad k_+ = (t+L)/2, \quad r_* \equiv k_- \pmod{L}, \quad 0 \leq r_* < L, \quad (30.7)$$

and set

$$\delta = \begin{cases} 1, & \eta = 1 \text{ and } (-1)^{L(t+1)/2} z^{Lt} = -1, \\ 0, & \text{otherwise.} \end{cases} \quad (30.8)$$

The exponent $L(t+1)/2$ is an integer whenever it is used. Then

$$\chi_{t,c}(X) = \prod_{r=0}^{L-1} (X - \lambda_r)^{e_r}, \quad e_r = d_r + 1 - \delta \mathbf{1}_{r=r_*}. \quad (30.9)$$

When $\eta = 0$, the subtracted term means zero and no r_* is needed. A factor with exponent zero is omitted. In particular,

$$R(t, c) = \frac{L(3t^2 - L^2 + 13 - 3\eta)}{12} - \delta. \quad (30.10)$$

Remark 30.2 (The meaning of one defect). The baseline in (30.10) is $R_{\text{grp}} = \sum_r (d_r + 1)$, not the larger generic order $R_{\text{gen}}(t)$. Merging many rates can already cause a large decrease from R_{gen} . The theorem says that *further cancellation after this grouping* decreases the order by either zero or one.

Corollary 30.3 (An elementary version of the defect test). *In the resonant range $3 \leq L \leq t$, the defect is one exactly in one of the following two disjoint cases:*

$$\begin{aligned} t, L \text{ even}, \quad L &\equiv 2 \pmod{4}; \\ t, L \text{ odd}, \quad z^L &= (-1)^{(t-1)/2}. \end{aligned} \quad (30.11)$$

There is no defect when t and L have different parity.

The proof of the theorem occupies Sections 31–35. The remaining parameter is governed by a known product.

Theorem 30.4 (Central root; Cigler's product with Catalan normalization). *For $a, b \in \mathbb{N}_0$ put*

$$B_{a,b}(N) = \prod_{i=1}^a \prod_{j=1}^b \frac{N+i+j-1}{i+j-1}. \quad (30.12)$$

Every empty product is one. For $m \in \mathbb{N}_0$,

$$H_N^{(2m)}(2) = B_{m,m}(N), \quad (30.13)$$

$$H_N^{(2m+1)}(2) = (-1)^{N(N+1)/2} B_{m,m+1}(N). \quad (30.14)$$

Their minimal polynomials are

$$\chi_{2m,2}(X) = (X-1)^{m^2+1}, \quad (30.15)$$

$$\chi_{2m+1,2}(X) = (X^2+1)^{m(m+1)+1}. \quad (30.16)$$

The even formula includes the trivial multiplier $t = 0$.

Together these results classify every single repeated nonendpoint root, not only rational roots or a bounded list of multiplicities. The endpoints $c = 0, 4$ are intentionally outside the theorem; they require endpoint jets and have different triangular degree laws.

30.1 Immediate generating-function and scaling consequences

If the distinct characteristic roots and their exponents are (λ_r, e_r) , the reduced denominator of $\sum_{N \geq 0} H_N^{(t)}(c) u^N$ is exactly

$$\prod_r (1 - \lambda_r u)^{e_r}. \quad (30.17)$$

There is a numerator of degree less than $R(t, c)$ and constant term one. Thus the theorem determines all poles and their exact orders as well as the recurrence. For the multiplier $\gamma(x-c)^t$, $\gamma \neq 0$, all determinants are multiplied by γ^N . Its monic minimal polynomial is $\gamma^{R(t,c)} \chi_{t,c}(X/\gamma)$.

31 Catalan orthogonality and the confluent determinant

We derive the needed determinant identity rather than assuming the correctness of an unrefereed repository lemma. These are classical arguments; their role here is normalization and self-containment.

31.1 A convenient monic orthonormal family

Define

$$d\mu(x) = \frac{1}{2\pi} \sqrt{\frac{4-x}{x}} dx \quad (0 < x < 4). \quad (31.1)$$

Its moments are C_n . For example, the substitution $x = 4v$ gives a beta integral; integration by parts gives successive moment ratio $2(2n+1)/(n+2)$, with zeroth moment one, exactly the Catalan ratio.

Let

$$p_0(x) = 1, \quad p_1(x) = x - 1, \quad p_{n+1}(x) = (x-2)p_n(x) - p_{n-1}(x) \quad (n \geq 1). \quad (31.2)$$

All p_n are monic. At $x = 2 + 2\cos\theta$, $0 < \theta < \pi$,

$$p_n(x) = \frac{\sin((n+\frac{1}{2})\theta)}{\sin(\theta/2)}, \quad d\mu = \frac{2}{\pi} \sin^2(\theta/2) d\theta, \quad (31.3)$$

where the orientation of the integral is reversed as necessary. The elementary sine orthogonality integral therefore gives

$$\int_0^4 p_n(x) p_m(x) d\mu(x) = \mathbf{1}_{n=m}. \quad (31.4)$$

A triangular monic change of polynomial basis has determinant one. Consequently the unweighted size- N Catalan Hankel determinant is one.

31.2 The determinant formula and its sign

For variables $x_1, \dots, x_N$ set $\Delta_N(x) = \prod_{i < j} (x_j - x_i)$. Expansion of two determinants and integration term by term gives

$$\det \left(\int x^{i+j} q(x) d\mu(x) \right)_{0 \leq i, j < N} = \frac{1}{N!} \int \Delta_N(x)^2 \prod_{j=1}^N q(x_j) d\mu(x_j). \quad (31.5)$$

This is the usual Andréief identity in the particular form needed here. It follows directly because the double sum over permutations contains $N!$ equal copies of the determinant on the left.

Take distinct $a_1, \dots, a_t$ and $q(x) = \prod_{i=1}^t (x - a_i)$. The mixed Vandermonde obeys

$$\Delta_{N+t}(x, a) = \Delta_N(x) \Delta_t(a) \prod_{j=1}^N \prod_{i=1}^t (a_i - x_j). \quad (31.6)$$

Replace its monomial columns by the monic $p_0, \dots, p_{N+t-1}$ without changing the determinant. In its Laplace expansion along the first N rows, integrate against $\det(p_{i-1}(x_j))_{j,i=1}^N$. Orthogonality kills every choice of N columns other than $p_0, \dots, p_{N-1}$. That choice contributes $N!$ times the remaining evaluation determinant, with positive Laplace sign. Combining (31.5) and (31.6) yields

$$\det \left(\int x^{i+j} \prod_{u=1}^t (x - a_u) d\mu(x) \right) = (-1)^{Nt} \frac{\det(p_{N+j}(a_i))_{1 \leq i \leq t, 0 \leq j < t}}{\Delta_t(a)}. \quad (31.7)$$

The sign is caused by $x - a$ in the multiplier and $a - x$ in the mixed Vandermonde. The left side is polynomial in all a_i , so the quotient has a well-defined confluent limit.

Lemma 31.1 (Confluent Christoffel identity). *For every $N, t \in \mathbb{N}_0$ and every $c \in \mathbb{C}$,*

$$H_N^{(t)}(c) = (-1)^{Nt} \det_{0 \leq r, j < t} \left(\frac{p_{N+j}^{(r)}(c)}{r!} \right). \quad (31.8)$$

Proof. In (31.7), let $a_i = c + \varepsilon_i$ and expand each evaluation row by Taylor's formula. The least possible distinct Taylor orders are $0, 1, \dots, t-1$. Their determinant is $\Delta_t(\varepsilon)$ times the determinant of the normalized jets $p_{N+j}^{(r)}(c)/r!$. All higher Taylor terms vanish after division by the Vandermonde and passage to the common limit. This proves the formula, including the empty determinant cases. $\square$

No division by $H_N^{(t)}(c)$ occurred. Some of those weighted determinants can be zero; the proof and the classification remain valid.

[Merge note: Lemma 31.1 is Part II's (15.8) for $q = (x - c)^t$, derived here a second time; Part I's Lemma 4.1 and Proposition 4.2 are further special cases, and the measure and orthogonality above repeat Part I's Section 3 and Part II's Section 15. The recursion (31.2) is Part II's (15.1). In (31.7) the manuscript prints the multiplier as $\prod_{a=1}^t (x - a_a)$; the product index is written u here (as in Part II), since i is a row index.]

32 An exact subset expansion for every sector

Fix $t \geq 1$ and abbreviate

$$T = \binom{t}{2}, \quad A_u = \prod_{j=0}^{u-1} j!, \quad K_t(z) = \frac{(z/(z^2 - 1))^T}{(z - 1)^t A_t}. \quad (32.1)$$

For a subset $S \subseteq J := \{0, \dots, t-1\}$ define

$$\sigma(S) = \sum_{j \in S} j, \quad \Delta(S) = \prod_{\substack{i, j \in S \\ i < j}} (j - i), \quad S^c = J \setminus S. \quad (32.2)$$

Empty products equal one.

Theorem 32.1 (Canonical sector formula). *For $0 \leq k \leq t$, put $h = t - k$ and define*

$$F_{t,k}(N; z) = \sum_{\substack{S \subseteq J \\ |S|=k}} (-1)^{T+h-\sigma(S)} z^{2\sigma(S)+k-T} \Delta(S) \Delta(S^c) \prod_{\substack{i \in S \\ j \in S^c}} (2N + i + j + 1), \quad (32.3)$$

$$P_{t,k}(N; z) = K_t(z) F_{t,k}(N; z). \quad (32.4)$$

Then the all-index identity

$$H_N^{(t)}(c) = \sum_{k=0}^t P_{t,k}(N; z) \lambda_k^N \quad (N \geq 0) \quad (32.5)$$

holds. Each $P_{t,k}$ is polynomial in N of degree at most $k(t - k)$. The definition remains meaningful when the numerical rates coincide.

Proof. Locally write $z = e^u$ and $x = 2 + 2 \cosh u$. Formula (31.2) gives

$$p_n(x) = \frac{z^{n+1} - z^{-n}}{z - 1} = g(u) (e^{(n+1/2)u} - e^{-(n+1/2)u}), \quad g(u) = \frac{1}{z^{1/2} - z^{-1/2}}. \quad (32.6)$$

Also $du/dx = z/(z^2 - 1)$. For a Wronskian of t functions, change of variable multiplies its determinant by $(du/dx)^T$: row r has leading derivative term $(du/dx)^r \partial_u^r$, and all other terms are combinations of earlier

rows. A common factor g in all functions similarly multiplies the Wronskian by g^t , by the Leibniz rule and a triangular row transformation. The normalization of row r by $r!$ contributes A_t^{-1} .

Set $v_j = N + j + \frac{1}{2}$. After these row transformations the Wronskian in (31.8) has columns

$$(v_j^r e^{v_j u} - (-v_j)^r e^{-v_j u})_{r=0}^{t-1}.$$

Expand each column by its two summands. Let S be the plus columns and S^c the minus columns. The exponential factor is $z^{(2k-t)N+2\sigma(S)-T+(2k-t)/2}$. The remaining determinant is the Vandermonde in the signed v_j . Within each sign block its magnitude is $\Delta(S)\Delta(S^c)$; between the blocks it contributes $\prod_{i \in S, j \in S^c} (v_i + v_j)$.

There are h signs from choosing the minus summand, $\binom{h}{2}$ from pairs of minus columns, and one for each plus column preceding a minus column. The number of the latter pairs is $k(t-1) - \sigma(S) - \binom{k}{2}$. Their total is exactly $T + h - \sigma(S)$. Finally $g(u)^t = z^{t/2}/(z-1)^t$ cancels all half-integral powers, leaving $z^{2\sigma(S)+k-T}$ outside $z^{(2k-t)N}$. The sign $(-1)^{Nt}$ in the Christoffel identity is absorbed into λ_k^N . This proves (32.3)–(32.5). There are kh linear factors in each product, giving the stated bound. $\square$

Remark 32.2 (No branch choice in the final formula). The auxiliary local logarithm and square root in the proof disappear from (32.3). Only integer powers of z remain. Since the result is a rational identity in z , valid off $0, \pm 1$, no global logarithm or continuation convention is required.

This expression is useful computationally: $F_{t,k}$ has Laurent polynomial coefficients in z with integer coefficients. In a cyclotomic integer ring it can be evaluated without any division. The common nonzero factor $K_t(z)$ has no effect on degrees or on cancellation among sectors.

33 The first two sector coefficients and reflection

The exact degree law alone is inherited from the preceding repository work. The additional coefficient calculated here distinguishes a one-degree loss from an arbitrarily long cancellation. [Merge note: the degree law is Part II's Theorem 14.2; Theorem 33.2 below re-proves it for one root, and its leading coefficient (33.3) is the one-root case of Part II's Theorem 18.1 (Section 28.5).]

33.1 A signed finite Vandermonde identity

Lemma 33.1. *For $h = t - k$ and an indeterminate q ,*

$$W_{t,k}(q) := \sum_{\substack{S \subseteq J \\ |S|=k}} (-1)^{\sigma(S)} \Delta(S) \Delta(S^c) q^{\sigma(S)} = (-1)^{\binom{k}{2}} A_k A_h q^{\binom{k}{2}} (1-q)^{kh}. \quad (33.1)$$

Proof. Take the $t \times t$ matrix with rows indexed by $j = 0, \dots, t-1$. Its first k columns are $j^a q^j$, $0 \leq a < k$, and its last h columns are j^b , $0 \leq b < h$. The convention $j^0 = 1$ includes $j = 0$. Laplace expansion along the first k columns gives the left side of (33.1) multiplied by $(-1)^{\binom{k}{2}}$.

To compute the same determinant, start with columns of derivatives of w^j at the two nodes $q, 1$. The ordinary confluent Vandermonde is $A_k A_h (1-q)^{kh}$. Replacing ∂_w^a by $(w \partial_w)^a$ is triangular in the derivative order and multiplies the determinant by $q^{0+1+\dots+(k-1)}$ in the first block and one in the second. Its columns are exactly those of our matrix. Equating the two evaluations proves the identity. $\square$

For $q \neq 0, 1$, logarithmic differentiation gives

$$\frac{q W'_{t,k}(q)}{W_{t,k}(q)} = \binom{k}{2} - \frac{khq}{1-q}. \quad (33.2)$$

This is an algebraic weighted mean of $\sigma(S)$, not a positivity or probability assertion. The weights have signs and can be complex.

33.2 Leading coefficient and the decisive correction

Theorem 33.2 (Exact degree and first centered correction). *Let $h = t - k$, $d = kh$, and $w = N + t/2$. The leading coefficient of $P_{t,k}$ is*

$$\ell_{t,k}(z) = (-1)^{h(h+1)/2} \frac{2^{kh} A_k A_h z^{k^2}}{A_t (z-1)^t (z^2-1)^{\binom{k}{2} + \binom{h}{2}}}, \quad \ell_{t,k}(z) \neq 0. \quad (33.3)$$

In particular $\deg_N P_{t,k} = kh$. If $d \geq 1$, then

$$P_{t,k}(N; z) = \ell_{t,k}(z) \left(w^d + \frac{kh(h-k)(z^2+1)}{4(z^2-1)} w^{d-1} \right) + Q_{t,k}(w), \quad \deg Q_{t,k} \leq d-2. \quad (33.4)$$

A polynomial of degree at most -1 means zero. If $d = 0$, the sector is the constant $\ell_{t,k}(z)$.

Proof. Every product in (32.3) has leading coefficient 2^d . Its weighted sum is, by (33.1),

$$[N^d]F_{t,k} = (-1)^{T+h+\binom{k}{2}} 2^d A_k A_h z^{k^2-T} (1-z^2)^d.$$

Multiply by $K_t(z)$. Since $T = \binom{k}{2} + \binom{h}{2} + kh$, collecting the signs and powers gives exactly (33.3). All factors are nonzero under the standing hypotheses, proving the exact degree.

For the next coefficient, rewrite the mixed factor as

$$2N + i + j + 1 = 2w + i + j + 1 - t.$$

For a fixed subset S , the sum of its d offsets equals

$$\begin{aligned} B(S) &= \sum_{i \in S, j \in S^c} (i + j + 1 - t) \\ &= (h-k)\sigma(S) + kT - kh(t-1). \end{aligned} \quad (33.5)$$

Indeed $\sum_{j \in S^c} j = T - \sigma(S)$, which proves the second line by direct summation. The relative coefficient of w^{d-1} in this product is $B(S)/2$.

The weighted sum uses the same weights as $W_{t,k}(z^2)$. By (33.2), its mean offset is

$$\begin{aligned} \bar{B} &= (h-k) \left(\binom{k}{2} - \frac{khq}{1-q} \right) + kT - kh(t-1) \\ &= \frac{kh(h-k)(q+1)}{2(q-1)}, \quad q = z^2. \end{aligned} \quad (33.6)$$

Dividing by two gives the coefficient in (33.4). Only lower powers remain, proving the assertion. $\square$

It is useful to isolate the correction as

$$b_{t,k}(z) = \frac{k(t-k)(t-2k)(z^2+1)}{4(z^2-1)}. \quad (33.7)$$

Then $b_{t,t-k} = -b_{t,k}$. Except for the trivial factors $k = 0, t$ or $k = t/2$, its only zero away from the endpoints is $z^2 = -1$, namely $c = 2$.

33.3 An exact reflection of the whole sector polynomial

Proposition 33.3 (Centered sector reflection). *For every $0 \leq k \leq t$, with $h = t - k$,*

$$P_{t,h}(N; z) = (-1)^{t(t+1)/2} z^{(h-k)t} P_{t,k}(-N - t; z). \quad (33.8)$$

Consequently,

$$\frac{\ell_{t,h}(z)}{\ell_{t,k}(z)} = (-1)^{(h-k)(t+1)/2} z^{(h-k)t}. \quad (33.9)$$

The exponent in the sign is always an integer.

Proof. In the subset formula replace S by the complement of $\{t - 1 - j : j \in S\}$. This new subset has size h and sum $T - k(t - 1) + \sigma(S)$. Its two Vandermonde magnitudes are unchanged, its power of z gains $(h - k)t$, and replacing N by $-N - t$ changes the sign of every one of the kh mixed linear factors. The remaining sign simplifies to $(-1)^{t(t+1)/2}$. This is a termwise bijection of the two sums and proves (33.8). Alternatively the same reflection comes from $p_{-n-1} = -p_n$ and reversal of the columns of the Wronskian.

Comparison of the leading coefficients gives $(-1)^{t(t+1)/2+kh} z^{(h-k)t}$. Its sign equals the one in (33.9); the difference of the two exponents is even. One can also obtain (33.9) immediately by dividing the products in (33.3). $\square$

The center $-t/2$ is therefore natural, rather than a convenient numerical fit. At a colliding symmetric pair, the two leading terms are either equal or opposite, and the first centered corrections have opposite relative signs. That is the algebraic source of the classification.

34 Why the noncentral cancellation has exactly one defect

34.1 Minimal annihilators of exponential polynomials

Lemma 34.1. *Suppose $f(N) = \sum_{j=1}^b Q_j(N) \alpha_j^N$, where the α_j are distinct and nonzero and each Q_j is a nonzero polynomial of degree d_j . The minimal annihilator of f is*

$$\prod_{j=1}^b (X - \alpha_j)^{d_j+1}. \quad (34.1)$$

The same conclusion holds for eventual annihilators. Its generating function has reduced denominator $\prod_j (1 - \alpha_j u)^{d_j+1}$.

Proof. On $Q(N) \alpha^N$, the operator $E - \alpha$, where $(Ef)(N) = f(N + 1)$, gives $\alpha^{N+1}(Q(N + 1) - Q(N))$. It lowers the polynomial degree by exactly one until the term becomes zero. In contrast, $E - \beta$ with $\beta \neq \alpha$ preserves the degree and has a nonzero leading coefficient. Distinct exponential blocks are independent: applying the annihilators of all but one block isolates a nonzero polynomial multiple of the remaining exponential. A nonzero polynomial cannot vanish at every sufficiently large integer. It follows that any annihilator, even an eventual one, must contain every factor in (34.1) to the displayed multiplicity. Their product clearly suffices.

For generating functions, $(u d/du)^a (1 - \alpha u)^{-1}$ is the series of $N^a \alpha^N$. A polynomial multiplier of degree d therefore gives a pole of exact order $d + 1$ at $u = \alpha^{-1}$. Distinct poles cannot cancel. This proves the denominator assertion. $\square$

[Merge note: Lemma 34.1 restates Part I's Lemma 6.1 and Part II's Lemma 19.1; the manuscript's proof is kept as a third proof.]

The nonresonant case of Theorem 30.1 now follows from Theorems 32.1 and 33.2. It remains to determine the degree of each grouped polynomial

$$G_r(N) = \sum_{k \in I_r} P_{t,k}(N; z). \quad (34.2)$$

34.2 The geometry of a residue class

Lemma 34.2 (The unique possible top tie). *Assume $2 \leq L \leq t$. In every class I_r , the maximum of $k(t - k)$ has a unique maximizer except possibly in one class. A tied class exists exactly when $t \equiv L \pmod{2}$. Its two maximizers are k_-, k_+ from (30.7); they are consecutive members of that class, separated by L , and*

$$d_* := k_-(t - k_-) = k_+(t - k_+) = (t^2 - L^2)/4. \quad (34.3)$$

Any other index in that class has degree at most $d_* - 2L^2$.

Proof. The identity

$$k(t - k) = \frac{t^2 - (2k - t)^2}{4} \quad (34.4)$$

shows that one must choose the members of the arithmetic progression I_r nearest $t/2$. There are at most two. A tie requires two consecutive members to be symmetric about $t/2$, so their difference is L and their sum is t . They are consequently the two indices in (30.7), and exist exactly under the parity condition. Their values $2k - t$ are $-L, L$. Any further index in that same class has $|2k - t| \geq 3L$, so (34.4) decreases its degree by at least $((3L)^2 - L^2)/4 = 2L^2$. $\square$

A group with a unique maximizer has degree d_r , since its unique leading coefficient is nonzero by (33.3). Thus the only remaining issue is the one tied group.

34.3 The tied group and the nonvanishing next term

Assume the parity condition and use $k = k_-$, $h = k_+$, so $h - k = L$. By (33.9), the ratio of their leading coefficients is

$$\rho = (-1)^{L(t+1)/2} z^{Lt}. \quad (34.5)$$

Since $(z^L)^2 = 1$, this is either 1 or -1 . When $\rho = 1$, the leading coefficient of G_{r_*} is $2\ell_{t,k} \neq 0$ and there is no defect.

Suppose $\rho = -1$. If $d_* = 0$, then $t = L$, the two indices are $0, t$, and they are the only indices in their class. Their sectors are opposite constants, so $G_{r_*} = 0$. The class contributes no characteristic factor rather than the naive exponent one.

If $d_* > 0$, apply (33.4) in the centered variable $w = N + t/2$. The leading terms cancel, while the coefficient of w^{d_*-1} in their sum is

$$\ell_{t,k} b_{t,k} + (-\ell_{t,k})(-b_{t,k}) = \frac{\ell_{t,k} d_* L (z^2 + 1)}{2(z^2 - 1)}. \quad (34.6)$$

It is nonzero when $c \neq 2$. The lower members of the same residue class cannot affect this coefficient, by Lemma 34.2. Therefore $\deg G_{r_*} = d_* - 1$ exactly.

We have now proved all the exponents in (30.9), including the disappearing constant case. Applying Lemma 34.1 proves their minimality and the generating-function assertion. The total order formula will follow from a finite sum of squares.

Corollary 34.3 (Only a real characteristic root can lose multiplicity). *Whenever the noncentral defect occurs, its characteristic root is*

$$\lambda_* = (-1)^t z^{-L} = (-1)^t z^L \in \{1, -1\}. \quad (34.7)$$

No other characteristic root loses multiplicity beyond grouping.

Proof. Substitute $2k_- - t = -L$ into (30.4) and use $z^{2L} = 1$. The degree proof above excludes every other group. $\square$

35 Closed orders, exceptional parameters, and reflection

35.1 Summing the grouped degrees

For each residue r , select a nearest centered value $a_r = 2k_- - t$ at a maximizer. In a tie either choice gives the same square. When t and L have different parity, their squares are the squares of the complete list

$$-(L-1), -(L-3), \dots, L-3, L-1.$$

When the parities agree, they are the squares of

$$-L, -L+2, \dots, L-4, L-2.$$

The finite interval $[-t, t]$ contains all chosen representatives because $t \geq L$. Summing squares in these arithmetic progressions gives

$$\sum_{r=0}^{L-1} a_r^2 = \frac{L(L^2-1)}{3} + L\eta. \quad (35.1)$$

Using (34.4),

$$\begin{aligned} R_{\text{grp}}(t, L) &= \sum_{r=0}^{L-1} (d_r + 1) = L + \frac{Lt^2 - \sum_r a_r^2}{4} \\ &= \frac{L(3t^2 - L^2 + 13 - 3\eta)}{12}. \end{aligned} \quad (35.2)$$

The loss established in the preceding section is exactly δ , proving (30.10).

For completeness, Corollary 30.3 follows without any complex arithmetic. If t, L are even, $z^{Lt} = 1$ and the parity of $L(t+1)/2$ is that of $L/2$. If both are odd, $z^{Lt} = z^L$ and the condition $\rho = -1$ is equivalent to $z^L = (-1)^{(t-1)/2}$. This proves all the sign cases.

35.2 An exact finite exceptional locus

Corollary 35.1 (Every nongeneric nonendpoint parameter). *For fixed $t \geq 1$, the parameters $c \neq 0, 4$ at which $R(t, c) < R_{\text{gen}}(t)$ are exactly*

$$\mathcal{E}_t = \left\{ 2 + 2 \cos \frac{\pi a}{L} : 2 \leq L \leq t, \quad 1 \leq a < L, \quad \gcd(a, L) = 1 \right\}. \quad (35.3)$$

In particular all exceptional parameters are real and lie in $(0, 4)$, and

$$|\mathcal{E}_t| = \sum_{L=2}^t \varphi(L), \quad (35.4)$$

where $\varphi(L)$ is the number of integers $1 \leq a < L$ coprime to L . At the parameter represented by a/L , one has $z^L = (-1)^a$.

Proof. A collision of two rates means $z^{2(k-j)} = 1$ with $1 \leq |k-j| \leq t$. This is equivalent to finite $L \leq t$. Then z is a root of unity, so $c = 2 + z + z^{-1}$ is real in $(0, 4)$. Choose the reciprocal of z as needed to write $z = e^{\pi i a/L}$ with $1 \leq a < L$ and $\gcd(a, L) = 1$. Conversely any such parameter has exactly the stated order L .

At a collision, grouping replaces at least two positive contributions $d_k + 1$ by at most their maximum, strictly decreasing the total generic order. Further cancellations cannot undo that decrease. In the absence of collisions, the nonzero leading coefficients give the full generic order, including $t = 1, c = 2$. Finally $\cos \theta$ is injective on $(0, \pi)$, and the fractions a/L in (35.3) are reduced. No two listed parameters coincide. Counting them proves (35.4). $\square$

Thus the algebraic position of a root, not its numerical proximity to another parameter, determines whether an exact recurrence drops order. There is no open interval of exceptional values.

35.3 The missing sign datum and reflection of the root

Corollary 35.2 (Order under $c \mapsto 4 - c$). *For $c \neq 0, 2, 4$, the orders at c and $4 - c$ differ exactly when $L \leq t$ and L, t are both odd. In that case they differ by one:*

$$R(t, 4 - c) - R(t, c) = 2\delta - 1. \quad (35.5)$$

In every other case they are equal.

Proof. The replacement $z \mapsto -z$ sends c to $4 - c$ without changing z^2 or L . For odd L it changes the sign of z^L ; for even L it does not. Corollary 30.3 shows that the two defects are complementary precisely when t, L are odd and $L \leq t$. Their grouped orders are identical. In the nonresonant case both orders are generic. $\square$

The example $t = L = 3, z = e^{2\pi i/3}$ versus $z = e^{\pi i/3}$, is the smallest instance. The two input roots are $c = 1$ and $c = 3$. The orders 7 and 6 show that t and $\text{ord}(z^2)$ alone cannot parametrize the answer. The additional sign is invariant under the allowed choice $z \leftrightarrow z^{-1}$, so it is genuine information about c , not an artifact of a branch convention.

35.4 Quadratic quasipolynomial growth in the multiplicity

Corollary 35.3. *Fix a noncentral cyclotomic parameter with $L \geq 3$. For $t \geq L$, $R(t, c)$ is a quadratic quasipolynomial in t , with period dividing four and leading coefficient $L/4$. In particular,*

$$R(t, c) = \frac{L}{4}t^2 + O_L(1), \quad (35.6)$$

whereas the nonresonant order is $t^3/6 + O(t)$.

Proof. In (30.10), η depends only on parity and δ depends only on t modulo four once L and z^L are fixed. The remaining term is $Lt^2/4$ plus a constant depending on L . Equation (29.1) gives the comparison. $\square$

Here the asymptotic variable is the root multiplicity t , not the Hankel size N . This distinction matters: the determinant itself is an exact quasipolynomial in N at a cyclotomic parameter, but its recurrence order is a different function.

36 The central exception and the known product

36.1 Reduction to middle-binomial moments

Set $y = x - 2$ in (31.1). Then

$$d\mu = (1 - y/2) d\alpha(y), \quad d\alpha(y) = \frac{dy}{\pi\sqrt{4 - y^2}}, \quad -2 < y < 2. \quad (36.1)$$

The arcsine moments are zero in odd degree and $\int y^{2r} d\alpha = \binom{2r}{r}$, as follows from $y = 2 \cos \theta$. Consequently the centered Catalan moments are

$$\mu_n := \int (x - 2)^n d\mu(x) = (-1)^n \binom{n}{\lfloor n/2 \rfloor}. \quad (36.2)$$

Indeed the odd moment is $-\frac{1}{2} \binom{2r+2}{r+1} = -\binom{2r+1}{r}$.

A monic translation of the polynomial basis leaves the determinant unchanged, giving $H_N^{(t)}(2) = \det(\mu_{i+j+t})$. If

$$D_t(N) = \det_{0 \leq i, j < N} \left(\binom{i+j+t}{\lfloor (i+j+t)/2 \rfloor} \right),$$

row and column signs yield

$$H_N^{(t)}(2) = (-1)^{Nt} D_t(N). \quad (36.3)$$

Cigler's Theorem 1 [9] states $D_t(N) = (-1)^t \binom{N}{2} B_{\lfloor t/2 \rfloor, \lceil t/2 \rceil}(N)$. Equation (36.3) is therefore precisely (30.13)–(30.14). In particular, the uniform central formula requested by the repository [Section 24.1] is already a result established in a 2021 preprint after this change of normalization.

36.2 A self-contained condensation proof

We include the proof to keep the classification independent of an external determinant evaluation. It uses the same classical condensation mechanism as Cigler's argument, not a new method. Write $h_N^{(t)} = \det(\mu_{i+j+t})$ and set $h_0^{(t)} = 1$. The Desnanot–Jacobi determinant identity gives, for $N \geq 2$,

$$h_N^{(t)} h_{N-2}^{(t+2)} = h_{N-1}^{(t)} h_{N-1}^{(t+2)} - (h_{N-1}^{(t+1)})^2. \quad (36.4)$$

It is a polynomial determinant identity and does not require any minor to be nonzero.

We verify the proposed products in (30.13) and (30.14), and then induct on N simultaneously for all $t \geq 0$. At $N = 0$ the products equal one. At $N = 1$, cancellation in (30.12) gives

$$B_{a,b}(1) = \prod_{i=1}^a \frac{i+b}{i} = \binom{a+b}{a}, \quad (36.5)$$

so the products give exactly the moments (36.2).

For the induction put $n = N - 1 \geq 1$. Four ratios suffice. Direct cancellation of the rectangular products yields

$$\frac{B_{m,m}(n) B_{m+1,m+1}(n)}{B_{m,m+1}(n)^2} = \frac{n+2m+1}{2m+1}, \quad (36.6)$$

$$\frac{B_{m,m}(n+1) B_{m+1,m+1}(n-1)}{B_{m,m+1}(n)^2} = \frac{n}{2m+1}, \quad (36.7)$$

$$\frac{B_{m,m+1}(n) B_{m+1,m+2}(n)}{B_{m+1,m+1}(n)^2} = \frac{n+2m+2}{2(n+m+1)}, \quad (36.8)$$

$$\frac{B_{m,m+1}(n+1) B_{m+1,m+2}(n-1)}{B_{m+1,m+1}(n)^2} = \frac{n}{2(n+m+1)}. \quad (36.9)$$

For example, adding the last column of a rectangle gives

$$\frac{B_{a,b+1}(n)}{B_{a,b}(n)} = \prod_{i=1}^a \frac{n+i+b}{i+b}, \quad (36.10)$$

and increasing n telescopes within each row:

$$\frac{B_{a,b}(n+1)}{B_{a,b}(n)} = \prod_{i=1}^a \frac{n+i+b}{n+i}. \quad (36.11)$$

Applying (36.10) and (36.11), or their inverses, to the four ratios cancels all interior factors and gives the displayed right sides. These identities also cover $m = 0$.

For even $t = 2m$, divide the product version of (36.4) by $B_{m,m+1}(n)^2$. The squared odd sign is one, and the identity becomes

$$\frac{n}{2m+1} = \frac{n+2m+1}{2m+1} - 1.$$

For odd $t = 2m+1$, write $s_N = (-1)^{N(N+1)/2}$. One has $s_N s_{N-2} = -1$ and $s_{N-1}^2 = 1$. Division by $B_{m+1,m+1}(n)^2$ reduces the same identity to

$$-\frac{n}{2(n+m+1)} = \frac{n+2m+2}{2(n+m+1)} - 1.$$

Thus the proposed products satisfy the condensation equation.

By the induction hypothesis, the actual denominator $h_{N-2}^{(t+2)}$ in solving (36.4) for $h_N^{(t)}$ is the proposed signed product, which is nonzero at nonnegative integer $N-2$. Solving is therefore legitimate. It proves the product at size N for every t , completing the induction. The nonzero denominator is a conclusion established earlier in this induction, not an assumption about arbitrary weighted Hankel minors.

36.3 Minimality and why the one-defect law excludes this point

The product $B_{a,b}(N)$ has degree ab and nonzero leading coefficient. The even product therefore has minimal polynomial $(X-1)^{m^2+1}$ by Lemma 34.1. In the odd case,

$$s_N = \frac{1+i}{2} i^N + \frac{1-i}{2} (-i)^N. \quad (36.12)$$

Both coefficients are nonzero. Multiplication by $B_{m,m+1}(N)$ gives both characteristic roots $i, -i$ with multiplicity $m(m+1)+1$, proving Theorem 30.4 in full.

For $t = 2m \geq 2$, the two grouped rates at $z = i$ are $1, -1$. Their uncorrected exponents would be $m^2 + 1$ and m^2 , so $R_{\text{grp}} = 2m^2 + 1$. The exact product has only the rate 1: the entire -1 group vanishes, and the actual order is $m^2 + 1$. The additional loss is m^2 , not one. Formula (34.6) correctly signals the problem because $z^2 + 1 = 0$ here.

For odd t , $L = 2$ has different parity from t , so there is no tied maximum at all. The unique-maximum argument already gives the correct odd central order. The product evaluation provides much more: all values and the exact period-four sign.

37 A root-free minimal-recurrence algorithm

The classification need not be implemented by factoring the quadratic for z , recognizing a root of unity, or computing Hankel determinants. There is a direct exact algorithm over the coefficient field of c .

37.1 Detecting a resonance by a short scalar recurrence

Put $v = c - 2$ and define

$$U_0 = 1, \quad U_1 = v, \quad U_{a+1} = vU_a - U_{a-1}. \quad (37.1)$$

These are scalar values of the Chebyshev polynomials of the second kind, $U_a(v/2)$; we suppress the argument to emphasize computation. From the recurrence,

$$U_a = \frac{z^{a+1} - z^{-a-1}}{z - z^{-1}}. \quad (37.2)$$

Since $z \neq \pm 1$, the first zero $U_{L-1} = 0$ occurs exactly at $L = \text{ord}(z^2)$, if that order is finite. Moreover,

$$\sigma := z^L = -U_{L-2} \in \{1, -1\}. \quad (37.3)$$

Indeed $U_L = (z^{L+1} - z^{-L-1})/(z - z^{-1}) = z^L$, while the recurrence at the zero gives $U_L = -U_{L-2}$.

Testing $U_1, \dots, U_{t-1}$ therefore decides whether the recurrence is resonant at the given multiplicity. There is no need to decide whether z has some much larger finite order: all $L > t$ have the same nonresonant formula. This requires $O(t)$ exact field operations and zero tests.

37.2 Linear and quadratic factors in the original coefficient field

Define another scalar sequence

$$S_0 = 2, \quad S_1 = v, \quad S_{a+1} = vS_a - S_{a-1}, \quad S_a = z^a + z^{-a}, \quad (37.4)$$

and set

$$Q_a(X) = X^2 - (-1)^t S_a X + 1, \quad E_a = \frac{t^2 - a^2}{4} + 1 \quad (a \equiv t \pmod{2}). \quad (37.5)$$

These factors have coefficients in $\mathbb{Q}(c)$, or in any characteristic-zero coefficient field containing c .

Theorem 37.1 (Root-free factorization). *For $c \neq 0, 4$, test (37.1) through U_{t-1} . If no zero occurs, then*

$$\chi_{t,c}(X) = (X - 1)^{\mathbf{1}_{2|t}(t^2/4+1)} \prod_{\substack{1 \leq a \leq t \\ a \equiv t \pmod{2}}} Q_a(X)^{E_a}. \quad (37.6)$$

The first factor is omitted when t is odd. If the first zero occurs at $L - 1$ with $L = 2$, use (30.15)–(30.16). If $3 \leq L \leq t$, compute σ by (37.3) and δ by (30.8), replacing z^{Lt} by σ^t . Then

$$\chi_{t,c}(X) = (X - 1)^{\mathbf{1}_{2|t}(t^2/4+1)} \prod_{\substack{1 \leq a < L \\ a \equiv t \pmod{2}}} Q_a(X)^{E_a} \cdot (X - (-1)^t \sigma)^{\eta(E_L - \delta)}. \quad (37.7)$$

The last factor is omitted when $\eta = 0$ or its exponent is zero; E_L is only used when $\eta = 1$. The nonconstant displayed factors have pairwise disjoint roots, and each displayed quadratic has distinct roots.

Proof. In the nonresonant case, pair the sectors $k, t - k$ and put $a = |2k - t|$. Their rates are $(-1)^t z^a$ and $(-1)^t z^{-a}$, whose quadratic is (37.5). Their common exponent is E_a . When t is even, the unpaired sector $k = t/2$ has rate one and exponent $t^2/4 + 1$. Distinctness follows from $L > t$.

In the resonant case, use the centered representatives in Section 35. The pairs $\pm a$ with $0 < a < L$ give exactly the quadratics in (37.7). The possible center $a = 0$ gives the first linear factor. The tied endpoints $a = \pm L$ occur exactly when $\eta = 1$ and give the single rate $(-1)^t \sigma$, with its one-defect exponent. No two different interior pairs coincide: equality of rates would force their corresponding k indices to agree modulo L , contrary to the choice of distinct centered representatives. An interior quadratic cannot have a double root because that would require $z^{2a} = 1$ with $0 < a < L$. If both linear factors occur, L is even, whence $z^L = -1$; they are therefore also distinct. Theorem 30.1 supplies minimality. $\square$

The theorem outputs $O(t)$ linear or quadratic factors with integer exponents after $O(t)$ scalar field operations. This is an arithmetic operation count for the *factored* answer; it is not a bit-complexity bound and not the cost of expanding a degree- $\Theta(t^3)$ polynomial. Exact equality tests in the input field are essential. Approximate floating-point tests cannot reliably detect an exact resonance.

This improves the task-specific route considerably over building a long prefix of large Hankel determinants and solving for an unknown recurrence. It does not replace the repository's more general algorithm for a multiplier with several different roots [Theorem 20.2 with the bound (20.6)].

38 Worked examples and a complete rational-parameter corollary

38.1 Two cubic multipliers with the same cyclotomic order

For $t = 3, c = 1$, choose $z = e^{2\pi i/3}$. Then $L = 3, z^L = 1$, and $\delta = 0$. The tied constant sectors retain the rate -1 . For $t = 3, c = 3$, choose $z = e^{\pi i/3}$. Then $L = 3, z^L = -1$, and $\delta = 1$. The tied constant sectors at rate 1 cancel entirely. The root-free factors are particularly simple:

$$\chi_{3,1}(X) = (X + 1)(X^2 - X + 1)^3, \quad R(3, 1) = 7, \quad (38.1)$$

$$\chi_{3,3}(X) = (X^2 + X + 1)^3, \quad R(3, 3) = 6. \quad (38.2)$$

For both inputs the generic order would be eight.

There are concise formulas for the determinants themselves. For $n \geq 0$,

$$\begin{aligned} H_{3n}^{(3)}(3) &= 2n + 1, \\ H_{3n+1}^{(3)}(3) &= -(n + 1)(18n + 13), \\ H_{3n+2}^{(3)}(3) &= (n + 1)(18n + 23). \end{aligned} \quad (38.3)$$

For $c = 1$, the six formulas are

r	$H_{6n+r}^{(3)}(1)$	$n = 0$	$n = 1$	$n = 2$
0	$(4n + 1)^2$	1	25	81
1	$(2n + 1)(4n + 1)$	1	15	45
2	$-(2n + 1)(4n + 3)$	-3	-21	-55
3	$-(4n + 3)^2$	-9	-49	-121
4	$-2(n + 1)(4n + 3)$	-6	-28	-66
5	$2(n + 1)(4n + 5)$	10	36	78

These are exact identities, not extrapolations. The sector theorem shows that each displayed progression is a polynomial of degree at most two: the rates have periods three and six, respectively, and all sector degrees are at most two. The first three values in each progression determine its quadratic. Evaluating them from (31.8) gives the displayed formulas. The accompanying code additionally verifies each identity through $N = 59$.

For reference, their first twelve values are

$$\begin{aligned} H_N^{(3)}(1) : & \quad 1, 1, -3, -9, -6, 10, 25, 15, -21, -49, -28, 36, \dots, \\ H_N^{(3)}(3) : & \quad 1, -13, 23, 3, -62, 82, 5, -147, 177, 7, -268, 308, \dots \end{aligned}$$

The difference between their minimal orders persists even though both are low-degree quasipolynomial sequences.

38.2 A noncentral loss below a positive degree

Take $c = 2 + \sqrt{3}$, $z = e^{\pi i/6}$, $L = 6$, and $t = 8$. The tied indices are 1, 7, of degree seven, at rate -1 . Since $L \equiv 2 \pmod{4}$, their leading terms cancel. Here $z^2 + 1 \neq 0$, so (34.6) proves that the grouped polynomial has degree six, not less. The full answer is

$$\chi_{8, 2+\sqrt{3}}(X) = (X-1)^{17}(X+1)^7(X^2+X+1)^{13}(X^2-X+1)^{16}. \quad (38.4)$$

Its order is 82, compared with grouped order 83 and generic order 93. This example illustrates the genuinely positive-degree part of the one-defect theorem, rather than just cancellation of two constants. The same recurrence holds at $c = 2 - \sqrt{3}$.

38.3 All rational nonendpoint roots

Corollary 38.1. *If $c \in \mathbb{Q} \setminus \{0, 1, 2, 3, 4\}$, then $R(t, c) = R_{\text{gen}}(t)$ for every $t \geq 1$. For $c = 1$ and $c = 3$, $R(1, c) = 2$. For $t \geq 2$ their orders are*

$$\begin{array}{c|cc} & c = 1 & c = 3 \\ \hline t \text{ even} & 3t^2/4 + 1 & 3t^2/4 + 1 \\ t \text{ odd} & (3t^2 + 1)/4 - \mathbf{1}_{t \equiv 1 \pmod{4}} & (3t^2 + 1)/4 - \mathbf{1}_{t \equiv 3 \pmod{4}}. \end{array} \quad (38.5)$$

For $c = 2$, the exact orders are

$$R(t, 2) = \begin{cases} t^2/4 + 1, & t \text{ even}, \\ (t^2 + 3)/2, & t \text{ odd}. \end{cases} \quad (38.6)$$

Proof. If z is a root of unity, $z + z^{-1}$ is an algebraic integer. A rational algebraic integer is an integer: substitution of a reduced fraction into its monic integer equation forces its denominator to be one. Also $-2 < z + z^{-1} < 2$ when $c \neq 0, 4$ is real and cyclotomic. Thus the only rational possibilities are $c = 1, 2, 3$. Every other rational nonendpoint input is nonresonant. At $c = 1, 3$ use $L = 3$ and $z^L = 1, -1$, respectively, in (30.10). The case $t = 1$ precedes the resonant range and is stated separately. The central formula is Theorem 30.4. $\square$

t	$R(t, 1)$	$R(t, 2)$	$R(t, 3)$	$R_{\text{gen}}(t)$
1	2	2	2	2
2	4	2	4	4
3	7	6	6	8
4	13	5	13	15
5	18	14	19	26
6	28	10	28	42
7	37	26	36	64
8	49	17	49	93
9	60	42	61	130
10	76	26	76	176
11	91	62	90	232
12	109	37	109	299

All entries follow from the proved formulas. The machine-readable version extends through $t = 20$.

39 Exact verification and reproducibility

The universal results above follow from the displayed proofs. The verification suite is an independent finite audit of formulas, signs, normalizations, and algorithmic implementations. No assertion of universality is inferred merely from the number of successful tests.

39.1 Arithmetic models and independent routes

The program `code/verify.py` (shipped here as `code/03-cyclotomic-verify.py`) uses Python's standard library only. Cyclotomic arithmetic takes place in

$$\mathbb{Z}[z]/(\Phi_M(z)), \quad (39.1)$$

with coefficient vectors in the usual basis of degree less than $\deg \Phi_M$. The monic polynomials Φ_M are constructed by exact successive division of $x^M - 1$ by the previously constructed factors for its proper divisors. All arithmetic is integral; comparisons do not use tolerances or approximate complex roots. Negative powers of z are reduced using $z^M = 1$.

The raw sector formula (32.3) is evaluated as an exact polynomial in N . Its degree, leading coefficient, first centered correction, and reflected polynomial are compared with the theorems. The sectors are then added according to equal rates, and their actual polynomial degrees are compared with every predicted characteristic multiplicity. The root-free algorithm is tested separately: it does not access the cyclotomic order or the spectral generator when constructing its factors from c . Its factors are checked against the actual grouped rates and multiplicities.

The original small Hankel determinants in the same rings are computed by the Leibniz permutation formula, not by the sector determinant. To compare with the rational sector expression, the test clears the common denominator. Explicitly it checks

$$H_N^{(t)}(c)(z^2 - 1)^T(z - 1)^t A_t = (-1)^{Nt} z^T \sum_{k=0}^t F_{t,k}(N; z) z^{(2k-t)N}. \quad (39.2)$$

Thus no inversion in a number field is needed even in this independent comparison.

For integer c , a second route evaluates the original moment matrix by pivoted fraction-free Bareiss elimination and compares it with the $t \times t$ normalized-jet determinant. The jets are generated from (31.2): if $u_{n,r} = p_n^{(r)}(c)/r!$, then

$$u_{n+1,r} = (c - 2)u_{n,r} + u_{n,r-1} - u_{n-1,r}, \quad u_{n,-1} = 0, \quad (39.3)$$

with $u_{0,0} = 1$, $u_{1,0} = c - 1$, $u_{1,1} = 1$, and other initial jets zero. This is independent of the subset expansion.

39.2 Parameter ranges and the recorded run

The principal cyclotomic grid uses primitive orders $3 \leq M \leq 24$ and root multiplicities $1 \leq t \leq 12$. A computation in $\mathbb{Z}[z]/\Phi_M$ is exact at a primitive root and respects all its Galois conjugates, rather than sampling a floating-point angle. The original cyclotomic determinant comparisons use $M \leq 14$, $t \leq 8$, and sizes $N = 0, \dots, 4$.

The integer determinant comparison uses $c \in \{-2, 0, 1, 2, 3, 4, 5\}$, $t = 1, \dots, 10$, and $N = 0, \dots, 10$. The endpoint values here only audit the classical determinant identity; they are not applications of the nonendpoint classification. The central product is checked for $t = 0, \dots, 16$ and $N = 0, \dots, 16$. The scalar sign and square-sum identities are checked on an additional grid with $M \leq 60$ and $t \leq 100$.

Exact assertion category	Checks passed
Canonical sector degrees	1,980
Product leading coefficients	1,980
First centered corrections	1,452
Full sector reflections	1,980
Grouped minimal multiplicities	1,569
Closed total-order formula	264
Root-free detection, factor degrees, and multiplicities	2,097
Independent cyclotomic Hankel determinants	480
Original integer matrices versus jet determinants	770
Central product evaluations	289
Four condensation ratio identities	252
Centered residue-class square sums	4,495
Simplified defect signs	4,396
Worked residue-class formulas	120
Total	22,124

The run completed successfully. The individual counts and the parameter-by-parameter recurrence orders are stored as JSON, and the rational examples are also stored as CSV. A check is one explicit assertion as counted in the program; the number is not a measure of proof strength or a substitute for peer review.

[Merge note: the recorded run is shipped as `data/03-cyclotomic-verification_summary.json` (status PASS and the twenty counts behind the table; the table's condensation row adds four counts of 63, and its root-free row adds $264 + 264 + 1,569$) and `data/03-cyclotomic-verification_run.txt` (the console transcript); the parameter-by-parameter orders, the 264 principal cases with their multiplicities by residue class, as `data/03-cyclotomic-cyclotomic_orders.json`; and the rational examples, the orders at $c = 1, 2, 3$ and the generic order for $t = 1, \dots, 20$, as `data/03-cyclotomic-orders_c1_c2_c3.csv` (all-CRLF as written by Python's csv module). All four were delivered in `data/` without the prefix.]

39.3 Rebuilding the article and the audit

From the package directory, run

```
python3 code/verify.py
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
```

The first command regenerates the data files. A smaller main grid is available with `-quick`; it is explicitly marked as such in its output. The supplied full-run transcript is `data/verification_run.txt`. The $\text{\LaTeX}$ source uses ordinary TeX Live packages, including `newtxtext` and `newtxmath`. The source bibliography is embedded, so no separate bibliography processor or network access is needed.

In this report these commands must not be run in place. Here `code/verify.py` is Part I's verifier, and the shipped `code/03-cyclotomic-verify.py` writes by default into the report's `data/`, adding the three unprefixed files `verification_summary.json`, `cyclotomic_orders.json` and `orders_c1_c2_c3.csv` beside the shipped ones; the transcript is its console output, not a file it writes. The delivered Makefile, shipped as `code/03-cyclotomic-Makefile`, has the same defect when run from the report directory: its `verify` target runs Part I's `code/verify.py`, which rewrites Part I's data files; its `quick` target passes that verifier options it does not know, and its `pdf` and `clean` targets build and clean the

merged `article.tex` in place. The article is built as described in the report's README. Restore the delivered layout in a scratch copy instead (POSIX shell, run from the report directory):

```
W=/path/to/scratch/cyclotomic
mkdir -p "$W/code" "$W/recorded"
cp code/03-cyclotomic-verify.py "$W/code/verify.py"
for f in data/03-cyclotomic-*; do
  cp "$f" "$W/recorded/${f#data/03-cyclotomic-}"; done
cd "$W" && python3 code/verify.py --output data > verification_run.txt
```

The run writes three files to `$W/data` and the transcript to `$W/verification_run.txt`; compare them with the copies in `$W/recorded`. Run this way on 29 September 2026 (Python 3.14.4, about 30 s), it passed; the CSV was byte-identical and the two JSON files and the transcript agreed with the recorded ones apart from Windows line endings. The quick grid is `python3 code/verify.py -quick -output data_quick`. The delivery README's example of the root-free routine, run in the same scratch directory, reproduces (38.4):

```
import sys
sys.path.insert(0, "code")
from verify import CyclotomicRing, root_free_factors

ring = CyclotomicRing(12)
c = ring.add(ring.integer(2), ring.add(ring.zp(1), ring.zp(-1)))
L, factors = root_free_factors(8, c, ring) # c = 2 + sqrt(3)
assert L == 6
assert sum((len(p) - 1) * e for p, e in factors) == 82
```

It returns pairs (polynomial coefficients in ascending order, exponent), and reads neither the spectral generator nor the cyclotomic order. The subset expansion in the program is deliberately exponential in t and serves testing; the root-free routine is the compact algorithm.

40 Further research questions and proposed next steps

The following are precise continuations of the present results. They are research tasks relative to this article; they are not all asserted to be globally unsolved problems in the published literature.

40.1 Several distinct repeated roots

For $q(x) = \prod_i (x - c_i)^{t_i}$, equal rates are described by multiplicative relations among several z_i . The degree function is $\sum_i k_i(t_i - k_i)$, so maximizing labels lie on a higher-dimensional lattice rather than a single arithmetic progression. Can one classify when several maximal labels cancel, and bound the additional loss in terms of the dimension or shape of that maximizing set? The one-dimensional proof shows exactly which ingredient is missing: a tractable sum of leading and first surviving coefficients over all maximal labels.

40.2 Endpoint factors coupled to a cyclotomic interior root

Investigate $x^a(x - 4)^b(x - c)^t$. The repository's degree law already [Theorem 14.2] adds fixed endpoint triangular contributions. The open structural issue is how those endpoint blocks alter the ratio of two tied interior leading coefficients and the first centered correction. Determine which endpoint multiplicities preserve a one-defect law and which produce higher-order cancellation.

40.3 Higher centered coefficients and a symmetric-function formula

The first correction arose from a logarithmic derivative of a signed Vandermonde. Higher corrections involve elementary symmetric functions of the mixed offsets $i + j + 1 - t$. Find a closed differential or symmetric-function expression for all centered coefficients of $P_{t,k}$. A particularly useful target is a formula exposing their parity under (33.8) and their factors at $z^2 = -1$. The goal is structural information, not merely a larger interpolation.

40.4 Uniform limits near a disappearing characteristic factor

The present formulas are exact at fixed parameters. They do not provide a uniform large- N approximation as c approaches a cyclotomic value, especially the center where whole sectors vanish. Determine the correct scaling of $c - c_0$ with N near a defect and identify the transition between the generic and specialized exponential-polynomial expansions. The exact subset formula is a finite starting point with no asymptotic error to hide.

40.5 Degrees and zeros on individual size progressions

At a cyclotomic parameter, restrict $H_N^{(t)}(c)$ to one residue class of N modulo a common rate period. Different characteristic rates now become the same, so additional cancellations can occur: (38.3) has a linear progression alongside two quadratic ones. Classify the exact degree and leading coefficient on each progression, and determine which progressions can vanish identically. This would lead to a precise description of the zero set of the Hankel determinant at cyclotomic parameters.

40.6 Other eventually constant Jacobi recurrences

Replace (31.2) by a family with a different finite initial Jacobi segment and the same constant tail. The exponential branches survive, but their amplitudes change. Determine a condition on those amplitudes under which only one extra multiplicity can be lost at a single-root cyclotomic resonance. Such a theorem would separate the universal lattice mechanism from the special Catalan boundary condition.

40.7 Effective arithmetic and reduction modulo primes

The root-free algorithm uses exact characteristic-zero equality. Determine a bit-complexity bound for algebraic inputs, including the cost of field arithmetic and coefficient growth, and study which primes preserve the characteristic-zero minimal polynomial on reduction. In positive characteristic, finite differences may lower polynomial degree abnormally and factorial factors may vanish, so neither the degree lemma nor the minimality theorem transports unchanged.

40.8 A focused formalization in Lean

The shortest reusable formalization targets are the exponential-block minimality lemma, the nearest-representative and square-sum lemmas, and the purely algebraic implication from the two sector coefficients to the one-defect law. These can be separated from integration and orthogonal polynomials. A second stage would formalize the signed confluent Vandermonde and the Christoffel identity, replacing the coefficient hypotheses by proved formulas. This staged plan must keep imported hypotheses visible rather than labeling the entire analytic argument formalized after only its final combinatorics has been checked.

41 Conclusion

The single repeated nonendpoint-root problem admits a complete sign-sensitive answer. When the rates are distinct, their nonzero sector leading coefficients give the known cubic generic order. At a cyclotomic resonance, the parabolic sector-degree function has at most one tied maximizing pair. Its leading ratio is a sign; when that sign is negative, the first centered correction proves that the loss is exactly one, except at the central root. The latter is an entire-group cancellation controlled by Cigler's known product.

This yields the minimal characteristic polynomial at every such parameter, not merely an annihilating bound. It also explains why t and the order of z^2 alone do not suffice, counts the exceptional parameters, and gives a root-free factorization computed directly from c by two short scalar recurrences. The proofs and exact audit are provided for independent checking. They establish the claims made here within ordinary mathematics; they do not establish priority or machine-checked formal status.

42 Normalization, boundary, and proof audit

This section is Appendix A of the manuscript.

Size rather than last index. N is the matrix size. An indexing convention with rows $0, \dots, n$ requires $N = n + 1$. The value at $N = 0$ is part of the sequence and equals one.

Multiplier sign. The multiplier is $(x - c)^t$, not $(c - x)^t$. This is the source of $(-1)^{Nt}$ in the Christoffel determinant and $(-1)^t$ in every characteristic root. Reversing the multiplier sign would scale the entire sequence by $(-1)^{Nt}$.

Jet factorials. The determinant uses $p_n^{(r)}(c)/r!$. Accordingly A_t appears once, in K_t , and not a second time in the confluence denominator. The subset Vandermondes use actual index differences, with their products of factorials kept explicitly.

Two different orders of a root of unity. The theorem uses $L = \text{ord}(z^2)$, while the audit ring uses a primitive M th root z , so $L = M/\text{gcd}(M, 2)$. For odd L , z can have order L or $2L$. These give opposite signs of z^L and can give different recurrence orders. This distinction is essential.

Tied constants. When $t = L$ and the defect sign is negative, $d_* = 0$. There is no degree -1 nonzero polynomial; the whole constant group is zero and its characteristic exponent is zero. This case is not covered by dividing the correction by d_* and must be handled separately, as in Section 34.

What the computation does not prove. Finite success at $t \leq 12$ does not prove an unbounded-multiplicity claim. The general mechanism is the explicit coefficient identity and the maximizing-pair lemma. Likewise, inspecting a finite set of sources does not prove historical novelty.

No unsupported nonvanishing assumptions. The Christoffel derivation divides only by a Vandermonde before confluence, not by weighted Hankel minors. The central condensation proof divides by a smaller minor only after its nonzero product formula has been established by induction. Minimality uses the nonzero coefficients actually calculated, not genericity by analogy.

43 Source and package concordance

This section is Appendix B of the manuscript.

The older root-collisions manuscript is now Part II of the repository article cited in Section 29. Its standalone Section 12.1 is the cyclotomic-classification question answered here; its Section 12.2 asks for structural cancellation criteria, addressed here for the single repeated root. The generic degree law and finite recurrence-recovery algorithm belong to that prior work. Its certified central fourth-power example is subsumed by the already known general product.

[Merge note: the standalone Sections 12.1 and 12.2 are Sections 24.1 and 24.2 here; the generic degree law and the recovery algorithm are Theorems 14.2 and 20.2; the certified central fourth-power example is (21.8).]

The present paper does not claim a general solution of multi-root resonance, uniform collision asymptotics, or positive-characteristic minimality. Those remain separate tasks. No code was added to the ProveIt repository and no Lean compiler was run for this article. [Merge note: this describes the manuscript at its pin. Its code and data have since been placed in this report, byte-identical, under the prefix 03-cyclotomic- (placement commit 3609d0473); nothing has been formalized.]

The archive contains `article.tex`, `article.pdf`, `README.md`, `SOURCES.md`, `Makefile`, `code/verify.py`, and the generated data directory. The JSON records distinguish actual grouped minimal multiplicities from generic order bounds. No external source PDFs or font files are included.

[Merge note: of these, `article.tex`, `article.pdf` and `README.md` are not shipped here; they survive in the arrival commit 8315d24e3. `SOURCES.md`, `Makefile`, `code/verify.py` and the four data files are shipped with the prefix 03-cyclotomic- (the `Makefile` in `code/`), as listed in Section 39.]

Part IV. Two-endpoint confluence

Part IV was added on 29 September 2026 in batch 53 of ProveIt's incoming-report intake, from one manuscript written against the pinned state of Parts I–III. It answers the endpoint half of Part II's research question “Uniform endpoint and root-collision limits” (Section 24.4): for any fixed numbers of roots approaching 0 and 4, with arbitrary collisions inside each cluster, it proves an exact even convergent expansion of the normalized determinant, its complete first correction, and a sharp outward scale. The question's other half, collisions of roots away from the endpoints, remains open. Parts I–III are unchanged except for dated pointers to this addition (Sections 11, 22 and 24.4), a line on the title page (whose vertical spaces were reduced by 12 mm in all so that it still fits on one page) and a paragraph of the abstract, a contents entry, three bibliography entries used only by Part IV (with a label column widened for the longer list), a clause in the closing note on source dates, and two new labels, on Part II's Section 24.4 and Part III's Section 40.2. Section 44 records the provenance and fixes the notation of the addition.

44 Provenance and conventions of Part IV

44.1 One manuscript, one addition

Part IV is batch-53 manuscript 02, *Two-Endpoint Confluence for Catalan Hankel Determinants*, subtitled *An exact even finite-size expansion, the first interaction law, and a sharp critical scale*, an AI-assisted research article prepared for Vladimir Reshetnikov and dated 29 September 2026 (20 A4 pages as delivered). Unlike the sources of Parts I–III, it does not name the assistant that prepared it. It arrived as the archive `Two_Endpoint_Catalan_Confluence.zip` (inner directory `Two_Endpoint_Catalan_Confluence/`, main file `article.tex`) in ProveIt commit 148a906aa and was placed in 1dc874990. It was written against ProveIt commit 1ee53d57de253d16cdaad79d1f54bbd95d76d682, reading this report's `article.tex` (Git blob e5a4d81408c80f0385ae66c26fbd11bba7dfdb09) and `README` through the GitHub connector; its source notes name Part II's Sections 22 and 24.4, and Part III, as the relevant material. The report's `article.tex` is unchanged between the pin and the placement commit (the same blob), so every statement the manuscript makes about “the repository”, “the source” or “the report” refers to Parts I–III as printed above.

The manuscript's program and recorded outputs are shipped beside the earlier files with the prefix `04-endpoint-confluence-` (Part I's unprefix files count as 01, Part II's as 02, Part III's as 03), its dependency file under `data/`, and its source notes and proof audit, delivered as `SOURCES.md` and `PROOF_AUDIT.md`, as `04-endpoint-confluence-SOURCES.md` and `04-endpoint-confluence-PROOF_AUDIT.md` at the report root. Its manuscript, delivery `README` and PDF are not shipped; they survive in the arrival commit. Section 60 lists the shipped files.

The manuscript contributes, in the order printed here: the repository question, what is classical, and a guide to the proof (Section 45); the two-endpoint model and its confluent cosine and sine kernels (Section 46); the four main statements, namely the exact even expansion, the first interaction law, the endpoint values and the sharp outward scale (Section 47); a self-contained Christoffel reduction (Section 48); the exact centred hyperbolic alternant that generates every coefficient (Section 49); the proof of the expansion, with an explicit remainder bound (Section 50); the derivation of the first correction (Section 51); positive Schur expansions of the outward profiles (Section 52); the proof of the sharp scale (Section 53); the answer in the question's own uncentred scaling, and exact spatial displacements (Section 54); worked examples (Section 55); its computations and verification boundary (Section 56); ten research directions (Section 57); a conclusion; a sign and normalization audit (Section 59); and its archive and reproduction notes (Section 60). Every result, proof, example, remark, limitation and question of the manuscript is printed. Where it re-derives something Parts I–III already prove, the overlap is marked (Section 44.5).

The manuscript's own abstract reads:

The Catalan-Hankel report in *ProveIt* asks for uniform limits when a repeated root approaches either endpoint of the Catalan moment interval. We solve that endpoint problem for arbitrary fixed multiplicities and, more generally, for two simultaneous clusters with arbitrary internal collisions. For total multiplier degree d , the effective size is $\nu = N + d/2$. After removing the endpoint sign and the power $\nu^{\binom{m}{2} + \binom{\ell+1}{2}}$, the determinant is the restriction of a holomorphic *even* function of $1/\nu$. Its expansion is convergent, locally uniform in all scaled root coordinates, and has an explicit Cauchy remainder bound. The leading coefficient factors into two confluent cosine/sine determinants. We derive the complete first correction, including a universal mixed derivative with coefficient $-1/4$. A centered hyperbolic Vandermonde identity generates every higher coefficient. Positive Schur expansions of the outward profiles then prove an exact scale criterion: the determinant is asymptotic to its fixed-endpoint value if and only if all outward displacements are $o(N^{-2})$; if the largest scaled displacement tends to infinity, the ratio diverges. Written proofs are accompanied by 810 exact-arithmetic and high-precision checks, clearly separated by their evidentiary status.

Status (manuscript 02). This is an unrefereed, AI-assisted manuscript with conventional written proofs, not a Lean or Rocq formalization. It resolves the endpoint portion of Part II's moving-root question, not its nonendpoint collision problem, and it does not resolve hard-edge universality for arbitrary weights or limits with multiplicities that grow with N : the multiplicities m and ℓ are fixed, and the constants and the analytic radius are not claimed to be uniform in them. The if-and-only-if scale criterion is restricted to outward, nonnegative displacements. Classical identities are credited, not presented as new: the fixed-size Christoffel reduction and its confluence (Krattenthaler [3], re-proved in the present normalization), characteristic-polynomial determinants, integrable kernels and hard-edge limits from random matrix theory [10, 11], and the Mehler–Heine phenomenon of which the one-variable cosine and sine limits are elementary half-integer cases [12]. The proposed contribution is the explicit combined package: the simultaneous two-endpoint, collision-uniform theorem, the centre $N + d/2$, the convergent expansion without odd powers, the first interaction operator, the all-orders coefficient generator, the explicit error bound and the sharp outward criterion. The literature check was bounded, not exhaustive, and worldwide priority for every identity or corollary is not established; no claim of settling a famous external conjecture is made. The proofs do not assume the report's sector-degree or resonance theorems. The 810 recorded checks (710 exact, 100 at 100 digits) count executions, not universal theorems; the high-precision checks are diagnostics, not outward-rounded interval certificates, and the shipped program verifies the leading term and the first correction; it is not an arbitrary-order coefficient engine. No proof assistant was run, nothing in Part IV is formalized, and its placement beside a Lean project confers no formal status. (Sources: the manuscript, its delivery README, and its source notes and proof audit, 04-endpoint-confluence-SOURCES.md and 04-endpoint-confluence-PROOF_AUDIT.md.)

44.2 Reading conventions

Inside Sections 45–60, “this article”, “the present article” and “here” mean manuscript 02 as printed in Part IV; “the repository”, “the repository's question”, “the source” and “the report” mean Parts I–III at the pin 1ee53d57d. The manuscript cites this report's own section numbers, which did not change: its “Part II, Section 24.4” is Section 24.4 and its “Section 22” is Section 22. “The archive” means the delivered package, whose program and data are shipped under the prefixed names listed in Section 60. The manuscript's two appendices are printed as Sections 59 and 60, so that Part I's appendices keep their letters. Its Sections 1–14 are Sections 45–58, so a manuscript number $n.j$ of a section, theorem, lemma, corollary or equation is printed as $(n + 44).j$ (for example, its Theorem 3.1 is Theorem 47.1), and its Appendices A and B are Sections 59 and 60. The manuscript cites Krattenthaler's Ramanujan Journal paper through arXiv:2101.04225v5, pointing to its Theorem 1 and “the confluent discussion in Section 5”; that is Part I's reference [3]. Its DLMF reference is Section 18.11(ii), not the Section 18.5 of Part I's [4], and is listed separately as [12]. Bracketed paragraphs beginning “Merge note” are this merge's, not the manuscript's.

44.3 Notation of Part IV

Part IV keeps the manuscript's symbols. Within Sections 45–60 every symbol has its Part IV meaning, except in merge notes and where a sentence names Part I, II or III or cites one of their equations. The multiplier has m roots near 0 and ℓ near 4 and no other roots; with all of them at the endpoints it is Part II's $q = \gamma x^m (x - 4)^\ell \prod_i (x - c_i)^{t_i}$ with $\gamma = 1$ and $s = 0$, and $H_N(q)$ is Part II's (14.1). Table 3 lists the symbols that a reader of Parts I–III is likely to misread. No symbol was renamed and no normalization changed. Two readings need particular care: the letter t is the variable of the kernels in Sections 46 to 53, but the multiplicity of one repeated root in (45.2) and Section 54, as in Part II's question; and the derivative columns of the kernels (46.8)–(46.9) are *not* divided by factorials, whereas repeated-root rows are (Section 59).

Table 3: Symbols and notions of Part IV. The last column gives the meaning of the same letter in Part I, II or III, or the tempting false reading.

Symbol or notion	Meaning in Part IV	Not to be confused with
$H_N(q)$	the weighted determinant (45.1), Part II's (14.1)	Part I's $H_N^{(m)} = H_N(x^m)$, whose superscript is a shift; Part III's $H_N^{(t)}(c) = H_N((x - c)^t)$.
$m, \ell, d = m + \ell$	numbers of roots near 0 and near 4; the degree of the multiplier	Part II's multiplicities m, ℓ of the exact roots 0, 4 (the same when the roots sit there); d is Part II's $D = \deg q$ here, not Part I's $d = \binom{m}{2}$ nor Part II's sector degrees d_k .
b_r, ρ, κ	$\binom{r}{2}, b_m + b_\ell, b_m + \binom{\ell+1}{2}$ (46.1)	κ is Part II's κ (14.5); b_r is not Part I's parameter b ; ρ is not Part II's modulus bound in (22.1) nor Part III's leading ratio (34.5).
$\nu = N + d/2, h = 1/\nu$	centred size and its inverse; the expansion is in h^2	Part II's block multiplicities ν_u ; Part I's moments ν_{i+j} ; Part III's $h = t - k$ (compare its centred size $w = N + t/2$).
t, s	variables of C, S and of the kernels; but t is a multiplicity in (45.2) and Section 54	Part I's generating-function variable t ; Part II's number s of interior roots; Part III's multiplicity t .
$C(t, y), S(t, y); C, S$ in (55.1)	$\cos(t\sqrt{y})$ and $\sin(t\sqrt{y})/\sqrt{y}$ as entire series (46.2)–(46.3); abbreviations of $C(1, y), S(1, y)$	the Catalan numbers C_n (always subscripted); Part II's $S = 2^D$ and matrix C ; Part III's subsets S .
$y_i, v_j; c_i^\pm; q_N$	scaled coordinates (negative real means outward); the moving roots (46.5); the size-dependent multiplier	Part II's fixed interior roots c_i and fixed multiplier q .
$Z_N, Z(h; \mathbf{y}, \mathbf{v}), A_j$	normalized determinant (46.7), its even interpolation, its coefficients	Part II's amplitudes $A_\pm(z)$; Part III's superfactorials A_u ; Part I's constant A_m .
$K_m^-, K_\ell^+; K_n^\varepsilon$	confluent cosine and sine kernels (46.8), (46.9); $\varepsilon = 0, 1$ selects them in Theorem 52.1	Part I's $K = (m - 1)^2$; Part III's common sector factor $K_t(z)$.
$\mathcal{A}_{m,\ell}$	endpoint value $A_0(\mathbf{0}, \mathbf{0})$ (47.5)	it equals Part II's leading coefficient for $q = x^m(x - 4)^\ell$ (Section 44.4).
$a, b; \mathbf{a}, \mathbf{b}$	b_m and $\binom{\ell+1}{2}$ in Corollary 47.3; limits of scaled displacements in Theorem 47.4; a fixed positive number in Section 53	Part I's parameters a, b of $x^m(a + bx)$, which appear in the dated note of Section 11 and in Section 44.4.
$\epsilon_{i,N}, \eta_{j,N}; \varepsilon$	outward displacements at 0 and 4; the kernel type 0 or 1	Part III's parity indicator η .
R_N, M_N	ratio to the fixed-endpoint determinant (47.8); N^2 times the largest displacement	Part II's Christoffel matrix M_N (15.7) and bound R ; Part III's minimal order $R(t, c)$.

Symbol or notion	Meaning in Part IV	Not to be confused with
$M(h; y, v), Q, E(h; u, w)$	the matrices (50.3), (49.1) and the quotient (50.5)	Part II's M_N ; Part I's $E_m(N)$ and the shift E ; the multiplier q .
$\mathcal{R}, \mathcal{R}_2$	hyperbolic generator (49.3) (constant term 1) and its h^2 coefficient (49.5)	Part II's maximal order $\mathcal{R}_0$; Part IV has no $\mathcal{R}_0$.
$\alpha_j; u_i, w_a; U, W; Y, V$	centred exponents $j - (d - 1)/2$; auxiliary variables and their sums; coordinate sums $\sum y_i, \sum v_j$	Part III's arcsine measure α and values U_a ; Part II's nodes w_a ; Part II's matrix V .
ϕ_h, c_h, s_h, D_h	the even entire functions of Section 50	Part III's sign s_N ; Part I's $D_N^{(m)}(a, b)$.
$\mathcal{D}_\pm$	Euler operators (47.4)	—
J_0, J_1, J_2, J_{11}	kernel minors with raised derivative orders (Section 51)	Part III's index set J .
k_j, λ, s_λ	strictly increasing exponents, their partition (52.1), the Schur polynomial	Part II's labels k and rates λ_k ; Part III's rates λ_k ; Part I's $\lambda_\pm$.
$R, r, T, B_{m,\ell,R}$	coordinate radius, disk radius, bound and constant of Theorem 50.2	Part III's $T = \binom{t}{2}$ and product $B_{a,b}(N)$; Part II's $T = \sum_i t_i$ and B_0, B_4 .
$F_t^\pm, \tau_t, \kappa_t, f_\varepsilon$	repeated-root profiles (54.1); κ_t is $\binom{t}{2}$ at 0 and $\binom{t+1}{2}$ at 4; $f_0 = C, f_1 = S$	—
$p_n, \mu; a_{n,r}$	monic orthogonal polynomials (48.2) and the Catalan measure (48.1); Taylor jets of p_n	the same as Part II's p_n, μ (Part I's $p_n^{(0)}, \mu_0$); Part III's centred moments μ_n .
Δ	the Vandermonde $\prod_{i < j} (u_j - u_i)$	Part II's Δ_q , with multiplicities; Part III's $\Delta(S)$.
$\text{sinc}, \text{sinhc}, (x)_r$	$\sin z/z, \sinh z/z$ (both 1 at 0), the falling factorial	—

44.4 What Part IV answers, and what stays open

Part II's Section 24.4 asked, for $c_N = 2 \pm 2 \cosh(\tau/N)$ and $q_N = (x - c_N)^t$, for the locally uniform limit of $H_N(q_N)$ normalized by $(-1)^t N^{t(t+1)/2}$ at the right endpoint and by $N^{t(t-1)/2}$ at the left, with its value at $\tau = 0$, a uniform confluent determinant limit and error bounds; and, in a second version, for two distinct nonendpoint roots approaching each other at rate N^{-1} . Part IV answers the first version. Corollary 54.1 gives the limits $F_t^+(\tau)$ and $F_t^-(\tau)$, entire and even in τ , their values at $\tau = 0$ (54.5), and the N^{-1} term, which is only the centring $N \mapsto N + t/2$; Theorem 50.2 gives an explicit remainder bound. It does so inside a larger statement: m roots near 0 and ℓ near 4 at once, any collision pattern within each cluster, coordinates in compact complex sets (Theorems 47.1 and 47.2), and a sharp outward scale (Theorem 47.4). For $t = 1$ the question noted that the limits are elementary hyperbolic functions; they are $F_1^-(\tau) = \cosh \tau$ and $F_1^+(\tau) = 2 \sinh \tau / \tau$ (Section 55). The second version, collisions away from the endpoints, is *not* answered: it is Part IV's first research question (Section 57.1), and Section 24.4 stays open for it.

The same results settle, for endpoint roots, the caveat that closes Part II's Section 22: its fixed-parameter estimate cannot be used with moving parameters, and Part IV proves the double-scaling limit directly, on the scale N^{-2} , without using that estimate. They also treat the limit $a \rightarrow 0$ that Part I's Section 11 names, at fixed m . For $b \neq 0$ and $\varepsilon = a/b$, $D_N^{(m)}(a, b) = b^N H_N(x^m(x + \varepsilon))$, a cluster of $m + 1$ roots at the endpoint 0, m of them fixed there. For real $\varepsilon \geq 0$ the root $-\varepsilon$ is an outward displacement, so by Theorem 47.4, with $\ell = 0$ and $m + 1$ roots, $D_N^{(m)}(a, b)/D_N^{(m)}(0, b)$ tends to 1 exactly when $\varepsilon = o(N^{-2})$, to the profile ratio (47.10) when $N^2 \varepsilon$ converges, and to infinity when $N^2 \varepsilon \rightarrow \infty$. Joint limits with m growing with N remain open (Section 57, "Multiplicities growing with determinant size"). Part III's research directions on endpoint factors coupled to a cyclotomic interior root (Section 40.2) and on uniform limits near a disappearing characteristic factor concern interior roots and are not addressed; Part IV's second research question (endpoint clusters in the presence of fixed interior roots) is close to the first of them.

Consistency with Parts II and III (observations of this merge, checked by a separate exact computation not shipped with the report).

1. The endpoint value (47.5) is Part II's leading coefficient (18.8) for $q = x^m(x-4)^\ell$ ($s = 0$, $\gamma = 1$): there $\mathcal{W} = ((-1, m), (1, \ell))$, $\Delta_q = 4^{m\ell}$ and $\mathcal{V}(\mathcal{W}) = (-1)^{b_m} 2^{m\ell} \prod_{h < m} h! \prod_{h < \ell} h!$, and $B_0(m)B_4(\ell)\mathcal{V}(\mathcal{W})/\Delta_q$ simplifies to $\mathcal{A}_{m,\ell}$; exact arithmetic agrees for $m, \ell \leq 3$. The sign $(-1)^{N\ell}$ removed in (46.7) is Part II's rate $(-1)^\ell$. So Corollary 47.3 is a second route to the case $s = 0$ of Theorem 18.1, and it adds the next coefficient. More is true at exact endpoints: Part II's single sector polynomial $P(N)$ of degree κ satisfies $v^{-\kappa}P = Z(1/v)$, and a polynomial in h that agrees with the even series $Z(h)$ of Theorem 47.1 at infinitely many points accumulating at 0 has only even powers; hence $P = \sum_j A_j(\mathbf{0}, \mathbf{0})v^{\kappa-2j}$ contains only powers of $v = N + d/2$ of the parity of κ . For example $H_N((x-4)^2) = v(4v^2-1)/3$ with $v = N+1$, and $H_N(x^2(x-4)^2) = v^2(v^2-1)/12$ with $v = N+2$.
2. The product (55.4) for $H_N(x^2(x-4)^2)$ is the same polynomial as Part II's (21.8) for $H_N((x-2)^4)$, which is Part III's central product $B_{2,2}(N)$; the coincidence of the two determinants is observed (both formulas are proved, and they agree for $N \leq 7$ by direct evaluation), not explained here.
3. The small determinants of Section 59 ($H_N(x)$, $H_N(x-4)$, $H_N(x(x-4))$, $H_N(x^2)$, $H_N((x-4)^2)$) agree with direct evaluation for $N \leq 7$, and the table of Section 55 agrees with the case both-double recorded in data/04-endpoint-confluence-verification.json.

44.5 Where the merge had to choose

1. *Placement.* Part IV is printed after Part III and before Part I's appendices, so that no section, equation, theorem or table number of Parts I–III moves (checked against the .aux file of a build of the committed text). Its sections continue the numbering (Sections 44 to 60), with the concordance of Section 44.2.
2. *Labels.* Every label of Part IV carries the prefix tec:; without it, seven of the manuscript's labels, thm:main, lem:Christoffel, eq:Christoffel, eq:measure, eq:R, eq:M and eq:Q, would have collided with Part I's bare labels. Labels were added to the manuscript's sixteen sections and to its first research subsection, and to Part II's Section 24.4 and Part III's Section 40.2 so that they can be cited; no existing label was changed.
3. *Overlaps with Parts I–III.* The manuscript re-derives, without assuming the report correct, several things already proved there. Printed with a pointer and kept: the determinant (45.1) (Part II's (14.1)); the Catalan measure, the monic orthogonal polynomials, their trigonometric form (48.3) and $H_N(1) = 1$ (a fourth route, after Part I's Section 3, Part II's Section 15, whose (15.3) is (48.3), and Part III's Section 31); the Gram integral identity (48.6) and the Christoffel identity (Lemma 48.1 is Part II's Lemma 15.1 for $\gamma = 1$, and Part III's Lemma 31.1 and Part I's Lemma 4.1 are other special cases); the endpoint jets (56.1), which are Part II's (16.9) and (16.10) (the signs $(-1)^{n+r}$ and $(-1)^{n-r}$ agree); and the endpoint value of Corollary 47.3 (Section 44.4). Everything else in Part IV (the evenness and convergence, the first interaction law, the alternant generator, the remainder bound, the Schur positivity and the sharp scale) has no counterpart in Parts I–III, which are fixed-parameter throughout. No statement of Parts I–III was changed.
4. *Bibliography.* The manuscript's Krattenthaler reference is Part I's [3]. Its references to Strahov–Fyodorov, Akemann–Fyodorov and DLMF Section 18.11(ii) are added to the common bibliography as [10], [11] and [12]. Its reference to the pinned repository report is replaced by Parts I–III themselves and the pin recorded in Section 44.1; the sentence that cited it (Section 45) now points there, and gains a pointer to Section 22 and a merge note. The manuscript's references to Part II's Section 24.4 are printed as cross-references to Section 24.4; the numbers are unchanged.
5. *File names.* Sections 56 and 60 give the shipped names beside the delivered ones in merge notes. The manuscript's reproduction command is printed as delivered but must not be run in place here, because the program would overwrite Part I's recorded outputs; a recipe on a scratch copy follows it.

6. *Front matter.* The manuscript's title page, abstract, scope box and contents are replaced by Section 44.1, which prints the abstract and the status.
7. *Notation and macros.* The manuscript's symbols are kept, with the scoping convention and Table 3 of Section 44.3; nothing was renamed. Its macros `\Del`, `\bv`, `\ba`, `\bb`, `\bu`, `\bw`, `\sinc`, `\sinhc` and `\fall` were added to the preamble with the manuscript's definitions; its `\by` and `\CC` are the report's, and its `\repo` is spelled out. No mathematical text of the manuscript was changed.

45 The repository question and the contribution

45.1 The endpoint moving-root question

For $C_n = \binom{2n}{n}/(n+1)$ and $q(x) = \sum_r q_r x^r$, put

$$H_N(q) = \det \left(\sum_r q_r C_{i+j+r} \right)_{0 \leq i, j < N}, \quad H_0(q) = 1. \quad (45.1)$$

The report *Shifted Catalan Hankel Polynomials* in *ProveIt* contains three parts: a shifted linear-multiplier analysis, an extension to arbitrary polynomial multipliers, and a single-root cyclotomic classification. Its Part II, Section 24.4, asks for the locally uniform limit associated with

$$c_N = 2 + 2 \cosh(\tau/N), \quad q_N(x) = (x - c_N)^t, \quad \frac{H_N(q_N)}{(-1)^{tN} N^{t(t+1)/2}}, \quad (45.2)$$

and for the corresponding left-endpoint limit with exponent $t(t-1)/2$. It specifically requests confluent determinant limits and error bounds. The same subsection separately asks about collisions away from the endpoints; that separate problem is not solved here.

The source was read at repository commit 1ee53d57de253d16cdaad79d1f54bbd95d76d682. The mathematical source file had blob e5a4d81408c80f0385ae66c26fbd11bba7dfdb09. The path, relevant sections, and comparison are recorded in Section 44.1 and the accompanying `SOURCES.md`. In particular, the source's fixed-parameter asymptotics (Section 22) expressly warn that they cannot simply be substituted into a moving-root limit.

[Merge note: the manuscript's `SOURCES.md` is shipped as `04-endpoint-confluence-SOURCES.md`; its bibliography item for the pinned report is replaced by Parts I–III themselves and the pin recorded in Section 44.1. Section 24.4 of Part II and Section 22 of Part II, as the manuscript cites them, are Sections 24.4 and 22 here: the numbers did not change.]

The results here address a larger fixed-degree problem. Any number m of roots may approach 0, any number ℓ may approach 4, and roots inside either cluster may coincide in any pattern. The approach may be from outside the interval, inside it, or in complex directions. Uniformity includes all these collisions on compact sets of the scaled coordinates.

45.2 What is classical, and what is established here

The reduction of a polynomially modified moment determinant to a fixed-size determinant of orthogonal polynomials is classical. Krattenthaler gives a detailed account, four proofs, a confluent version, and recurrence applications [3]. We use that identity, with a self-contained proof for the present normalization. The broader appearance of characteristic-polynomial determinants, integrable kernels, and hard-edge limits is likewise established in random matrix theory; see Strahov–Fyodorov and Akemann–Fyodorov [10, 11]. The one-variable cosine and sine limits are elementary half-integer cases of the Mehler–Heine phenomenon [12].

The contribution is the following explicit package for the repository's problem: a simultaneous two-endpoint, collision-uniform theorem; the exact center $N + d/2$; a genuinely convergent expansion containing no odd powers; a closed first interaction operator; an all-orders coefficient generator; an elementary, explicit error bound; and a necessary-and-sufficient outward stability criterion. These conclusions are proved directly, without assuming the repository's newer sector-degree or resonance claims. A bounded literature check does not establish priority for every identity or corollary in this package.

45.3 A guide to the proof

The decisive algebra is not a delicate large determinant estimate. Centering the columns turns an exponential alternant into a product of hyperbolic sines within each cluster and hyperbolic cosines between the clusters. After the forced zero is divided out, that product is even in the inverse size. A fixed-dimensional Christoffel formula transports this identity to Catalan determinants. The first correction follows from the quadratic term of the product; positivity of the limiting profiles then settles the sharp scale question.

46 Notation and the two-endpoint model

Fix nonnegative integers m, ℓ with $d = m + \ell \geq 1$. Write

$$b_r = \binom{r}{2}, \quad \rho = b_m + b_\ell, \quad \kappa = b_m + \binom{\ell+1}{2} = \rho + \ell, \quad \nu = N + \frac{d}{2}, \quad h = \nu^{-1}. \quad (46.1)$$

For a tuple $\mathbf{u}$, let $\Delta(\mathbf{u}) = \prod_{i < j} (u_j - u_i)$; empty products and empty determinants are 1.

All square-root expressions below denote entire power series in the squared variable, so no branch choice is part of the definition. Set

$$C(t, y) = \cos(t\sqrt{y}) = \sum_{k \geq 0} \frac{(-1)^k t^{2k} y^k}{(2k)!}, \quad (46.2)$$

$$S(t, y) = \frac{\sin(t\sqrt{y})}{\sqrt{y}} = \sum_{k \geq 0} \frac{(-1)^k t^{2k+1} y^k}{(2k+1)!}. \quad (46.3)$$

In particular $S(t, 0) = t$. For $\mathbf{y} \in \mathbb{C}^m$ and $\mathbf{v} \in \mathbb{C}^\ell$, define the moving roots and multiplier by

$$c_i^-(h) = 2 - 2\cos(h\sqrt{y_i}), \quad 1 \leq i \leq m, \quad (46.4)$$

$$c_j^+(h) = 2 + 2\cos(h\sqrt{v_j}), \quad 1 \leq j \leq \ell, \quad (46.5)$$

$$q_N(x) = \prod_{i=1}^m (x - c_i^-(h)) \prod_{j=1}^\ell (x - c_j^+(h)). \quad (46.6)$$

Thus $c_i^- = h^2 y_i + O(h^4)$ and $c_j^+ = 4 - h^2 v_j + O(h^4)$. Negative real coordinates mean outward motion. The normalized determinant is

$$Z_N(\mathbf{y}, \mathbf{v}) = (-1)^{N\ell} \nu^{-\kappa} H_N(q_N). \quad (46.7)$$

Define the two entire, symmetric kernel functions by

$$K_m^-(t; \mathbf{y}) = (-1)^{b_m} \frac{\det[\partial_t^r C(t, y_i)]_{1 \leq i \leq m, 0 \leq r < m}}{\Delta(\mathbf{y})}, \quad (46.8)$$

$$K_\ell^+(s; \mathbf{v}) = (-1)^{b_\ell} \frac{\det[\partial_s^r S(s, v_j)]_{1 \leq j \leq \ell, 0 \leq r < \ell}}{\Delta(\mathbf{v})}. \quad (46.9)$$

The quotients are defined at collisions by analytic continuation. More concretely, a block of r repeated coordinates z is replaced by rows of normalized derivatives $(1/a!) \partial_z^a$, $0 \leq a < r$; the remaining denominator is the product of cross-block differences to the products of their multiplicities. Set $K_0^\pm = 1$.

47 Main results

47.1 An exact even expansion

Theorem 47.1 (Two-endpoint confluence). *There are entire functions $A_j(\mathbf{y}, \mathbf{v})$, symmetric separately in each cluster, such that for all sufficiently large N ,*

$$Z_N(\mathbf{y}, \mathbf{v}) = \sum_{j=0}^{\infty} \frac{A_j(\mathbf{y}, \mathbf{v})}{v^{2j}}. \quad (47.1)$$

The series converges locally uniformly in the root coordinates and uniformly for $|h|$ in a sufficiently small closed disk. It is the Taylor series of an even holomorphic function $Z(h; \mathbf{y}, \mathbf{v})$ with $Z_N = Z(v^{-1}; \mathbf{y}, \mathbf{v})$. In particular,

$$A_0(\mathbf{y}, \mathbf{v}) = 2^{\ell-m\ell} K_m^-(1; \mathbf{y}) K_\ell^+(1; \mathbf{v}). \quad (47.2)$$

All statements include repeated roots, zero coordinates, and mixed collision patterns. For every fixed J the remainder after A_J is $O(v^{-2J-2})$ locally uniformly, including after any fixed number of derivatives in the coordinates.

The theorem is stronger than the existence of a formal asymptotic series. It constructs an analytic function of the inverse size. The absence of odd powers is exact, not a cancellation observed through a finite order. The restriction to fixed m, ℓ is important: the constants and the analytic radius are not claimed to be uniform in growing multiplicities.

47.2 The complete first correction

In the next statement every kernel and its derivatives are evaluated at $t = s = 1$. Put $Y = \sum_i y_i$ and $V = \sum_j v_j$.

Theorem 47.2 (First interaction law). *The coefficient of v^{-2} in (47.1) is*

$$A_1 = 2^{\ell-m\ell} \left\{ \frac{(m+3\ell+1)Y + (\ell+3m-1)V}{24} K_m^- K_\ell^+ - \frac{(\partial_t^2 K_m^-) K_\ell^+ + K_m^- (\partial_s^2 K_\ell^+)}{24} - \frac{(\partial_t K_m^-)(\partial_s K_\ell^+)}{4} \right\}. \quad (47.3)$$

This identity is entire in every coordinate; it does not divide by the leading profile, so it remains valid at zeros of A_0 .

At leading order the clusters decouple, up to the scalar $2^{\ell-m\ell}$. The first correction has a mixed derivative whose coefficient is always $-1/4$. It is therefore incorrect, in general, to multiply two one-endpoint finite-size expansions and expect the simultaneous two-endpoint correction.

An equivalent form uses the Euler operators

$$\mathcal{D}_- = b_m + 2 \sum_i y_i \partial_{y_i}, \quad \mathcal{D}_+ = \binom{\ell+1}{2} + 2 \sum_j v_j \partial_{v_j}. \quad (47.4)$$

On the product of the kernels, replace ∂_t by $\mathcal{D}_-$, ∂_t^2 by $\mathcal{D}_-(\mathcal{D}_- - 1)$, and analogously for s . This eliminates auxiliary variables from the correction formula.

47.3 The endpoint value

Corollary 47.3. *At $y = \mathbf{0}$, $v = \mathbf{0}$,*

$$A_0(\mathbf{0}, \mathbf{0}) = \mathcal{A}_{m,\ell} := 2^{\ell-m\ell+b_m+b_\ell} \prod_{r=0}^{m-1} \frac{r!}{(2r)!} \prod_{r=0}^{\ell-1} \frac{r!}{(2r+1)!} > 0. \quad (47.5)$$

Writing $a = b_m$ and $b = \binom{\ell+1}{2}$, one also has

$$\frac{A_1(\mathbf{0}, \mathbf{0})}{\mathcal{A}_{m,\ell}} = -\frac{a(a-1) + b(b-1) + 6ab}{24}. \quad (47.6)$$

47.4 A sharp scale criterion

For nonnegative real outward displacements define

$$\tilde{q}_N(x) = \prod_{i=1}^m (x + \epsilon_{i,N}) \prod_{j=1}^{\ell} (x - 4 - \eta_{j,N}), \quad (47.7)$$

$$R_N = \frac{H_N(\tilde{q}_N)}{H_N(x^m(x-4)^\ell)}, \quad M_N = N^2 \max\{\epsilon_{i,N}, \eta_{j,N}\}. \quad (47.8)$$

The denominator is nonzero and R_N is positive.

Theorem 47.4 (Exact outward stability threshold). *With fixed m, ℓ as above,*

$$R_N \longrightarrow 1 \iff M_N \longrightarrow 0. \quad (47.9)$$

If $N^2\epsilon_{i,N} \rightarrow a_i$ and $N^2\eta_{j,N} \rightarrow b_j$ with finite nonnegative limits, then

$$R_N \longrightarrow \frac{K_m^-(1; -\mathbf{a})}{K_m^-(1; \mathbf{0})} \frac{K_\ell^+(1; -\mathbf{b})}{K_\ell^+(1; \mathbf{0})}. \quad (47.10)$$

This limit exceeds 1 if at least one limit coordinate is positive. If $M_N \rightarrow \infty$, then $R_N \rightarrow \infty$.

The equivalence is restricted to outward, nonnegative displacements. Inside the interval, oscillations and zeros prevent the same monotonicity argument and may invalidate such an equivalence.

48 The Catalan moment functional and Christoffel reduction

48.1 Orthogonal polynomials with unit norms

The Catalan numbers are moments of the probability measure

$$d\mu(x) = \frac{1}{2\pi} \sqrt{\frac{4-x}{x}} dx, \quad 0 < x < 4. \quad (48.1)$$

Indeed, $x = 4u$ gives $\int x^k d\mu = (2/\pi)4^k B(k+1/2, 3/2) = C_k$. Define monic polynomials by

$$p_0 = 1, \quad p_1 = x - 1, \quad p_{n+1} = (x - 2)p_n - p_{n-1} \quad (n \geq 1). \quad (48.2)$$

Their elementary trigonometric expressions are

$$p_n(2 + 2\cos\theta) = \frac{\sin((n+1/2)\theta)}{\sin(\theta/2)}, \quad (48.3)$$

$$p_n(2 - 2\cos\theta) = (-1)^n \frac{\cos((n+1/2)\theta)}{\cos(\theta/2)}. \quad (48.4)$$

The identities follow by checking $n = 0, 1$ and the recurrence; the apparent singularities at special angles are removable where appropriate.

Under $x = 2 + 2 \cos \theta$, the measure is $(2/\pi) \sin^2(\theta/2) d\theta$ on $0 < \theta < \pi$. Thus the functions in (48.3) are orthonormal, since the half-integer sine functions have squared integral $\pi/2$ and distinct ones are orthogonal. In particular $\int p_n p_k d\mu = \delta_{nk}$ and $H_N(1) = 1$. These facts fix all normalizing constants used below.

48.2 A proof of the fixed-size determinant identity

Lemma 48.1 (Christoffel identity). For $q(x) = \prod_{a=1}^d (x - c_a)$,

$$H_N(q) = (-1)^{Nd} \frac{\det[p_{N+j}(c_i)]_{1 \leq i \leq d, 0 \leq j < d}}{\Delta(c_1, \dots, c_d)}. \quad (48.5)$$

The quotient is a symmetric polynomial in the c_a and hence extends to all collisions by normalized derivative rows.

Proof. Expanding two determinants and integrating term by term yields the Gram integral identity

$$H_N(q) = \frac{1}{N!} \int_{(0,4)^N} \Delta(x_1, \dots, x_N)^2 \prod_{a=1}^N q(x_a) d\mu(x_a). \quad (48.6)$$

The same identity holds for complex q , by finite multilinearity. The enlarged Vandermonde in the ordered tuple $(x_1, \dots, x_N, c_1, \dots, c_d)$ is

$$\Delta(\mathbf{x})\Delta(\mathbf{c}) \prod_{a,i} (c_i - x_a).$$

Replace its monomial columns by the monic columns $p_0, p_1, \dots$; the determinant does not change. Multiply by $\Delta(\mathbf{x}) = \det[p_{j-1}(x_i)]_{1 \leq i, j \leq N}$ and integrate. Laplace expansion on the first N rows followed by orthogonality kills every choice of N columns except the first N . The surviving integral is $N!$, and the remaining determinant is $\det[p_{N+j}(c_i)]$. Comparing with (48.6) gives (48.5), including the sign $(-1)^{Nd}$. An alternating polynomial is divisible by the Vandermonde. Taylor expansion of rows in a colliding block proves the normalized derivative rule and its factorials. $\square$

This proof is included to make the present article self-contained, not as a claim of a new Christoffel identity. The general identity and its confluence are explicitly treated in [3].

49 The centered hyperbolic alternant

The following finite determinant is the algebraic engine of the article. Let $\alpha_j = j - (d-1)/2$, $0 \leq j < d$. Introduce independent auxiliary variables $u_1, \dots, u_m, w_1, \dots, w_\ell$ and the $d \times d$ matrix

$$E(h; \mathbf{u}, \mathbf{w}) = \begin{bmatrix} [(-1)^j e^{\alpha_j h u_i}]_{i,j} \\ [e^{\alpha_j h w_a}]_{a,j} \end{bmatrix}. \quad (49.1)$$

We use $\sinh z = \sinh(z)/z$, with value 1 at zero.

Lemma 49.1 (Exact centered alternant). One has the identity of entire functions

$$\det E = (-1)^{b_m} 2^{m\ell} h^\rho \Delta(\mathbf{u}) \Delta(\mathbf{w}) \mathcal{R}(h; \mathbf{u}, \mathbf{w}), \quad (49.2)$$

where

$$\mathcal{R} = \prod_{i < i'} \sinh\left(\frac{h}{2}(u_{i'} - u_i)\right) \prod_{a < a'} \sinh\left(\frac{h}{2}(w_{a'} - w_a)\right) \prod_{i,a} \cosh\left(\frac{h}{2}(w_a - u_i)\right). \quad (49.3)$$

Consequently $\mathcal{R}$ is even in h , symmetric within each tuple, and its constant term is 1.

Proof. Factor $e^{-h(d-1)u_i/2}$ from each lower row and $e^{-h(d-1)w_a/2}$ from each upper row. The remaining determinant is the Vandermonde at nodes $-e^{hu_i}$ and e^{hw_a} . Within the first cluster its factors are $-2e^{h(u_i+u_{i'})/2} \sinh(h(u_{i'} - u_i)/2)$; within the second cluster they are the analogous positive factors. Between clusters they are $2e^{h(u_i+w_a)/2} \cosh(h(w_a - u_i)/2)$. Every variable occurs in exactly $d - 1$ exponential factors, so the centering cancels all exponentials. Extract h times each within-cluster difference. There are ρ such differences and $m\ell$ cross pairs. This gives (49.2), including its sign and power of 2. $\square$

49.1 The quadratic generator

The first nonconstant term is

$$\mathcal{R} = 1 + h^2 \mathcal{R}_2 + O(h^4), \quad (49.4)$$

where, writing $U = \sum_i u_i$ and $W = \sum_a w_a$,

$$24\mathcal{R}_2 = (m + 3\ell) \sum_i u_i^2 - U^2 + (\ell + 3m) \sum_a w_a^2 - W^2 - 6UW. \quad (49.5)$$

Indeed, a within-cluster pair contributes its squared difference divided by 24, and a cross pair contributes its squared difference divided by 8. Summing the differences gives the displayed formula.

49.2 The all-orders meaning

Expand the right side of (49.3) to any prescribed order. The coefficient of h^{2j} is a finite symmetric polynomial with rational coefficients. In the alternant on the left, expanding each row in powers of h identifies the coefficients with minors of $[(-1)^j \alpha_j^r / r!]$ and $[\alpha_j^s / s!]$. Equivalently, expand $\mathcal{R}$ in the two Schur bases; a Schur partition records exactly which derivative orders replace the consecutive orders $0, \dots, m - 1$ or $0, \dots, \ell - 1$.

This supplies an all-orders coefficient algorithm. It can also be read as an identity of constant-coefficient differential operators: replace u_i, w_a by derivatives in separate row variables and apply the result to a product of the functions C and S . Each finite Taylor coefficient uses only finitely many derivatives. Convergence is justified below by holomorphy, rather than by a purely formal interchange of infinite operators.

50 Proof of the even, collision-uniform expansion

50.1 An exact normalized formula

Put

$$\phi_h(y) = \frac{2 - 2 \cos(h\sqrt{y})}{h^2}, \quad \phi_0(y) = y, \quad (50.1)$$

$$c_h(y) = \cos(h\sqrt{y}/2), \quad s_h(y) = \text{sinc}(h\sqrt{y}/2). \quad (50.2)$$

All three functions are entire in (h, y) and even in h . Define

$$M(h; \mathbf{y}, \mathbf{v}) = \begin{bmatrix} [(-1)^j C(1 + \alpha_j h, y_i)]_{i,j} \\ [S(1 + \alpha_j h, v_a)]_{a,j} \end{bmatrix}. \quad (50.3)$$

Using (48.3)–(48.4) in (48.5) yields

$$Z(h; \mathbf{y}, \mathbf{v}) = \frac{2^\ell (-1)^{b_\ell} h^{-\rho} \det M}{\prod_i c_h(y_i) \prod_a s_h(v_a) \Delta(\phi_h(\mathbf{y})) \Delta(\phi_h(\mathbf{v})) \prod_{i,a} [4 - h^2(\phi_h(y_i) + \phi_h(v_a))]} \quad (50.4)$$

At $h = (N + d/2)^{-1}$ the right side equals Z_N .

For completeness, the powers and signs are as follows. The m lower rows supply $(-1)^{Nm}$, which combines with the Christoffel sign to leave $(-1)^{N\ell}$. The upper rows each supply $2/h$. The lower within-cluster Vandermonde is $h^{2b_m} \Delta(\phi_h(y))$; the upper one is $(-1)^{b_\ell} h^{2b_\ell} \Delta(\phi_h(v))$. Multiplication by h^κ leaves $h^{-\rho}$, as in (50.4). No additional factorials belong in this formula.

50.2 Removal of the apparent singularities

Lemma 50.1. *The function*

$$Q(h; y, v) = \frac{\det M(h; y, v)}{h^\rho \Delta(y) \Delta(v)} \quad (50.5)$$

is entire and even in h , and entire in every coordinate.

Proof. Taylor-expand each row in h . In a nonzero determinant term the selected derivative orders within one cluster must be distinct, since equal orders would give proportional rows in the corresponding coefficient matrix. Their smallest possible sum is b_m for the lower cluster and b_ℓ for the upper cluster. Thus the determinant vanishes to order at least ρ . Its quotient by h^ρ is entire, jointly with the coordinates. Alternation in the coordinates of either cluster then removes the two Vandermondes. One may justify joint removal by successive analytic divided differences, which also gives the stated confluent rule.

For parity, reverse the d columns. Since $\alpha_{d-1-j} = -\alpha_j$, one obtains

$$\det M(-h) = (-1)^{b_d+m(d-1)} \det M(h) = (-1)^\rho \det M(h).$$

The last equality follows because $b_d + m(d-1) - \rho = m(m-1) + 2m\ell$ is even. Dividing by h^ρ proves evenness. Alternatively, every odd coefficient vanishes in (49.2). $\square$

Write

$$D_h(y) = \prod_{i < i'} \frac{\phi_h(y_{i'}) - \phi_h(y_i)}{y_{i'} - y_i}, \quad (50.6)$$

with derivatives at coincidences, and similarly for v . Then (50.4) is $2^\ell (-1)^{b_\ell} Q$ divided by $\prod c_h \prod s_h D_h(y) D_h(v)$ and the cross factors. At $h = 0$ these denominator factors are respectively 1, 1, 1, 1 and $4^{m\ell}$. They remain nonzero on a sufficiently small disk, uniformly on any compact coordinate set. This proves holomorphy and convergence of the even Taylor series.

The least-order term in the row expansion uses precisely the derivative orders $0, \dots, m-1$ and $0, \dots, \ell-1$. By the constant term of (49.2), its coefficient is

$$(-1)^{b_m} 2^{m\ell} \det[\partial_t^r C(1, y_i)] \det[\partial_s^r S(1, v_a)].$$

Substitution in (50.4) gives (47.2). All Taylor coefficients are entire in the coordinates, because every denominator has a nonzero constant in h and every numerator coefficient is an entire alternating quotient. Uniform convergence of differentiated series follows from Cauchy's estimates on a slightly larger compact set. This proves Theorem 47.1.

50.3 An explicit remainder bound

Theorem 50.2 (A conservative effective bound). *Suppose $|y_i|, |v_a| \leq R$, $R \geq 0$. Let*

$$r = \frac{1}{4\sqrt{R+1}}, \quad T = 1 + \frac{d-1}{2}r, \quad (50.7)$$

$$B_{m,\ell,R} = 2^{\ell+d+\rho} d! r^{-\rho} T^d \exp(dT\sqrt{R+1}). \quad (50.8)$$

Then Z is holomorphic on a neighborhood of $|h| \leq r$ and has modulus at most $B_{m,\ell,R}$ on $|h| = r$. For $|h| < r$,

$$\left| Z(h) - \sum_{j=0}^J A_j h^{2j} \right| \leq B_{m,\ell,R} \frac{(|h|/r)^{2J+2}}{1 - (|h|/r)^2}. \quad (50.9)$$

Proof. Here $|h\sqrt{y}| \leq 1/4$. The power-series bounds give $|c_h(y) - 1| < 1/2$, $|s_h(y) - 1| < 1/2$, and

$$|\phi'_h(y) - 1| = |\operatorname{sinc}(h\sqrt{y}) - 1| \leq \frac{\sinh(1/4)}{1/4} - 1 < \frac{1}{50}.$$

A divided difference of ϕ_h is an integral of its derivative on the straight segment between its nodes. It consequently has distance less than $1/50$ from 1. Also $|\phi_h(y)| \leq (51/50)R$, so each cross factor has modulus greater than 3. In particular all denominator factors in the rewritten version of (50.4) are nonzero; for an upper bound their inverse product is at most $2^{d+\rho}$.

To bound the determinant divided by the coordinate Vandermondes, replace rows by successive divided differences within each cluster. Every such row entry is bounded by the supremum of its original entire row function on $|y| \leq R + 1$, by Cauchy's derivative estimate and the integral formula for divided differences. On that larger disk, $|1 + \alpha_j h| \leq T$, and both C and S have modulus at most $T \exp(T\sqrt{R+1})$. The determinant is therefore bounded by $d!$ times the d th power of this number. On $|h| = r$, division by h^ρ costs $r^{-\rho}$. The factor 2^ℓ in (50.4) gives (50.8). Finally apply Cauchy's coefficient bound to the even Taylor series and sum its geometric tail. $\square$

The constants deliberately favor transparency over sharpness. They are explicit analytic bounds, whereas the numerical tests later in the article are not interval certificates.

51 Derivation of the first interaction coefficient

51.1 The five relevant minors

For a cluster of size n , denote the determinant with column derivative orders $0, \dots, n-1$ by J_0 . Let J_1, J_2, J_{11} use, respectively,

$$(0, \dots, n-2, n), \quad (0, \dots, n-2, n+1), \quad (0, \dots, n-3, n-1, n).$$

Set $J_{11} = 0$ if $n < 2$. The symbols refer to either the C or S cluster. They may be divided by the same Vandermonde and multiplied by the same sign as the base kernel.

The coefficient of degree two in the centered alternant gives the following ratios to the base coefficient. The row coefficients use $\alpha_j^r/r!$, so the factorials in this table are already included.

Derivative selection	Coefficient ratio
Lower J_2 , upper J_0	$(m + 3\ell - 1)/24$
Lower J_{11} , upper J_0	$-(m + 3\ell + 1)/24$
Lower J_0 , upper J_2	$(\ell + 3m - 1)/24$
Lower J_0 , upper J_{11}	$-(\ell + 3m + 1)/24$
Lower J_1 , upper J_1	$-1/4$

A selection involving a nonexistent cluster is simply absent. To verify the table directly, use $s_{(2)} - s_{(1,1)} = \sum u_i^2$ and $s_{(2)} + s_{(1,1)} = (\sum u_i)^2$ in (49.5). The term $UW = s_{(1)}(\mathbf{u})s_{(1)}(\mathbf{w})$ gives the mixed coefficient. There are no degree-one selections.

51.2 Reduction to ordinary kernel derivatives

Every row function satisfies $\partial_t^2 f(t, y) = -yf(t, y)$. Differentiating J_0 column by column leaves only the last column, hence

$$J'_0 = J_1, \quad J''_0 = J_2 + J_{11}, \quad J_2 - J_{11} = -\left(\sum y_i\right)J_0. \quad (51.1)$$

For the last identity, sum the determinants obtained by applying two derivatives to one column. The differential equation says that this sum is the coefficient of a diagonal row multiplication by $-y_i$, hence $-(\sum y_i)J_0$. All terms except the last two columns vanish; the next-to-last term is $-J_{11}$ because its two last columns are reversed. These identities are polynomial determinant identities and hold even where $J_0 = 0$.

It follows from the table that the normalized numerator correction is

$$-\frac{(K_m^-)''K_\ell^+ + K_m^-(K_\ell^+)''}{24} - \frac{(m+3\ell)Y + (\ell+3m)V}{24}K_m^-K_\ell^+ - \frac{(K_m^-)'(K_\ell^+)'}{4}. \quad (51.2)$$

51.3 The geometric denominator correction

The remaining factors must not be omitted. Their expansions are

$$\begin{aligned} \prod_i c_h(y_i)^{-1} &= 1 + \frac{h^2 Y}{8} + O(h^4), \\ \prod_a s_h(v_a)^{-1} &= 1 + \frac{h^2 V}{24} + O(h^4), \\ D_h(y)^{-1} &= 1 + \frac{h^2(m-1)Y}{12} + O(h^4), \\ D_h(v)^{-1} &= 1 + \frac{h^2(\ell-1)V}{12} + O(h^4), \\ \prod_{i,a} [4 - h^2(\phi_h(y_i) + \phi_h(v_a))]^{-1} &= 4^{-m\ell} \left(1 + \frac{h^2(\ell Y + mV)}{4} + O(h^4) \right). \end{aligned}$$

For example $\phi_h(y) = y - h^2 y^2/12 + O(h^4)$ proves the two divided Vandermonde expansions even at collisions. Adding these contributions to (51.2) gives exactly (47.3). This proves Theorem 47.2.

Finally, rescaling the kernel determinants gives

$$K_m^-(t; \mathbf{y}) = t^{b_m} K_m^-(1; t^2 \mathbf{y}), \quad (51.3)$$

$$K_\ell^+(s; \mathbf{v}) = s^{\binom{\ell+1}{2}} K_\ell^+(1; s^2 \mathbf{v}). \quad (51.4)$$

The identities first follow for nonzero t, s by extracting powers from rows, columns and Vandermondes, then extend to zero by holomorphy. Differentiation at 1 proves the Euler-operator form.

52 Positive Schur expansions of the outward profiles

52.1 A coefficientwise-positive formula

For a strictly increasing tuple $0 \leq k_0 < \dots < k_{n-1}$, define the partition

$$\lambda_i = k_{n-i} - (n-i), \quad 1 \leq i \leq n. \quad (52.1)$$

Let s_λ denote the Schur polynomial, equivalently the quotient $\det[a_i^{k_j}]/\Delta(\mathbf{a})$ in this ordering.

Theorem 52.1 (Positive outward expansion). *For $\varepsilon = 0, 1$, write K_n^ε for K_n^- and K_n^+ , respectively. Then*

$$K_n^\varepsilon(1; -\mathbf{a}) = 2^{b_n} \sum_{0 \leq k_0 < \dots < k_{n-1}} \frac{\Delta(k_0, \dots, k_{n-1})}{\prod_{j=0}^{n-1} (2k_j + \varepsilon)!} s_\lambda(\mathbf{a}). \quad (52.2)$$

The series converges locally uniformly on $\mathbb{C}^n$ and all its coefficients in the variables a_i are nonnegative real numbers. Its constant term is

$$K_n^\varepsilon(1; \mathbf{0}) = 2^{b_n} \prod_{j=0}^{n-1} \frac{j!}{(2j + \varepsilon)!}. \quad (52.3)$$

For $n \geq 1$, the coefficient of $a_1 + \dots + a_n$ is strictly positive.

Proof. At negative coordinates, the series for the row functions has positive coefficients in a . Apply Cauchy–Binet to its power-series factorization. For chosen powers $k_0 < \dots < k_{n-1}$, the second determinant has entries

$$\frac{(2k_i + \varepsilon)_r}{(2k_i + \varepsilon)!}, \quad 0 \leq i, r < n.$$

The numerator is the Vandermonde in $2k_i + \varepsilon$, because the falling factorials are monic polynomials of their degrees. It equals $2^{b_n} \Delta(\mathbf{k})$. The sign in the definition of the kernel cancels the sign of $\Delta(-\mathbf{a})$, leaving (52.2).

This computation may first be carried out for truncated series. Their entire row functions converge with all derivatives on compact sets, so the determinant divided differences converge there as well. Alternatively, on a positive diagonal $a_i = A$ the series is monotone and bounded by the confluent determinant limit; its positive coefficients then majorize the series on $|a_i| \leq A$. This gives local uniform convergence. Schur polynomials have nonnegative coefficients: the branching identity

$$s_\lambda(a_1, \dots, a_n) = \sum_{\mu \text{ interlaces } \lambda} a_n^{|\lambda| - |\mu|} s_\mu(a_1, \dots, a_{n-1})$$

proves this by induction, starting with a monomial in one variable. The identity follows from the bialternant formula after clearing its Vandermonde. The smallest tuple is $k_j = j$, which gives the constant term since $\Delta(0, 1, \dots, n-1) = \prod_j j!$. The tuple $(0, 1, \dots, n-2, n)$ gives $\lambda = (1)$ and a strictly positive coefficient. For $n = 1$ the same assertion uses $k_0 = 1$. $\square$

52.2 Consequences and a useful relative error bound

For nonnegative $\mathbf{a}$,

$$K_n^\varepsilon(1; -\mathbf{a}) \geq K_n^\varepsilon(1; \mathbf{0}) > 0. \quad (52.4)$$

The inequality is strict if at least one coordinate is positive, and every coordinate derivative of the profile is nonnegative. In particular, $A_0(-\mathbf{a}, -\mathbf{b}) \geq \mathcal{A}_{m,\ell}$. Thus the absolute error in Theorem 50.2, divided by $\mathcal{A}_{m,\ell}$, is a valid relative-to-leading-term bound on an outward compact set.

Formula (52.3) and the homogeneity laws immediately give (47.5) and (47.6). This proves Corollary 47.3. The two different triangular exponents are now visible directly: the cosine determinant contributes b_m , whereas the sine-divided-by-root determinant contributes $b_\ell + \ell$.

53 Proof of the sharp outward critical-scale theorem

53.1 Monotonicity at every finite size

Multiply (48.6) by $(-1)^{N\ell}$ for the outward multiplier. Its integrand contains

$$\prod_{a=1}^N \left\{ \prod_i (x_a + \epsilon_{i,N}) \prod_j (4 + \eta_{j,N} - x_a) \right\}.$$

Every factor is positive for $0 < x_a < 4$. The integral is positive, and increasing any displacement increases the integrand pointwise. Consequently $R_N \geq 1$ and it is nondecreasing in every displacement. No assumption of distinct roots is required.

53.2 Finite scaled coordinates

For an outward displacement $\epsilon \geq 0$, define its angular coordinate by

$$a(h, \epsilon) = h^{-2} \operatorname{arcosh}(1 + \epsilon/2)^2. \quad (53.1)$$

Then $2 - 2 \cos(h\sqrt{-a}) = -\epsilon$. Since $\operatorname{arcosh}(1 + \epsilon/2)^2 = \epsilon + O(\epsilon^2)$ as $\epsilon \rightarrow 0$, bounded $N^2\epsilon$ implies $a(h, \epsilon) - N^2\epsilon \rightarrow 0$. The same coordinate works for $4 + \eta$. Apply Theorem 47.1 to these bounded moving coordinates and divide by its value at zero. This proves (47.10), including $R_N \rightarrow 1$ when $M_N \rightarrow 0$.

53.3 Necessity and escape from the critical window

Suppose M_N does not tend to zero. Along a subsequence, one fixed coordinate has N^2 times its displacement at least some $a > 0$; there is a fixed coordinate after passing to a further subsequence because d is fixed. By finite-size monotonicity, R_N is bounded below by the ratio with that displacement replaced by a/N^2 and all other displacements set to zero. Its limit is strictly greater than 1 by Theorem 52.1. Hence R_N cannot tend to 1.

Now suppose $M_N \rightarrow \infty$. Fix $A > 0$. For all sufficiently large N , some displacement is at least A/N^2 . The same comparison bounds R_N below by the minimum of the finitely many one-coordinate comparison ratios. For each location, its limiting Schur series has a positive linear coefficient in A , so the minimum of those finitely many limits tends to infinity as $A \rightarrow \infty$. Given any desired lower bound, choose A first and then N sufficiently large. This proves $R_N \rightarrow \infty$ and completes Theorem 47.4.

The divergence conclusion is not a large-coordinate use of the compact asymptotic theorem. It follows by comparison with arbitrarily large *fixed* critical coordinates, which is why no unjustified uniformity at infinity is needed.

54 The original scaling and other coordinate choices

54.1 An explicit answer to the repository's question

Let $\tau_t = (-\tau^2, \dots, -\tau^2) \in \mathbb{C}^t$. Define

$$F_t^-(\tau) = K_t^-(1; \tau_t), \quad F_t^+(\tau) = 2^t K_t^+(1; \tau_t). \quad (54.1)$$

Both functions are entire in τ and even. At repeated coordinates they are the explicit $t \times t$ determinants

$$K_t^\varepsilon(1; (z, \dots, z)) = (-1)^{b_t} \det \left[\frac{1}{i!} \partial_z^i \partial_s^j f_\varepsilon(s, z) \right]_{s=1} \Big|_{0 \leq i, j < t}, \quad (54.2)$$

where $f_0 = C$ and $f_1 = S$.

Corollary 54.1 (Original uncentered formulation). *Locally uniformly for $\tau \in \mathbb{C}$,*

$$\frac{H_N((x - (2 - 2 \cosh(\tau/N)))^t)}{N^{t(t-1)/2}} = F_t^-(\tau) + O(N^{-1}), \quad (54.3)$$

$$\frac{H_N((x - (2 + 2 \cosh(\tau/N)))^t)}{(-1)^t N^{t(t+1)/2}} = F_t^+(\tau) + O(N^{-1}). \quad (54.4)$$

The endpoint values are

$$F_t^-(0) = 2^{b_t} \prod_{j=0}^{t-1} \frac{j!}{(2j)!}, \quad F_t^+(0) = 2^{t+b_t} \prod_{j=0}^{t-1} \frac{j!}{(2j+1)!}. \quad (54.5)$$

If κ_t is the corresponding triangular exponent, the first uncentered correction is explicitly

$$F_t^\pm(\tau) + \frac{t}{2N} \left(\kappa_t + \tau \frac{d}{d\tau} \right) F_t^\pm(\tau) + O(N^{-2}). \quad (54.6)$$

Proof. Put $\nu = N + t/2$. In the centered theorem the coordinates are $-(\nu/N)^2 \tau^2$, and the normalization changes by $(\nu/N)^{\kappa_t}$. Expand this analytic rescaling. The centered finite-size correction itself starts at ν^{-2} , so the entire N^{-1} term is exactly the coordinate and normalization change. All expansions are uniform on compact sets. Formula (54.5) follows from (52.3). $\square$

Thus the $O(N^{-1})$ term in the originally posed coordinates is real, but it is a centering effect. Replacing N by $N + t/2$ in both the angular scale and the normalization removes it and every later odd inverse power.

54.2 Exact spatial displacements also have an even expansion

Suppose the desired roots are $h^2 y_i$ and $4 - h^2 v_a$, rather than (46.5). Inverting ϕ_h near zero gives

$$\widehat{y} = y + \frac{h^2 y^2}{12} + O(h^4), \quad \phi_h(\widehat{y}) = y.$$

The inverse is analytic in h^2 and locally uniform in y . Therefore the exact spatial-coordinate model also has an even convergent expansion. Its first correction is

$$A_1^{\text{spatial}} = A_1 + \frac{1}{12} \left(\sum_i y_i^2 \partial_{y_i} + \sum_a v_a^2 \partial_{v_a} \right) A_0. \quad (54.7)$$

This demonstrates that evenness is not confined to a trigonometric notation for the roots. The center of the determinant size, however, is essential to the stated form.

55 Worked examples and checks on the constants

55.1 One root at the left endpoint

For $m = 1$, $\ell = 0$, $\nu = N + 1/2$ and $\kappa = 0$, the exact formula is

$$Z_N(y) = \frac{\cos \sqrt{y}}{\cos(h\sqrt{y}/2)}.$$

Hence $A_0 = \cos \sqrt{y}$ and $A_1 = (y/8) \cos \sqrt{y}$. At $y = -\tau^2$ this is $\cosh \tau$ divided by $\cosh(h\tau/2)$, and the limiting profile is $F_1^-(\tau) = \cosh \tau$. This also makes the sharp N^{-2} threshold visible without determinants.

55.2 One root at the right endpoint

For $m = 0, \ell = 1$,

$$Z_N(v) = \frac{2 \sin \sqrt{v}/\sqrt{v}}{\text{sinc}(h\sqrt{v}/2)}.$$

Thus $A_0 = 2 \sin \sqrt{v}/\sqrt{v}$ and $A_1 = vA_0/24$. The outward profile is $F_1^+(\tau) = 2 \sinh \tau/\tau$, with value 2 at zero. These two one-root examples verify both boundary normalizations.

55.3 One root at each endpoint

For $m = \ell = 1$, write $C = C(1, y)$ and $S = S(1, v)$. Then

$$A_0 = CS, \quad A_1 = \left(\frac{y}{4} + \frac{v}{6}\right)CS - \frac{1}{4}C_t(1, y)S_t(1, v). \quad (55.1)$$

The mixed term generally does not vanish. At $y = v = 0$ the exact unsigned endpoint determinant is $N + 1$, and the centered normalization with $v = N + 1$ is identically 1; accordingly $A_1(0, 0) = 0$.

55.4 Two coincident roots at an endpoint

The two repeated-root profiles from (54.2) simplify to

$$F_2^-(\tau) = \frac{1}{2} + \frac{\sinh(2\tau)}{4\tau}, \quad (55.2)$$

$$F_2^+(\tau) = \frac{\sinh(2\tau) - 2\tau}{\tau^3}. \quad (55.3)$$

The singularities at zero are removable; the values there are 1 and $4/3$, respectively. To derive them, evaluate the 2×2 determinant in (54.2), or differentiate the two elementary hyperbolic row functions before taking the coincident limit.

For example, with two roots at each endpoint, all exactly at the endpoints, one has

$$H_N(x^2(x-4)^2) = \frac{(N+1)(N+2)^2(N+3)}{12}. \quad (55.4)$$

This follows by inserting the endpoint jets in the four-row Christoffel determinant and factoring the resulting polynomial. Writing $v = N + 2$ gives $Z_N = (1 - h^2)/12$, agreeing with both (47.5) and (47.6).

55.5 A genuinely confluent two-cluster numerical example

Take $m = \ell = 2$, $y = (-1, -1)$ and $v = (-2, -2)$. The first two coefficients, computed from the confluent kernels, are

$$A_0 = 0.1741166983662596749147659 \dots, \quad A_1 = -0.7183984032902893460064576 \dots$$

The following are high-precision evaluations, not interval enclosures. Here $h = 1/(N + 2)$.

N	$ Z_N - A_0 $	$ Z_N - A_0 - h^2 A_1 $	$(Z_N - A_0 - h^2 A_1)/h^4$
16	$2.2027646 \cdot 10^{-3}$	$1.4514422 \cdot 10^{-5}$	1.52366601
32	$6.2030784 \cdot 10^{-4}$	$1.1440675 \cdot 10^{-6}$	1.52885861
64	$1.6484093 \cdot 10^{-4}$	$8.0651949 \cdot 10^{-8}$	1.53034944
128	$4.2503422 \cdot 10^{-5}$	$5.3595798 \cdot 10^{-9}$	1.53074959

The corrected residual displays the predicted fourth-order behavior. The proof of the error order is Theorem 47.1, not the table.

56 Computation, reproducibility, and the verification boundary

56.1 Two independent determinant paths

The accompanying code/verify.py (shipped as code/04-endpoint-confluence-verify.py) uses exact rational arithmetic for its algebraic tests. One path forms the original Catalan moment matrix from (45.1). A second path uses the fixed-size confluent Christoffel determinant. The latter obtains Taylor jets from (48.2): if $p_n(c + u) = \sum_r a_{n,r} u^r$, then

$$a_{n+1,r} = (c - 2)a_{n,r} + a_{n,r-1} - a_{n-1,r}, \quad a_{n,-1} = 0.$$

No trigonometric root extraction is needed for these exact tests. The endpoint jet identities checked separately are

$$\frac{p_n^{(r)}(0)}{r!} = (-1)^{n+r} \binom{n+r}{2r}, \quad \frac{p_n^{(r)}(4)}{r!} = \frac{2n+1}{2r+1} \binom{n+r}{2r}, \quad (56.1)$$

with zero values when $r > n$. Both follow from the recurrence, or the trigonometric formulas expanded at the endpoints.

The code also evaluates the normalized kernel jets by the convergent entire series

$$\frac{1}{q!} \partial_y^q \partial_t^r f_\varepsilon(t, y) = \sum_{\substack{k \geq q \\ 2k+\varepsilon \geq r}} \frac{(-1)^k \binom{k}{q} y^{k-q} t^{2k+\varepsilon-r}}{(2k+\varepsilon-r)!}. \quad (56.2)$$

This handles zero coordinates and repeated roots without dividing by small root differences. High-precision tests use 100 decimal digits. They include inward roots, complex roots, exact endpoints, and mixed repeated-root blocks of total degree up to eight in the endpoint coefficient checks.

56.2 Executed checks

The run recorded in data/verification.json and data/verification.txt completed the following checks.

[Merge note: those two records are shipped as data/04-endpoint-confluence-verification.json and data/04-endpoint-confluence-verification.txt. In this report the unprefix data/verification.json and data/verification.txt are Part I's recorded outputs; see Section 60 before rerunning anything.]

Check	Count
Exact original determinants versus confluent Christoffel determinants	104
Exact endpoint Taylor-jet identities	378
Exact leading constants of centered coefficient minors	35
Exact vanishing of degree-one centered minors	60
Exact second-order one-cluster minor ratios	108
Exact mixed minor ratios $-1/4$	25
High-precision first-correction residual checks	40
High-precision analytic evenness checks	6
Analytic continuation formula versus Christoffel evaluation	6
High-precision endpoint leading and correction constants	48
Total; all passed	810

The first six rows comprise 710 exact-arithmetic checks. The last four rows comprise 100 high-precision diagnostic checks. A check count measures executed identities, not distinct universal results. In particular, finite tests of the table of minors do not replace the all-parameter proof from (49.2).

56.3 Computing higher coefficients

For a specified order J , expand the finite product (49.3) through degree $2J$. Convert its separately symmetric coefficients to Schur polynomials, or directly to the coefficient minors of the centered exponential matrix. Replace the corresponding alternants by kernel determinants with the indicated increased derivative orders. Finally multiply by the truncated reciprocal series of c_h, s_h, D_h and the cross factors in (50.4). Only finite operations are needed for each J .

This is a constructive mathematical algorithm, but the shipped script is a verification implementation for the leading and first correction, not a fully optimized arbitrary-order engine. The bound in Theorem 50.2 can be applied once the desired exact coefficients or certified enclosures for them have been computed. The present high-precision implementation does not perform outward-rounded interval arithmetic.

56.4 A formalization route

No Lean or Rocq file is claimed for this article. A feasible formalization would begin with the polynomial Vandermonde identity underlying (49.2), the first coefficient table over $\mathbb{Q}$, and (51.1). Formal power series can express the exact parity and coefficient algorithm independently of analysis. A second layer would prove the Catalan moment functional and the Christoffel identity. Only the final layer needs holomorphic divided differences, uniform convergence, and the Cauchy remainder estimate. Keeping those layers separate would make the new algebra reusable for other moment families.

57 Further research questions and proposed next steps

The following are research directions, not assertions that every item is globally open in the published literature.

57.1 Nonendpoint collisions on the reciprocal-size scale

The unsolved second portion of the repository's question lets two nonendpoint roots approach at order N^{-1} . Construct a uniform counterpart of the present theorem with oscillatory sector amplitudes retained. A useful first target is two equal-multiplicity roots with spectral coordinates $ze^{\pm\tau/N}$ and $z \neq \pm 1$, allowing z on the unit circle. Here different exponential sectors can compete; the endpoint factorization alone does not settle their cancellation.

57.2 Endpoint clusters in the presence of fixed interior roots

Add a fixed polynomial multiplier whose roots avoid 0 and 4. Determine the correct effective size and the influence of its boundary amplitudes on evenness. A finite perturbation of the Jacobi recurrence may alter the amplitude without changing the local oscillatory scale. An exact criterion for disappearance of odd powers would separate structural symmetry from a choice of normalization.

57.3 Higher interaction operators

The quadratic generator produces the universal mixed term $-1/4$. Compute the quartic and sextic operators in a minimal basis of Euler operators and symmetric coordinate polynomials. Determine whether the logarithm of the coefficient generator yields a particularly simple connected-interaction expansion. Its elementary factors already show that only even orders can occur, but do not automatically give the most compact differential representation.

57.4 Multiplicities growing with determinant size

The theorem keeps m, ℓ fixed. Find a regime $m = m_N, \ell = \ell_N$ in which the factorized leading profile or the first correction remains uniform. The explicit bound here is deliberately too coarse to locate a sharp growth threshold. Such a threshold should account simultaneously for the number of pair interactions, growth of the confluent kernels, and the shrinking domain on which an effective relative error is small.

57.5 Other Jacobi endpoint exponents

Replace the Catalan weight by $x^\alpha(4-x)^\beta$ with $\alpha, \beta > -1$. Determine which effective size optimizes convergence, and whether an exact even analytic interpolation survives beyond the half-integer trigonometric cases. The appropriate limiting kernels are Bessel-type, so the classical hard-edge theory supplies context; the specific objective is an explicit collision-uniform coefficient and remainder calculus analogous to the one proved here.

57.6 Zeros of inward confluent profiles

For positive scaled coordinates, the kernel determinants can vanish. Classify real zeros of the one-parameter repeated-root profiles and establish simplicity or interlacing where true. Holomorphic convergence and a nonzero derivative at a limiting zero would then give quantitative locations of zeros of the finite Catalan determinant. The absolute first-correction formula remains meaningful there even when a relative formula would be singular.

57.7 Degeneration of finite recurrences

For each fixed multiplier the repository gives a finite exponential-sector recurrence description. Explain how those recurrences degenerate under the present moving-root limit and recover differential equations for the profile determinants. A rigorous bridge between coalescing characteristic roots and the centered even series would connect the repository's algebraic and asymptotic descriptions rather than treating them separately.

57.8 Sharper analytic bounds and stable numerical evaluation

Locate the nearest genuine singularities of the analytic interpolation in h , distinguishing removable singularities from denominator zeros. Use that geometry to improve (50.9) and select a practical truncation order. Develop interval implementations using divided differences or jets so that collisions are handled without loss of significance. This would convert the available analytic error estimates into end-to-end numerical certificates.

57.9 Rational modifications and Cauchy-transform blocks

Allow poles of a rational multiplier away from the integration interval, with selected poles approaching the endpoints on the same scale. Characteristic-polynomial ratio identities suggest determinant blocks involving Cauchy transforms [10, 11]. Determine which sign, centering, and evenness statements survive, and what replaces the monotonicity argument for the sharp outward stability criterion.

57.10 A machine-checked coefficient and confluence library

Formalize the centered alternant and the second-order interaction formula first, then implement a verified finite coefficient generator. A reusable library should state its divided-difference confluence conventions

explicitly and track every factorial. The final target is a checked interface connecting exact rational algebra, analytic remainder bounds, and certified evaluations of the confluent kernels.

58 Conclusion

The repository's endpoint moving-root question has an explicit, collision-uniform answer for every fixed multiplicity, with a simultaneous two-endpoint extension. The principal structural fact is exact evenness in $(N + d/2)^{-1}$ after the natural sign and power are removed. The two endpoints decouple in the leading term but interact at the next order through a computable differential operator. Positive Schur coefficients turn the asymptotic profile into a sharp necessary-and-sufficient criterion for stability under outward perturbations.

The scope remains precise: the work proves the endpoint results stated here, not a general solution of all root collisions, all recurrence resonances, or hard-edge universality for arbitrary weights. The result is a conventional mathematical manuscript with explicit tests; neither formal verification nor worldwide priority is asserted.

59 A sign and normalization audit

This section is Appendix A of the manuscript.

The following elementary checks are useful when porting the formulas. For one multiplier factor, (48.5) says $H_N(x - c) = (-1)^N p_N(c)$. Thus

$$H_N(x) = 1, \quad H_N(x - 4) = (-1)^N (2N + 1).$$

For two endpoint factors the same formula and $p_n(0) = (-1)^n$, $p_n(4) = 2n + 1$ give $H_N(x(x - 4)) = (-1)^N (N + 1)$. For two repeated lower factors the derivative determinant gives $H_N(x^2) = N + 1$; for two upper factors it gives

$$H_N((x - 4)^2) = \frac{(N + 1)(2N + 1)(2N + 3)}{3}.$$

These identities include $N = 0$. They fix the asymmetric triangular powers and the factor 2^ℓ at the right endpoint. They also verify that derivatives in a repeated-root block must be divided by their factorials, whereas the derivative columns in (46.8) and (46.9) are *not* factorial-normalized.

At a tuple of all zero coordinates, the kernel coefficient matrix has entries $(-1)^i (2i + \varepsilon)_j / (2i + \varepsilon)!$. Its determinant has sign $(-1)^{b_n}$ and nonzero magnitude $2^{b_n} \prod i! / (2i + \varepsilon)!$, exactly as used in (52.3). This is a direct independent derivation of the endpoint constant without evaluating a Selberg integral.

60 Archive contents and reproduction

This section is Appendix B of the manuscript.

The archive contains the TeX source and compiled PDF, `code/verify.py`, its recorded data, a reproduction README, source notes, and a dependency file. After installing `requirements.txt`, run `python code/verify.py`; the script writes to `data/` regardless of the working directory. Exact tests use the standard library; `mpmath` is used only for numerical diagnostics. Rebuild with `latexmk -pdf article.tex`, or repeated `pdflatex` runs. No external images or custom font files are required.

[Merge note: of these, the TeX source, the 20-page PDF and the reproduction README are not shipped here; they survive in the arrival commit 148a906aa. The program, its two recorded outputs, the source notes, the proof audit and the dependency file are shipped byte-identical with the prefix

04-endpoint-confluence-:

- `code/04-endpoint-confluence-verify.py` (delivered as `code/verify.py`);

- `data/04-endpoint-confluence-verification.json` (delivered as `data/verification.json`);
- `data/04-endpoint-confluence-verification.txt` (delivered as `data/verification.txt`);
- `data/04-endpoint-confluence-requirements.txt` (delivered as `requirements.txt`; it pins `mpmath==1.3.0`);
- `04-endpoint-confluence-SOURCES.md` and `04-endpoint-confluence-PROOF_AUDIT.md` at the report root (delivered as `SOURCES.md` and `PROOF_AUDIT.md`).

The command above must *not* be run in this directory. Here `code/verify.py` is Part I's verifier, and the shipped Part IV program, which takes no arguments, always writes `verification.json` and `verification.txt` into the `data/` directory beside its own `code/` directory: run in place as `code/04-endpoint-confluence-verify.py`, it would overwrite Part I's recorded `data/verification.json` and `data/verification.txt`. On Windows it also writes CRLF line endings, whereas the recorded files are LF. Restore the delivered layout in a scratch copy instead (a POSIX shell, from the report directory):]

```
W=/path/to/scratch/endpoint-confluence
mkdir -p "$W/code" "$W/recorded"
cp code/04-endpoint-confluence-verify.py "$W/code/verify.py"
for f in data/04-endpoint-confluence-verification.*; do
  cp "$f" "$W/recorded/${f#data/04-endpoint-confluence-}"; done
cd "$W" && python3 code/verify.py # writes $W/data; compare with $W/recorded
```

[Run this way on 29 September 2026 with Python 3.13.5 and `mpmath` 1.3.0, it printed `TOTAL: 810` in about two seconds, and both outputs equalled the recorded ones apart from Windows line endings. The checks are Python assertions, so do not pass `-O`.]

A The precise correction to the printed conjecture

This appendix distinguishes two kinds of index. The source uses a size- $(n + 1)$ determinant. We write $N = n + 1$. To avoid silently identifying incompatible shift conventions, write s for the source's symbol in its displayed coefficient formula, and reserve m for our actual Catalan shift.

For $s \geq 2$ and $0 \leq k \leq n + 1$, the coefficient expression printed in Conjecture 2 of [1] can be written as

$$T_{n,k,s} = \frac{C_s}{2s-2} \binom{s+k-2}{s-2} \binom{n+k+2s-2}{2k+2s-3} \times \prod_{j=0}^{\lfloor (s-1)/2 \rfloor - 1} \frac{\binom{2n+2s-2j-1}{2s-4j-5}}{\binom{2s-2j-1}{2s-4j-5}} \prod_{j=0}^{s-3} \frac{2s-j-2}{n+k+2s-j-2}. \quad (\text{A.1})$$

Only the natural, nonambiguous range $s \geq 2$ is needed here. Empty products have value 1.

Proposition A.1 (Correct shift). *For $N \geq 1$, $s \geq 2$, and $0 \leq k \leq N$,*

$$T_{N-1,k,s} = c_{N,s-1,k}. \quad (\text{A.2})$$

Consequently, the polynomial using $T_{N-1,k,s}$ is $D_N^{(s-1)}(a, b)$, not $D_N^{(s)}(a, b)$. Equivalently, to use that printed expression for an actual shift $m \geq 1$, replace its shift symbol by $s = m + 1$.

Proof. Taking a ratio of successive terms in (A.1) and cancelling factorials gives

$$\frac{T_{n,k+1,s}}{T_{n,k,s}} = \frac{(n+1-k)(n+k+s+1)}{2(k+1)(2k+2s-1)}. \quad (\text{A.3})$$

This agrees with (2.3) after $N = n + 1$ and $m = s - 1$. It remains to match the initial coefficient. Putting $n = N - 1$, $k = 0$, and simplifying the factorials gives

$$T_{N-1,0,s} = C_s \frac{(N+s-1)!}{N!s!} \prod_{j=0}^{p-1} \frac{\binom{2N+2s-2j-3}{2s-4j-5}}{\binom{2s-2j-1}{2s-4j-5}}, \quad p = \left\lfloor \frac{s-1}{2} \right\rfloor. \quad (\text{A.4})$$

At $N = 1$ this is $C_s = H_1^{(s)}$. Its ratio at successive N is

$$\begin{aligned} \frac{T_{N,0,s}}{T_{N-1,0,s}} &= \frac{N+s}{N+1} \prod_{j=0}^{p-1} \frac{(2N+2s-2j-1)(2N+2s-2j-2)}{(2N+2j+4)(2N+2j+3)} \\ &= \frac{(2N+2s)!(2N+1)!}{(2N+s)!(2N+s+1)!}. \end{aligned} \quad (\text{A.5})$$

In the product, the numerator factors form a consecutive interval of integers, as do the denominator factors; the last equality holds for either parity of s . The shifted-product formula gives the identical ratio

$$\frac{H_{N+1}^{(s)}}{H_N^{(s)}} = \prod_{i=1}^{s-1} \frac{(2N+2i+1)(2N+2i+2)}{(2N+i+1)(2N+i+2)} = \frac{(2N+2s)!(2N+1)!}{(2N+s)!(2N+s+1)!}.$$

Induction in N therefore proves $T_{N-1,0,s} = H_N^{(s)}$. The ratio (A.3) then proves (A.2) for every k . $\square$

The discrepancy is visible in the smallest nontrivial instance. With $N = 1$ and $s = 2$, the printed expression is

$$T_{0,0,2}b + T_{0,1,2}a = 2b + a,$$

whereas the determinant with the stated shift $s = 2$ is

$$D_1^{(2)}(a, b) = aC_2 + bC_3 = 2a + 5b.$$

At $a = b = 1$, these are 3 and 7. The expression $a + 2b$ is exactly $D_1^{(1)}(a, b)$, so the correction is systematic rather than an isolated numerical exception. The original PDF page was inspected as well as its HTML rendering; this is not an artifact of an HTML conversion. The denominator-exponent statement is separate and uses the actual shift m correctly.

B Compact reference and archive conventions

Symbol	Meaning
N	Determinant size, with $D_0 = 1$
m	Actual Catalan shift in $aC_{i+j+m} + bC_{i+j+m+1}$
$H_N^{(m)}$	Pure shifted Catalan determinant
d, e, K	$\binom{m}{2}$, $\binom{m}{2} + 1$, and $(m-1)^2$
P_m	Uniform numerator, not automatically reduced at exceptions
q	$1 - (a + 2b)t + b^2t^2$
$M_{m,\ell}$	Minor with column ℓ deleted from the binomial block
$Q_m(N; r)$	Signed generating polynomial of those minors
$r_{\pm}$	Roots of $r^2 - (a/b + 2)r + 1$
$\lambda_{\pm}$	$br_{\pm}$, the characteristic roots in determinant size
s in Appendix A	Source shift symbol, equal to actual shift plus one

[Added 28 September 2026, batch 39: this table covers Part I. Part II reuses several of these letters with other meanings, among them q , s , ℓ , d , K , P , M and r ; its symbols are listed in Table 1, and its data files in Section 27.]

The JSON files use ascending polynomial powers. In `coefficient_rows.json`, position k is the coefficient of $a^k b^{N-k}$. In `minor_polynomials.json`, position j is the coefficient of N^j . In `two_branch_polynomials.json`, the arrays `P_plus` and `P_minus` describe the polynomial multipliers of 4^N and 1^N for $(a, b) = (1, 2)$; they are not the ordinary-generating-function numerator P_m .

The source audit records exact URLs, source versions, and search limitations. The tests establish reproducibility of the computation, not priority or formal verification. The archive intentionally contains no third-party article PDFs, no font files, no compiler auxiliaries, and no Python bytecode.

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